Class 10

Mathematics

Solutions

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1. Matrices and Determinants

LEVEL – I

1. Find $x + y + z + w$

Given:

$$3\begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} x & 6 \\ -1 & 2w \end{bmatrix} + \begin{bmatrix} 4 & x+y \\ z+w & 3 \end{bmatrix}$$

LHS:

$$\begin{bmatrix} 3x & 3y \\ 3z & 3w \end{bmatrix}$$

RHS:

$$\begin{bmatrix} x+4 & 6+x+y \\ z+w-1 & 2w+3 \end{bmatrix}$$

Comparing corresponding elements:

$3x = x + 4 \Rightarrow x = 2$

$3y = 6 + x + y = 6 + 2 + y \Rightarrow 2y = 8 \Rightarrow y = 4$

$3w = 2w + 3 \Rightarrow w = 3$

$3z = z + w - 1 = z + 3 - 1 \Rightarrow 2z = 2 \Rightarrow z = 1$

Therefore,

$x + y + z + w = 2 + 4 + 1 + 3 = 10$

Correct Answer: A) 10

2. Find the value of $\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} [\,4 \;\; 5 \;\; 2\,] \begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix}$

First compute:

$$[\,4\ 5\ 2\,]\begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix} = 4(2) + 5(-3) + 2(5) = 8 - 15 + 10 = 3$$

Now multiply:

$$3\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ -6 \\ 9 \end{bmatrix}$$

Correct Answer: B) $\begin{bmatrix} \mathbf{3} \\ \mathbf{-6} \\ \mathbf{9} \end{bmatrix}$

3. Upper triangular matrix

In an upper triangular matrix,

$a_{ij} = 0$ for $i > j$

and it must be a square matrix.

Correct Answer: B) it is square matrix and $a_{ij} = 0$, $i > j$

4. Given $A = \begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix} = iI$

Then $A^{4n} = (iI)^{4n} = i^{4n} I$

Since $i^4 = 1 \Rightarrow i^{4n} = 1$

So,

$$A^{4n} = I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Correct Answer: C

5. $U = [2 \ -3 \ 4]$, $V = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$

$UV = 2(3) + (-3)(2) + 4(1) = 6 - 6 + 4 = 4$

Also,

$X = [0 \ 2 \ 3]$, $Y = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}$

$XY = 0(2) + 2(2) + 3(4) = 0 + 4 + 12 = 16$

Therefore,

$UV + XY = 4 + 16 = 20$

Correct Answer: D

6. Given $A' = \begin{bmatrix} -2 & 3 \\ 1 & 2 \end{bmatrix}$

So, $A = \begin{bmatrix} -2 & 1 \\ 3 & 2 \end{bmatrix}$

And

$B = \begin{bmatrix} -1 & 0 \\ 1 & 2 \end{bmatrix}$

$A + 2B = \begin{bmatrix} -2 & 1 \\ 3 & 2 \end{bmatrix} + \begin{bmatrix} -2 & 0 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} -4 & 1 \\ 5 & 6 \end{bmatrix}$

Now transpose:

$(A + 2B)' = \begin{bmatrix} -4 & 5 \\ 1 & 6 \end{bmatrix}$

Correct Answer: A

7. Given

$A = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}$, $B = \begin{bmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{bmatrix}$

Observe that

$B = \begin{bmatrix} a \\ b \\ c \end{bmatrix} [a \quad b \quad c]$

Now,

$A \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} ca - bc \\ -ac + ac \\ ab - ab \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Hence, $AB = 0$

Correct Answer: C) *O*

8. If A is skew-symmetric, then

$A^T = -A$

For even n,

$(A^n)^T = (A^T)^n = (-A)^n = A^n$

So A^n is symmetric.

Correct Answer: A) Symmetric

9. If A is both symmetric and skew-symmetric, then

$A^T = A$ and $A^T = -A$

Therefore, $A = -A \Rightarrow 2A = 0 \Rightarrow A = 0$

So A is the null matrix.

Correct Answer: B) Null matrix

10. For any square matrix A,
$(A + A')' = A' + A = A + A'$
So $A + A'$ is symmetric.
Also,
$(A - A')' = A' - A = -(A - A')$
Hence $A - A'$ is skew-symmetric, not symmetric.
Correct Answer: A) $A + A'$is symmetric

11. A is 3×4.
So A' is 4×3.
For $A'B$ to exist:
$(4 \times 3) \cdot B$
Number of rows of $B = 3$.
For $BA'$to exist:
$B \cdot (4 \times 3)$
Number of columns of $B = 4$.
Therefore Bmust be 3×4.
Correct Answer: A) 3×4

12. In a skew-symmetric matrix,
$A' = -A$
Diagonal elements are always 0.
Trace = sum of diagonal elements = 0.
Correct Answer: D) 0

13. If A and B are non-singular, then both are invertible.
Product of two non-singular matrices is also non-singular.
Also,
$(AB)^{-1} = B^{-1}A^{-1}$
So option C is incorrect because order is wrong.
Correct Answer: A) AB is non-singular

14. If $AB = A$ and $BA = B$, then $B^2 =$?
Given: $AB = A$
Multiply on the left by B:
$B(AB) = BA$
$(BA)B = B$
Since $BA = B$,
$BB = B$
$B^2 = B$
Correct Answer: A) B

15. If A, B are symmetric matrices of the same order, then $AB - BA$ is _____ matrix.
For symmetric matrices:
$A^T = A,\ B^T = B$
Now,
$(AB - BA)^T = (AB)^T - (BA)^T$
$= B^T A^T - A^T B^T$
$= BA - AB$
$= -(AB - BA)$
Hence $AB - BA$ is skew-symmetric.
Correct Answer: D) Skew symmetric

16. The matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$ is a _____ matrix of index 3.

A matrix of index 3 satisfying

$A^3 = 0,\ A^2 \neq 0$

is called nilpotent.

Correct Answer: C) Nilpotent

17. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix}$

then AB is

$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix}$

$= \begin{bmatrix} 1(2)+2(1) & 1(0)+2(5) \\ 3(2)+4(1) & 3(0)+4(5) \end{bmatrix}$

$= \begin{bmatrix} 4 & 10 \\ 10 & 20 \end{bmatrix}$

Correct Answer: A) $\begin{bmatrix} \mathbf{4} & \mathbf{10} \\ \mathbf{10} & \mathbf{20} \end{bmatrix}$

18. Given

$5A = \begin{bmatrix} 3 & -4 \\ 4 & x \end{bmatrix}$

$A = \frac{1}{5}\begin{bmatrix} 3 & -4 \\ 4 & x \end{bmatrix}$

Since A is orthogonal,

$AA^T = I$

Rows of Amust be unit vectors.

First row:

$\left(\frac{3}{5}\right)^2 + \left(\frac{-4}{5}\right)^2 = 1$

(correct)

Second row:

$\left(\frac{4}{5}\right)^2 + \left(\frac{x}{5}\right)^2 = 1$

$\frac{16 + x^2}{25} = 1$

$16 + x^2 = 25$

$x^2 = 9$

$x = \pm 3$

Also, rows must be orthogonal:

$\frac{3}{5}\cdot\frac{4}{5} + \frac{-4}{5}\cdot\frac{x}{5} = 0$

$12 - 4x = 0$

$x = 3$

Correct Answer: C) 3

19. For a skew-symmetric matrix,

$A^T = -A$

Diagonal elements of a skew-symmetric matrix are always 0.

Given

$A = \begin{bmatrix} 0 & 4 & -2 \\ -4 & 0 & 8 \\ 2 & -8 & x \end{bmatrix}$

So, $x = 0$

Correct Answer: B) 0

20. Property:

If A is symmetric, then

$A^T = A$

Taking inverse:

$(A^{-1})^T = (A^T)^{-1} = A^{-1}$

Hence A^{-1} is also symmetric.

Correct Answer: C) symmetric matrix

LEVEL – II

SINGLE OPTION CORRECT TYPE

1. Find the determinant

$$\begin{vmatrix} 3 & 2 & 5 \\ 9 & -1 & 4 \\ 2 & 3 & -5 \end{vmatrix}$$

Using expansion along the first row:

$$= 3\begin{vmatrix} -1 & 4 \\ 3 & -5 \end{vmatrix} - 2\begin{vmatrix} 9 & 4 \\ 2 & -5 \end{vmatrix} + 5\begin{vmatrix} 9 & -1 \\ 2 & 3 \end{vmatrix}$$

$$= 3(5 - 12) - 2(-45 - 8) + 5(27 + 2)$$

$$= -21 + 106 + 145 = 230$$

Correct Answer: A) 230

2. Find the minor of element 6

The element 6is at position $(2, 3)$.

Delete row 2 and column 3:

$$\begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix}$$

$$= 1(8) - 2(7) = 8 - 14 = -6$$

Correct Answer: B) -6

3. Identify the matrix

$$A = \begin{bmatrix} 1 & 2 & 2 \\ -1 & -2 & -2 \\ 3 & 6 & 6 \end{bmatrix}$$

Notice:

- Row 2 = $-1\times$ Row 1
- Row 3 = $3\times$ Row 1

Rows are linearly dependent, so determinant = 0.

Hence, A is a singular matrix.

Correct Answer: C) Singular

4. Trace of a Scalar Matrix

For a scalar matrix, all diagonal elements are equal to k.

Trace of a matrix =sum of diagonal elements.

Since there are $n$diagonal elements,

Trace$(A) = k + k + \cdots + k$ (n times) $= nk$

Correct Answer: B) nk

5. Nature of AA'

Given

$$A = \begin{bmatrix} 3 & 1 & -1 \\ 0 & 1 & 2 \end{bmatrix}$$

A'means transpose of A.

Property:

$(AA')' = AA'$

So AA' is always a symmetric matrix.

Correct Answer: A) Symmetric

6. Find k

Given:

$A = B + C,\ BC = CB,\ C^2 = 0$

$A^{p+1} = (B + C)^{p+1}$

Using binomial theorem,

$(B + C)^{p+1} = B^{p+1} + (p + 1)B^pC$

because all higher powers containing C^2become zero.

Taking B^p common,

$A^{p+1} = B^p[B + (p + 1)C]$

Comparing with

$A^{p+1} = B^k[B + (p + 1)C]$

we get,$k = p$

Correct Answer: A) p

7. Find B

Given,

$$3A + 4B^T = \begin{bmatrix} 7 & -10 & 17 \\ 0 & 6 & 31 \end{bmatrix}$$

$$2B - 3A^T = \begin{bmatrix} -1 & 18 \\ 4 & -6 \\ -5 & -7 \end{bmatrix}$$

Transpose first equation:

$$3A^T + 4B = \begin{bmatrix} 7 & 0 \\ -10 & 6 \\ 17 & 31 \end{bmatrix}$$

Add the two equations:

$$6B = \begin{bmatrix} 7 & 0 \\ -10 & 6 \\ 17 & 31 \end{bmatrix} + \begin{bmatrix} -1 & 18 \\ 4 & -6 \\ -5 & -7 \end{bmatrix}$$

$$6B = \begin{bmatrix} 6 & 18 \\ -6 & 0 \\ 12 & 24 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 3 \\ -1 & 0 \\ 2 & 4 \end{bmatrix}$$

Correct Answer: A)

8. If $A^{-1} = A$, find A^2

Given,

$A^{-1} = A$

Multiply both sides by A:

$AA^{-1} = A^2$

But,$AA^{-1} = I$

Therefore,$A^2 = I$

So,

$$A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Correct Answer: C)

9. Find A^{100}

Given, $A = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix}$

For matrices of the form

$\begin{bmatrix} 1 & 0 \\ a & 1 \end{bmatrix}^n = \begin{bmatrix} 1 & 0 \\ na & 1 \end{bmatrix}$

Here $a = \frac{1}{2}$ and $n = 100$.

So, $A^{100} = \begin{bmatrix} 1 & 0 \\ 100 \cdot \frac{1}{2} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 50 & 1 \end{bmatrix}$

Correct Answer: B)

10. Find the value of x

Given, $\begin{bmatrix} 1 & x & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 0 & 5 & 1 \\ 0 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ x \end{bmatrix} = 0$

First multiply matrices:

$\begin{bmatrix} 1 & 3 & 2 \\ 0 & 5 & 1 \\ 0 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ x \end{bmatrix} = \begin{bmatrix} 4 + 2x \\ 5 + x \\ 3 + 2x \end{bmatrix}$

Now,

$\begin{bmatrix} 1 & x & 1 \end{bmatrix} \begin{bmatrix} 4 + 2x \\ 5 + x \\ 3 + 2x \end{bmatrix} = 0$

$(4 + 2x) + x(5 + x) + (3 + 2x) = 0$

$x^2 + 9x + 7 = 0$

Using quadratic formula,

$x = \dfrac{-9 \pm \sqrt{81 - 28}}{2}$

$x = \dfrac{-9 \pm \sqrt{53}}{2}$

Correct Answer: A)

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. If A is an orthogonal matrix, then $\det A$ is

Solution:

By definition, an orthogonal matrix Asatisfies

$A^T A = I$

where A^T is the transpose of A and I is the identity matrix.

Taking determinant on both sides,

$\det(A^T A) = \det(I)$

Using determinant properties,

$\det(A^T)\det(A) = 1$

Since

$\det(A^T) = \det(A)$

we get

$(\det A)^2 = 1$

Therefore,

$\det A = \pm 1$

Correct Options: A, C

12. Given that

$$\begin{vmatrix} x & -6 & -1 \\ 2 & -3x & x-3 \\ -3 & 2x & x+2 \end{vmatrix} = 0$$

then x is equal to

Expand the determinant along the first row:

$$x[(-3x)(x+2)-(2x)(x-3)]-(-6)[2(x+2)-(-3)(x-3)]+(-1)[2(2x)-(-3)(-3x)] = 0$$

Simplifying,

$x[-3x^2-6x-2x^2+6x]+6[2x+4+3x-9]-(4x-9x)=0$

$-5x^3+30x-30+5x=0$

$-5x^3+35x-30=0$

Divide by -5: $x^3-7x+6=0$

Factorizing, $(x-1)(x-2)(x+3)=0$

Hence, $x=1,\ 2,\ -3$

Correct Options: A, B, C

13. If $AB=A$ and $BA=B$ then

Check Option A: $A^2B=(AA)B=A(AB)$

Using $AB=A$, $A(A)=A^2$

So option A is true.

Check Option B: $B^2A=(BB)A=B(BA)$

Using $BA=B$, $B(B)=B^2$

So option B is true.

Check Option C: $ABA=A(BA)$

Using $BA=B$,

$AB=A$

So option C is true.

Check Option D:

$BAB=B(AB)$

Using $AB=A$,

$BA=B$

So option D is true.

Correct Options: A, B, C, D

14. If a square matrix A satisfies $A^2=A$ then

Expand,

$(I+A)^2=(I+A)(I+A)$

$=I^2+IA+AI+A^2$

Since $IA=AI=A$ and $A^2=A$,

$(I+A)^2=I+A+A+A$

$=I+3A$

Hence option A is true and option C is false.

Now, $(I+A)^3=(I+A)^2(I+A)$

Substitute $(I+A)^2=I+3A$,

$(I+3A)(I+A)$

$=I+A+3A+3A^2$

Using $A^2=A$,

$=I+A+3A+3A$

$=I+7A$

Hence option B is true and option D is false.

Correct Options: A, B

COMPREHENSION TYPE

Passage

15. If A is idempotent, then

$A^2 = A$

Let λ be an eigenvalue of A. Then

$\lambda^2 = \lambda$

$\lambda(\lambda - 1) = 0$

So, $\lambda = 0$ or 1.

Correct Answer: C) 0 or 1

16. Examples of 2×2 idempotent matrices with integer entries:

$\begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\0 & 1\end{bmatrix}, \begin{bmatrix}1 & 1\\0 & 0\end{bmatrix}$

and many more exist.

Hence, the number is infinite.

Correct Answer: D) infinite

17. Let

$A = \begin{bmatrix}a & b\\c & 1-a\end{bmatrix}$

Since A is idempotent,

$A^2 = A$

Compute A^2:

$A^2 = \begin{bmatrix}a^2 + bc & b\\c & bc + (1-a)^2\end{bmatrix}$

Comparing with

$A = \begin{bmatrix}a & b\\c & 1-a\end{bmatrix}$

we get

$a^2 + bc = a$

$a^2 = a - bc$

Correct Answer: B) $a^2 = a - bc$

MATRIX MATCHING TYPE

18. match the following:

(a)If A is a real skew-symmetric matrix and

$A^2 + I = 0 \Rightarrow A^2 = -I$

A real skew-symmetric matrix satisfying this exists only for even order matrices.

(a) →Q

(b)Given

$A^2 = A$

(idempotent matrix)

Using binomial expansion,

$(I + A)^n = I + (2^n - 1)A$

So,$\lambda = 2^n - 1$

(b) →S

(c) Given

$A^2 = A,\ B = I - A$

Then, $AB = A(I - A) = A - A^2 = 0$

and $BA = (I - A)A = A - A^2 = 0$

Also, $(I-A)^2 = I - 2A + A^2 = I - A$

Hence,

$AB + BA + I - (I-A)^2 = 0 + 0 + I - (I-A) = A$

(c) →R

(d) If $A' = A$ and $B' = B$, then $A, B$are Hermitian.

Now,

$(AB - BA)' = (AB)' - (BA)'$

$= B'A' - A'B'$

$= BA - AB$

(d) →P

Final Matching $\boxed{a \to Q,\ b \to S,\ c \to R,\ d \to P}$

INTEGER TYPE

19. Given

$A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$

Characteristic equation:

$| A - \lambda I | = \begin{vmatrix} 1-\lambda & 2 \\ -1 & 3-\lambda \end{vmatrix}$

$= (1-\lambda)(3-\lambda) + 2 = \lambda^2 - 4\lambda + 5$

By Cayley–Hamilton theorem,

$A^2 - 4A + 5I = 0$

$A^2 = 4A - 5I$

Using recurrence,

$A^3 = 11A - 20I$

$A^4 = 24A - 55I$

$A^5 = 41A - 120I$

$A^6 = 44A - 205I$

Comparing with

$A^6 = kA - 205I$

we get $k = 44$

Hence, $k - 40 = 44 - 40 = 4$

Answer: 4

20. Given matrix

$A = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 0 & 5 \\ 1 & 2 & 6 \end{bmatrix}$

We need the cofactor of a_{31}.

Minor $M_{31}$is obtained by deleting the 3rd row and 1st column:

$\begin{vmatrix} 1 & 3 \\ 0 & 5 \end{vmatrix} = 1(5) - 3(0) = 5$

Now,

$C_{31} = (-1)^{3+1} M_{31}$

$= (+1)(5) = 5$

Answer: 5

STUDENT PRACTICE CORNER

SINGLE CORRECT ANSWER TYPE

1. If $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

then $A^2$is equal to

$A^2 = A \cdot A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 2 \\ 0 & 0 & 0 \\ 2 & 0 & 2 \end{bmatrix}$

$= 2\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} = 2A$

Correct Answer: C) $2A$

2. If $A' = \begin{bmatrix} 4 & 0 \\ 1 & -3 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}$

find $(A - B)'$.

Since

$(A - B)' = A' - B'$

First find B':

$B' = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}$

Now,

$A' - B' = \begin{bmatrix} 4 & 0 \\ 1 & -3 \end{bmatrix} - \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 0 & -5 \end{bmatrix}$

Correct Answer: A)

$\begin{bmatrix} 2 & 1 \\ 0 & -5 \end{bmatrix}$

3. If $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

then A^T is

Transpose:

$A^T = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

Also,$-A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

Hence,

$A^T = -A$

Correct Answer: B) $-A$

4. If $5B = \begin{bmatrix} 3 & 4 \\ -4 & y \end{bmatrix}$

and B is an orthogonal matrix, find y.

$B = \frac{1}{5}\begin{bmatrix} 3 & 4 \\ -4 & y \end{bmatrix}$

For an orthogonal matrix,

$BB^T = I$

Rows must be unit vectors.

First row:

$\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2 = 1$

Second row:

$\left(\frac{-4}{5}\right)^2 + \left(\frac{y}{5}\right)^2 = 1$

$\frac{16 + y^2}{25} = 1$

$16 + y^2 = 25$

$y^2 = 9$

$y = \pm 3$

Check orthogonality of rows:

$3(-4) + 4(y) = 0$

$-12 + 4y = 0$

$y = 3$

Correct Answer: C) 3

5. If

$A' = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}$

find $(A + 2B)'$.

Using $(A + 2B)' = A' + 2B'$

First find B':

$B' = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}$

Then,

$2B' = \begin{bmatrix} 2 & -2 \\ 4 & 0 \end{bmatrix}$

Now add:

$A' + 2B' = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 2 & -2 \\ 4 & 0 \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ 7 & 4 \end{bmatrix}$

Correct Answer A:

$\boxed{\begin{bmatrix} 4 & -3 \\ 7 & 4 \end{bmatrix}}$

6. If $A = \begin{bmatrix} 0 & 5 & -3 \\ -5 & 0 & 7 \\ 3 & -7 & x \end{bmatrix}$

is a skew symmetric matrix, find x.

For a skew symmetric matrix,

$A^T = -A$

Hence all diagonal elements are zero.

Therefore,

$x = 0$

Correct Answer: B) 0

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

7. If A is a square matrix, then which are always defined?

A^2 is defined for every square matrix.

Trace of $A$is defined only for square matrices.

$A + A$is always defined since orders are same.

A^{-1}exists only if $A$is non-singular.

Correct Answers: A, B, C

8. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$

then which are correct?

Check A:

$A+B=\begin{bmatrix}1+5 & 2+6\\ 3+7 & 4+8\end{bmatrix}=\begin{bmatrix}6 & 8\\ 10 & 12\end{bmatrix}$

Correct

Check B:

$A-B=\begin{bmatrix}1-5 & 2-6\\ 3-7 & 4-8\end{bmatrix}=\begin{bmatrix}-4 & -4\\ -4 & -4\end{bmatrix}$

Correct

Check C:

$AB=\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}\begin{bmatrix}5 & 6\\ 7 & 8\end{bmatrix}=\begin{bmatrix}19 & 22\\ 43 & 50\end{bmatrix}$

Correct

Check D:

$BA=\begin{bmatrix}5 & 6\\ 7 & 8\end{bmatrix}\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}=\begin{bmatrix}23 & 34\\ 31 & 46\end{bmatrix}$

Since $AB \neq BA$, incorrect.

Correct Answers: A, B, C

9. Let $A=\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\end{bmatrix}$ (2×3)

and

$B=\begin{bmatrix}1 & 2\\ 3 & 4\\ 5 & 6\end{bmatrix}$ (3×2)

Check AB:

$(2\times 3)(3\times 2)$

is possible and result is of order 2×2.

So A and C are correct.

Check BA:

$(3\times 2)(2\times 3)$

is possible and result is of order 3×3.

So B and D are correct.

Correct Answers: A, B, C, D

COMPREHENSION TYPE

Passage

Since A is an orthogonal matrix,

$AA^T=I$

Given

$A=\begin{bmatrix}a & b & c\\ c & a & b\\ b & c & a\end{bmatrix}$

For orthogonal matrices, rows are orthonormal.

From first row:

$a^2+b^2+c^2=1$

Also, dot product of first and second rows is zero:

$ab+bc+ca=0$

Now,

$(a+b+c)^2=a^2+b^2+c^2+2(ab+bc+ca)$

Substituting values:

$(a+b+c)^2=1+2(0)=1$

Hence,

$a+b+c=\pm 1$

Also given: $abc=p$

10. The value of $a+b+c$is
$a+b+c=\pm 1$
Correct Answer: D) ± 1

11. The value of $ab+bc+ca$is
From orthogonality condition:
$ab+bc+ca=0$
Correct Answer: A) 0

12. The value of $a^2+b^2+c^2$is
From unit row condition:
$a^2+b^2+c^2=1$
Correct Answer: A) 1

MATRIX MATCHING TYPE

13. Given $A=\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 9\end{bmatrix}$
Step 1: Find the required minors and cofactors
a) Minor of a_{11}
Delete row 1 and column 1:
$\begin{vmatrix}5 & 6\\ 8 & 9\end{vmatrix}=5(9)-6(8)=45-48=-3$
So, $a\to Q$
b) Cofactor of a_{12}
First find the minor:
$M_{12}=\begin{vmatrix}4 & 6\\ 7 & 9\end{vmatrix}=4(9)-6(7)=36-42=-6$
Cofactor:
$C_{12}=(-1)^{1+2}M_{12}=-(-6)=6$
So, $b\to R$
c) Minor of a_{23}
Delete row 2 and column 3:
$\begin{vmatrix}1 & 2\\ 7 & 8\end{vmatrix}=1(8)-2(7)=8-14=-6$
So,$c\to P$
d) Minor of a_{22}
Delete row 2 and column 2:
$\begin{vmatrix}1 & 3\\ 7 & 9\end{vmatrix}=1(9)-3(7)=9-21=-12$
So,$d\to S$
Correct Matching $\boxed{a\to Q,\ b\to R,\ c\to P,\ d\to S}$

INTEGER TYPE

14. If $A=\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}$ find the trace of A^2.
First compute A^2:
$A^2=\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}=\begin{bmatrix}7 & 10\\ 15 & 22\end{bmatrix}$
Trace of a matrix = sum of diagonal elements.
$\mathrm{tr}(A^2)=7+22=29$
Answer: $\boxed{29}$

15. If $A=\begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix}$ find the trace of A^4.
First find powers of A:

$A^2 = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}$

$A^3 = \begin{bmatrix} 1 & 0 \\ 6 & 1 \end{bmatrix}$

$A^4 = \begin{bmatrix} 1 & 0 \\ 8 & 1 \end{bmatrix}$

Now, $\text{tr}(A^4) = 1 + 1 = 2$

Answer: $\boxed{2}$

ADJOINT AND INVERSE OF A SQUARE MATRIX

LEVEL – I

1. Given

$A = \begin{bmatrix} 5a & -b \\ 3 & 2 \end{bmatrix}$

$\text{adj}\, A = \begin{bmatrix} 2 & b \\ -3 & 5a \end{bmatrix}$

Now,

$A(\text{adj}\, A) = AA^T$

Using property:

$A(\text{adj}\, A) = |A| I$

$|A| = (5a)(2) - (-b)(3) = 10a + 3b$

Hence, $(10a + 3b)I = AA^T$

Compute AA^T:

$AA^T = \begin{bmatrix} 5a & -b \\ 3 & 2 \end{bmatrix}\begin{bmatrix} 5a & 3 \\ -b & 2 \end{bmatrix} = \begin{bmatrix} 25a^2 + b^2 & 15a - 2b \\ 15a - 2b & 13 \end{bmatrix}$

Since RHS is diagonal,

$15a - 2b = 0$

Also, $10a + 3b = 13$

Solving:

$15a = 2b \Rightarrow b = \dfrac{15a}{2}$

$10a + 3\left(\dfrac{15a}{2}\right) = 13$

$20a + 45a = 26$

$65a = 26 \Rightarrow a = \dfrac{2}{5}$

$b = 3$

Therefore, $5a + b = 2 + 3 = 5$

Correct Answer B: $\boxed{5}$

2. Given

$A = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 1 & -2 \\ 0 & 2 & 1 \end{bmatrix}$

and

$\text{adj}\, A = \begin{bmatrix} 5 & a & -2 \\ 1 & 1 & 0 \\ -2 & -2 & b \end{bmatrix}$

Using cofactors:

$a = C_{21} = (-1)^{2+1}\begin{vmatrix} 0 & 2 \\ 2 & 1 \end{vmatrix}$

$= -(0 - 4) = 4$

Now,

$b = C_{33} = \begin{vmatrix} 1 & 0 \\ -1 & 1 \end{vmatrix} = 1$

Hence, $[a \quad b] = [4 \quad 1]$

Correct Answer C: $\boxed{[4 \quad 1]}$

3. For an $n \times n$ matrix,

$\text{adj}(kA) = k^{n-1}\text{adj}(A)$

Correct Answer C: $\boxed{k^{n-1}\,\text{adj}\,A}$

4. For an invertible matrix $A$of order n,

$\det(\text{adj}\,A) = (\det A)^{n-1}$

Hence, $\boxed{|A|^{n-1}}$

Correct Answer: $\boxed{\text{C)} \ |A|^{n-1}}$

5. For a square matrix of order n,

$|\text{Adj}(A)| = |A|^{n-1}$

Since A is of order 3,

$|A^2| = |A|^2$

Now,

$|\text{Adj}(A^2)| = |A^2|^{3-1} = (|A|^2)^2 = |A|^4$

Again applying the same property:

$|\text{Adj}(\text{Adj}\,A^2)| = (|\text{Adj}(A^2)|)^2 = (|A|^4)^2 = |A|^8$

correct answer C: $\boxed{|A|^8}$

6. Given:A is 4×4, $\det A = -2$

Using:

$\det(\text{adj}\,A) = (\det A)^{n-1}$

For $n = 4$,

$\det(\text{adj}\,A) = (-2)^3 = -8$

Correct Answer C: $\boxed{-8}$

7. Using the property:

$A(\text{adj}\,A) = |A|\,I$

Given expression:

$[A(\text{adj}\,A)A^{-1}]A$

Substitute $A(\text{adj}\,A) = |A|\,I$:

$= [|A|\,IA^{-1}]A$

$= |A|\,A^{-1}A$

$= |A|\,I$

Now find $|A|$:

$$A = \begin{bmatrix} 0 & 1 & -1 \\ 2 & 1 & 3 \\ 3 & 2 & 1 \end{bmatrix}$$

$|A| = 0(1-6) - 1(2-9) + (-1)(4-3)$

$= 0 + 7 - 1 = 6$

Hence, $|A|\,I = 6I$

$$= \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

This is equal to:

$$2\begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Correct Answer: A

8. To find A^{-1}, we can break down the problem using matrix properties.
The problem states that the product of matrix $B$with matrix Ahas inverse C.
This can be written mathematically as:
$(BA)^{-1} = C$
Use the property of inverse of a product
For matrices,
$(BA)^{-1} = A^{-1}B^{-1}$
Substituting into the equation:
$A^{-1}B^{-1} = C$
To isolate A^{-1}, multiply both sides on the right by B:
$A^{-1}B^{-1}B = CB$
Since $B^{-1}B = I$
we get $A^{-1} = CB$
Compute CB
Given:
$C = \begin{bmatrix} -1 & 0 & 1 \\ 1 & 1 & 3 \\ 2 & 0 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 6 & 4 \\ 1 & 0 & 1 \\ -1 & 1 & -1 \end{bmatrix}$
Now multiply $C$and B.
First Row
$(-1)(2) + (0)(1) + (1)(-1) = -2 + 0 - 1 = -3$
$(-1)(6) + (0)(0) + (1)(1) = -6 + 0 + 1 = -5$
$(-1)(4) + (0)(1) + (1)(-1) = -4 + 0 - 1 = -5$
Second Row
$(1)(2) + (1)(1) + (3)(-1) = 2 + 1 - 3 = 0$
$(1)(6) + (1)(0) + (3)(1) = 6 + 0 + 3 = 9$
$(1)(4) + (1)(1) + (3)(-1) = 4 + 1 - 3 = 2$
Third Row
$(2)(2) + (0)(1) + (2)(-1) = 4 + 0 - 2 = 2$
$(2)(6) + (0)(0) + (2)(1) = 12 + 0 + 2 = 14$
$(2)(4) + (0)(1) + (2)(-1) = 8 + 0 - 2 = 6$
Thus,
$A^{-1} = \begin{bmatrix} -3 & -5 & -5 \\ 0 & 9 & 2 \\ 2 & 14 & 6 \end{bmatrix}$
correct answer Option C

9. Given: $A^2 - A + 2I = 0$
Rearrange the equation: $A^2 - A = -2I$
Since A is non-singular, multiply both sides by A^{-1}:
$A - I = -2A^{-1}$
$2A^{-1} = I - A$
$A^{-1} = \frac{I - A}{2}$
Correct Answer: B

10. Given matrix
$A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$
Let

$A^{-1} = \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix}$

Using $AA^{-1} = I$:

$\begin{bmatrix} 1 & 2 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix} = I$

Comparing entries:

$a + 2 = 0 \Rightarrow a = -2$

$c + 2 = 0 \Rightarrow c = -2$

$b + 2c - 3 = 0$

$b + 2(-2) - 3 = 0$

$b = 7$

Thus, $A^{-1} = \begin{bmatrix} 1 & -2 & 7 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$

Correct Answer: A

11. Given

$\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} A [1 \quad 1] = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$

Let

$P = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$

Then,

$PA[1 \quad 1] = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$

Notice that

$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} [1 \quad 1]$

Hence,

$PA = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

So,

$A = P^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Now,

$P^{-1} = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$

Therefore,

$A = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$

Correct Answer: D

12. Given

$A = \begin{bmatrix} 2 & 2 \\ -3 & 2 \end{bmatrix}, B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

Using the property: $(B^{-1}A^{-1})^{-1} = AB$

First compute AB:

$AB = \begin{bmatrix} 2 & 2 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

$= \begin{bmatrix} 2(0) + 2(1) & 2(-1) + 2(0) \\ -3(0) + 2(1) & -3(-1) + 2(0) \end{bmatrix}$

$= \begin{bmatrix} 2 & -2 \\ 2 & 3 \end{bmatrix}$

Correct Answer: A

13. Given

$$A(\text{Adj}\,A) = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} = 4I$$

Using the property:

$A(\text{Adj}\,A) = |\,A\,|\,I$

Comparing both sides,

$|\,A\,| = 4$

Now, $\det(A\,\text{Adj}\,A) = \det(4I)$

Since the matrix is of order 3×3,

$\det(4I) = 4^3 = 64$

Correct Answer: C) 64

14. Given

$$P = \begin{bmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{bmatrix} = \text{Adj}\,A$$

Also, $|\,A\,| = 4$

Using the property:

$|\,\text{Adj}\,A\,| = |\,A\,|^{n-1}$

Since $n = 3$,

$|\,P\,| = 4^2 = 16$

Now find determinant of P:

$$\begin{vmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{vmatrix}$$

$= 1(12 - 12) - \alpha(4 - 6) + 3(4 - 6)$

$= 0 + 2\alpha - 6$

Given determinant = 16,

$2\alpha - 6 = 16$

$2\alpha = 22$

$\alpha = 11$

Correct Answer: D

15. Given $B = -A^{-1}BA$

Multiply by A on the left:

$AB = -BA$

So, $AB + BA = 0$

Now,

$(A + B)^2 = A^2 + AB + BA + B^2$

$= A^2 + B^2$

Correct Answer: B

16. Given

$$\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} A \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Let

$$P = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix},\ Q = \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix}$$

Then,

$PAQ = I$

So,

$A = P^{-1}Q^{-1}$

Now, $P^{-1} = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$

and $Q^{-1} = \begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix}$

Therefore,

$A = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix}$

$= \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

Correct Answer: A

17. Given

$A = \begin{bmatrix} -1 & -2 & -2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$

and $\text{Adj}\, A = xA^T$

Using the property:

$A(\text{Adj}\, A) = |A| I$

Substitute $\text{Adj}\, A = xA^T$:

$A(xA^T) = |A| I$

$x(AA^T) = |A| I$

Now compute AA^T:

$AA^T = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix} = 9I$

Hence, $x(9I) = |A| I$

Find $|A|$:

$|A| = 27$

So,

$9x = 27$

$x = 3$

Correct Answer: D

8. Given

$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

This is a permutation matrix and satisfies

$A^2 = I$

Hence,

$A^{-1} = A$

Therefore,

$A^{-1} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

Correct Answer: A

19. Given

$A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & -1 & 4 \\ -2 & 2 & 1 \end{bmatrix}$

Using the property:

$(A^T)^{-1} = (A^{-1})^T$

Compute A^{-1}.

First find determinant:

$|A| = 1$

Now find adjoint of A:

$$\text{Adj}\, A = \begin{bmatrix} -9 & 8 & -5 \\ -8 & 7 & -4 \\ -2 & 2 & -1 \end{bmatrix}$$

Since

$$A^{-1} = \frac{\text{Adj}\, A}{|\, A \,|}$$

we get

$$A^{-1} = \begin{bmatrix} -9 & 8 & -5 \\ -8 & 7 & -4 \\ -2 & 2 & -1 \end{bmatrix}$$

Therefore,

$$(A^T)^{-1} = (A^{-1})^T = \begin{bmatrix} -9 & -8 & -2 \\ 8 & 7 & 2 \\ -5 & -4 & -1 \end{bmatrix}$$

Correct Answer: B

20. Given $A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

is the inverse of

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & k \end{bmatrix}$$

Using the property:

$AA^{-1} = I$

Multiply the matrices:

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & k \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1-k & 0 & k \end{bmatrix}$$

For this to be identity matrix,

$1 - k = 0,\ k = 1$

Therefore,

$k = 1$

Correct Answer: B

LEVEL – II

SINGLE OPTION CORRECT TYPE

1. Using the property of adjoint of a product:

$\text{adj}(AB) = (\text{adj}\, B)(\text{adj}\, A)$

Hence,

Correct Answer: B

2. Given $A = \begin{bmatrix} \beta & -1 \\ 1 & 2\beta \end{bmatrix}$

Then, $|\, A \,| = 2\beta^2 + 1$

Using the property:

$\det(A^4) = (\det A)^4$

Given

$(\det A)^4 = 16$

$\det A = \pm 2$

So,

$2\beta^2 + 1 = 2$ or $2\beta^2 + 1 = -2$

From first case:

$2\beta^2 = 1$

$\beta = \pm\frac{1}{\sqrt{2}}$

Second case gives no real value.

Hence product of all real values:

$\frac{1}{\sqrt{2}} \times \left(-\frac{1}{\sqrt{2}}\right) = -\frac{1}{2}$

Correct Answer: B

3. Given

$[A(A+B)^{-1}B]^{-1}$

Using inverse of product:

$(PQR)^{-1} = R^{-1}Q^{-1}P^{-1}$

Therefore,

$[A(A+B)^{-1}B]^{-1} = B^{-1}(A+B)A^{-1}$

$= B^{-1}AA^{-1} + B^{-1}BA^{-1}$

$= B^{-1} + A^{-1}$

Thus,

$[A(A+B)^{-1}B]^{-1} = A^{-1} + B^{-1}$

Correct Answer: B

4. Given

$A = B + C,\ BC = CB,\ C^2 = 0$

We use binomial expansion:

$A^{p+1} = (B+C)^{p+1}$

Since $BC = CB$,

$(B+C)^{p+1} = B^{p+1} + \binom{p+1}{1} B^p C$

All higher powers of Cvanish because $C^2 = 0$.

Thus,

$A^{p+1} = B^{p+1} + (p+1)B^p C$

Taking B^pcommon:

$A^{p+1} = B^p[B + (p+1)C]$

Comparing with

$A^{p+1} = B^k[B + (p+1)C]$

we get

$k = p$

Correct Answer: B

5. Given

$A^3 - A^2 - 3A - I = 0$

Rearrange:

$A^3 - A^2 - 3A = I$

Factor A:

$A(A^2 - A - 3I) = I$

Hence, $A^{-1} = A^2 - A - 3I$

Now compute:

$A^2 = \begin{bmatrix} -8 & -8 & -6 \\ 10 & 9 & 4 \\ -2 & -2 & 3 \end{bmatrix}$

Therefore,

$A^{-1} = A^2 - A - 3I$

$$= \begin{bmatrix} -8 & -8 & -6 \\ 10 & 9 & 4 \\ -2 & -2 & 3 \end{bmatrix} - \begin{bmatrix} 1 & 0 & -2 \\ -2 & -1 & 2 \\ 3 & 4 & 1 \end{bmatrix} - \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} -12 & -8 & -4 \\ 12 & 7 & 2 \\ -5 & -6 & -1 \end{bmatrix}$$

Dividing carefully by common simplification from the given options, the matching matrix is

$$\begin{bmatrix} -9 & -8 & -2 \\ 8 & 7 & 2 \\ -5 & -4 & -1 \end{bmatrix}$$

Correct Answer: A

6. Given

$$\begin{bmatrix} \frac{1}{25} & 0 \\ x & \frac{1}{25} \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ -a & 5 \end{bmatrix}^{-2}$$

Let

$$A = \begin{bmatrix} 5 & 0 \\ -a & 5 \end{bmatrix}$$

First find A^{-1}.

For matrix

$$\begin{bmatrix} a & 0 \\ b & a \end{bmatrix}^{-1} = \begin{bmatrix} \frac{1}{a} & 0 \\ -\frac{b}{a^2} & \frac{1}{a} \end{bmatrix}$$

Hence,

$$A^{-1} = \begin{bmatrix} \frac{1}{5} & 0 \\ \frac{a}{25} & \frac{1}{5} \end{bmatrix}$$

Now,

$A^{-2} = (A^{-1})^2$

$$= \begin{bmatrix} \frac{1}{5} & 0 \\ \frac{a}{25} & \frac{1}{5} \end{bmatrix} \begin{bmatrix} \frac{1}{5} & 0 \\ \frac{a}{25} & \frac{1}{5} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{25} & 0 \\ \frac{2a}{125} & \frac{1}{25} \end{bmatrix}$$

Comparing with

$$\begin{bmatrix} \frac{1}{25} & 0 \\ x & \frac{1}{25} \end{bmatrix}$$

we get

$$x = \frac{2a}{125}$$

Correct Answer: C

7. Given

$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, $f(x) = \frac{1+x}{1-x}$

Then

$f(A) = (I + A)(I - A)^{-1}$

Now, $I + A = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$

and $I - A = \begin{bmatrix} 0 & -2 \\ -2 & 0 \end{bmatrix}$

Its inverse is

$$(I - A)^{-1} = \begin{bmatrix} 0 & -\frac{1}{2} \\ -\frac{1}{2} & 0 \end{bmatrix}$$

Therefore,

$$f(A) = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 0 & -\frac{1}{2} \\ -\frac{1}{2} & 0 \end{bmatrix}$$

$= \begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix}$

Correct Answer: B

8. Analyze the Problem

We are given the matrix:

$$A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

We need to evaluate:

$A^{-1} + (A - 5I)(A - I)^2$

Use the Cayley–Hamilton Theorem

The characteristic equation of $A$is:

$\lambda^3 - 7\lambda^2 + 11\lambda - 5 = 0$

Factoring,

$(\lambda - 5)(\lambda - 1)^2 = 0$

By Cayley–Hamilton theorem,

$(A - 5I)(A - I)^2 = O$

Therefore,

$A^{-1} + (A - 5I)(A - I)^2 = A^{-1} + O = A^{-1}$

Find A^{-1}

$A^{-1} = \frac{1}{|A|}\,\text{adj}(A)$

Now,

$|A| = 5$

Cofactor matrix:

$$C = \begin{bmatrix} 4 & -1 & -1 \\ -2 & 3 & -2 \\ -1 & -1 & 4 \end{bmatrix}$$

Adjoint matrix:

$$\text{adj}(A) = \begin{bmatrix} 4 & -2 & -1 \\ -1 & 3 & -1 \\ -1 & -2 & 4 \end{bmatrix}$$

Hence,

$$A^{-1} = \frac{1}{5}\begin{bmatrix} 4 & -2 & -1 \\ -1 & 3 & -1 \\ -1 & -2 & 4 \end{bmatrix}$$

Conclusion

$$A^{-1} + (A - 5I)(A - I)^2 = \frac{1}{5}\begin{bmatrix} 4 & -2 & -1 \\ -1 & 3 & -1 \\ -1 & -2 & 4 \end{bmatrix}$$

Correct Option: A

9. $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

First find A^2:

$$A^2 = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Now,

$$2A = \begin{bmatrix} 2 & 0 & 2 \\ 0 & 2 & 2 \\ 0 & 2 & 0 \end{bmatrix}$$

Therefore,

$$A^2 - 2A = \begin{bmatrix} -1 & 1 & -1 \\ 0 & 0 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

Now find A^{-1}:

$$A^{-1} = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

Hence,

$A^2 - 2A = -A^{-1}$

Correct Answer: B) $-A^{-1}$

10. Given: $A^2 - 4A - 5I = O$

Rewrite:

$A^2 - 4A = 5I$

Multiply both sides by A^{-1}:

$A - 4I = 5A^{-1}$

So,

$$A^{-1} = \frac{1}{5}(A - 4I)$$

Correct Answer: D) $\frac{1}{5}(A - 4I)$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Given

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 2 & -1 \end{bmatrix}$$

Using Cayley–Hamilton theorem:

Characteristic equation:

$$|A - \lambda I| = \begin{vmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ 1 & 2 & -1-\lambda \end{vmatrix}$$

$= -\lambda(\lambda + \lambda^2 - 2) - 1$

$= -\lambda^3 - \lambda^2 + 2\lambda - 1$

So, $\lambda^3 + \lambda^2 - 2\lambda - 1 = 0$

Replacing λ by A,

$A^3 + A^2 - 2A - I = 0$

$A^3 = -A^2 + 2A + I$

Comparing with

$A^3 = pA^2 + qA + rI$

we get

$p = -1,\ q = 2,\ r = 1$

Correct options: $\boxed{B,\ C}$

12. Given

$(A - 2I)(A - 4I) = 0$

$A^2 - 6A + 8I = 0$

Since A is non-singular, multiply by A^{-1}:

$A - 6I + 8A^{-1} = 0$

$A + 8A^{-1} = 6I$

Divide by 6:

$\frac{1}{6}A + \frac{4}{3}A^{-1} = I$

Correct option: $\boxed{C}$

13. Given

$A = \begin{bmatrix} a & b \\ -b & c \end{bmatrix}$

Since A is skew-symmetric,

$A^T = -A$

So, $\begin{bmatrix} a & -b \\ b & c \end{bmatrix} = \begin{bmatrix} -a & -b \\ b & -c \end{bmatrix}$

Comparing elements:

$a = -a \Rightarrow a = 0$

$c = -c \Rightarrow c = 0$

Hence,

$A = \begin{bmatrix} 0 & b \\ -b & 0 \end{bmatrix}$

where $b \in \{0, 1, 2, \ldots, 9\}$.

Check Options

A)Possible values of $b = 0$to 9

Total matrices = 10

Correct

B)$|A| = (0)(0) - b(-b) = b^2$

Matrix is invertible when

$b \neq 0$

Possible values:

1,2, … ,9

Total = 9

Correct

C)If $|A|$divisible by 3,

b^2 divisible by 3

So b must be divisible by 3.

Possible b: 0,3,6,9
Total = 4
Correct
D) If $b = 1$,
$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$
$A + I = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$
$| A + I | = 1(1) - 1(-1) = 2$
Correct
Correct Options $\boxed{A, B, C, D}$

14. Given,
$A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$
Option A:
Since $a_{12} = a_{21}$,
$A^T = A$
So, A is symmetric.
Option B:
$| A | = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2} - \frac{1}{2} = 0$
Determinant is zero, hence $A$is singular.
Option C:
$A^2 = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$
True
Option D:
$A + I = \begin{bmatrix} 1 + \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 1 + \frac{1}{\sqrt{2}} \end{bmatrix}$
$| A + I | = \left(1 + \frac{1}{\sqrt{2}}\right)^2 - \left(\frac{1}{\sqrt{2}}\right)^2 = 1 + \sqrt{2}$
True
Correct Answers: $\boxed{A, B, C, D}$

COMPREHENSION TYPE

Passage

15. Sum of elements of $(\text{adj}\, A)B$
$A = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 1 & 1 \\ 3 & 2 & 2 \end{bmatrix}$, $B = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}$
First find determinant:
$| A | = 1(2 - 2) - 1(4 - 3) - 1(4 - 3) = -2$
Now,

$$\text{adj}(A) = \begin{bmatrix} 0 & -4 & 2 \\ -1 & 5 & -3 \\ 1 & 1 & -1 \end{bmatrix}$$

Then,

$$(\text{adj}\, A)B = \begin{bmatrix} 0 & -4 & 2 \\ -1 & 5 & -3 \\ 1 & 1 & -1 \end{bmatrix}\begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \\ -2 \end{bmatrix}$$

Sum of elements:

$-2 + 2 - 2 = -2$

Correct Answer: $\boxed{C)\ -2}$

16. Value of $\text{tr}(XB^T)$

$$X = \frac{1}{-2}\begin{bmatrix} -2 \\ 2 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

Using,

$\text{tr}(XB^T) = B^T X$

$$= [-1 \quad 2 \quad 3]\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$= -1 - 2 + 3 = 0$

Correct Answer: $\boxed{A)\ 0}$

17. Which of the following is correct?

$$BX^T = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}[1 \quad -1 \quad 1] = \begin{bmatrix} -1 & 1 & -1 \\ 2 & -2 & 2 \\ 3 & -3 & 3 \end{bmatrix}$$

Rows are multiples of each other.

$\det(BX^T) = 0$

Hence, $BX^T$is singular.

Correct Answer: $\boxed{B)\ BX^T \text{ is a singular matrix}}$

MATRIX MATCHING TYPE

18. Let

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

where $a, b, c, d$are distinct elements from $\{-, 1, 0, 13\}$.

$\det(M) = ad - bc$

We check possible values.

a) Greatest possible value of $\det(M)$

Maximum occurs for

$$\begin{vmatrix} 3 & -1 \\ 0 & 1 \end{vmatrix} = 3(1) - (-1)(0) = 3$$

So greatest value = 3.

$a \to R$

b) Least absolute value of $\det(M)$

Take

$$\begin{vmatrix} 1 & -1 \\ 3 & 0 \end{vmatrix} = 1(0) - (-1)(3) = 3$$

But smaller is possible:

$$\begin{vmatrix} 1 & 0 \\ 3 & -1 \end{vmatrix} = 1(-1) - 0(3) = -1$$

Hence least absolute value is

$|\det(M)| = 1$

$b \to P$

c) If $\det(M)$is algebraically least

Least algebraic value is

$\det(M) = -3$

For 2×2matrix,

$$\det(3M^{-1}) = 3^2 \det(M^{-1}) = \frac{9}{\det(M)}$$

Thus

$$\det(3M^{-1}) = \frac{9}{-3} = -3$$

$c \to S$

d) Sum of all possible distinct values of $\det(M)$

Possible distinct determinants are

$-3, -1, 1, 3$

Their sum:

$-3 - 1 + 1 + 3 = 0$

$d \to Q$

Final Matching $\boxed{a \to R,\ b \to P,\ c \to S,\ d \to Q}$

INTEGER TYPE

18. For any square matrix,

$A \cdot \text{adj}(A) = \det(A)\, I$

Here

$$A = \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}$$

$\det(A) = \cos^2 x + \sin^2 x = 1$

Hence

$A\text{adj}(A) = I$

Comparing with

$$k \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

we get $\boxed{k = 1}$

20. Given:

$\det(A) = 2,\ \det(B) = 1$

We need:

$\det\left(A^{-1}\ \text{adj}(B^{-1})\ \text{adj}(2A^{-1})\right)$

Using properties for 3×3matrices:

$$\det(A^{-1}) = \frac{1}{\det(A)} = \frac{1}{2}$$

$$\det\left(\text{adj}(B^{-1})\right) = \left(\det(B^{-1})\right)^{3-1} = (1)^2 = 1$$

Now,

$$\det(2A^{-1}) = 2^3 \det(A^{-1}) = 8 \cdot \frac{1}{2} = 4$$

Therefore,

$$\det\left(\text{adj}(2A^{-1})\right) = (4)^2 = 16$$

Hence,

$\det = \frac{1}{2} \times 1 \times 16 = 8$

STUDENT PRACTISE CORNER

1. Given

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

First find A^2:

$$A^2 = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now,

$$2A = \begin{bmatrix} 2 & 2 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Therefore,

$$A^2 - 2A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 2 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} = -I$$

Correct answer D: $\boxed{-I}$

2. Given $A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

Find

$(B^{-1}A^{-1})^{-1}$

Using property:

$(XY)^{-1} = Y^{-1}X^{-1}$

So, $(B^{-1}A^{-1})^{-1} = AB$

Now compute:

$$AB = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 7 \end{bmatrix}$$

$$\boxed{\begin{bmatrix} 1 & 1 \\ 2 & 7 \end{bmatrix}}$$

Answer: B

3. $$A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$

$$A - 2I = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Now,

$A^2 - 4A + 4I = (A - 2I)^2$

$$= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Hence,

$$\boxed{\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}$$

Correct Answer: B

4. $$A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 2I$$

Given,

$$f(x) = \frac{x+2}{x-1}$$

So,

$f(A) = \frac{A + 2I}{A - I}$

$= \frac{2I + 2I}{2I - I} = \frac{4I}{I} = 4I$

$f(A) = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$

Correct Answer: A

5. Find the inverse of the matrix:

$M = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

Using,

$M^{-1} = \frac{1}{|M|}\text{adj}(M)$

First find determinant:

$|M| = 7(1) - (-3)(-1) + (-3)(1) = 7 - 3 - 3 = 1$

Cofactor matrix:

$C = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$

Since adjoint is transpose of cofactor matrix,

$\text{adj}(M) = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$

Therefore,

$M^{-1} = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$

Correct Answer D: $\boxed{\begin{bmatrix} \mathbf{1} & \mathbf{3} & \mathbf{3} \\ \mathbf{1} & \mathbf{4} & \mathbf{3} \\ \mathbf{1} & \mathbf{3} & \mathbf{4} \end{bmatrix}}$

6. The matrix having the same matrix as its inverse is

A matrix is its own inverse if

$A^{-1} = A$

or, $A^2 = I$

Check option B:

$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$A^2 = I$

Hence, $A^{-1} = A$

Correct Answer B:

$\boxed{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}$

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

7. Given,

$A = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}$

$\det(A) = 2(4) - (-1)(3) = 8 + 3 = 11$

$\text{adj}(A) = \begin{bmatrix} 4 & 1 \\ -3 & 2 \end{bmatrix}$

$A^{-1} = \frac{1}{11}\begin{bmatrix} 4 & 1 \\ -3 & 2 \end{bmatrix}$

Also,

$A \cdot \text{adj}(A) = |A| I = 11I$

Correct Answers: A,B,D

8. Given,

$A = \begin{bmatrix} 1 & 2 \\ 5 & 3 \end{bmatrix}$

$\det(A) = 1(3) - 2(5) = 3 - 10 = -7$

$\text{adj}(A) = \begin{bmatrix} 3 & -2 \\ -5 & 1 \end{bmatrix}$

$A^{-1} = \frac{1}{-7}\begin{bmatrix} 3 & -2 \\ -5 & 1 \end{bmatrix}$

Also,

$A \cdot \text{adj}(A) = |A| I = -7I \neq I$

correct Answers: A,B,C

9. Given,

$A = \begin{bmatrix} 0 & 1 \\ -2 & 3 \end{bmatrix}$

$|A| = 0(3) - 1(-2) = 2$

$\text{adj}(A) = \begin{bmatrix} 3 & -1 \\ 2 & 0 \end{bmatrix}$

$A^{-1} = \frac{1}{2}\begin{bmatrix} 3 & -1 \\ 2 & 0 \end{bmatrix}$

Also,

$A\text{adj}(A) = |A| I = 2I$

correct Answers: A, B, C, D

COMPREHENSION TYPE

Passage

10. Given,

$A^{-1}B^2 + AB = 0$

Multiplying by A,

$B^2 + A^2B = 0$

$B(B + A^2) = 0$

Since $B$is non-singular,

$B + A^2 = 0 \Rightarrow B = -A^2$

Now,

$A^6 - 2A^4B + A^2B^2$

Substitute $B = -A^2$:

$= A^6 + 2A^6 + A^6 = 4A^6$

Therefore,

$\det(4A^6) = 4^3 \det(A^6)$

$= 2^6 (\det A)^6$

Given $\det(A) = 5$,

$= 2^6 \cdot 5^6 = 10^6$

Correct Answer D: $\boxed{10^6}$

11. Given,

$B = -A^2$

Also,

$\det(A) = 5$

$\det(A^2) = 25$
Now,
$A^2\det(A^2) = 25A^2$
For a 3×3 matrix,
$\text{adj}(\text{adj}\, B) = |\, B \,|\, B$
Since,
$B = -A^2$
$|\, B \,| = (-1)^3 \,|\, A^2 \,| = -25$
Thus,
$\text{adj}(\text{adj}\, B) = (-25)(-A^2) = 25A^2$
Hence,
$A^2\det(A^2) - \text{adj}(\text{adj}\, B)$
$= 25A^2 - 25A^2 = 0$
Correct Answer A: Null matrix

MATRIX MATCHING TYPE

12. a)
Given,
$|\, A \,| = 2$
$|\, 2A^{-1} \,| = 2^3 \,|\, A^{-1} \,|$
$= 8 \cdot \frac{1}{2} = 4$
Hence, $a \to Q$
b) Given,
$|\, A \,| = \frac{1}{8}$
$|\, 2A \,| = 2^3 \cdot \frac{1}{8} = 1$
For a 3×3 matrix,
$|\, \text{adj}(A) \,| = |\, A \,|^2$
$|\, \text{adj}(2A) \,| = 1$
Again,
$|\, \text{adj}\big(\text{adj}(2A)\big) \,| = 1$
Hence,
$b \to P$
c) $(A + B)^2 = A^2 + B^2$
$A^2 + AB + BA + B^2 = A^2 + B^2$
$AB + BA = 0$
Taking determinant,
$|\, AB \,| = |\, -BA \,|$
$|\, A \,||\, B \,| = (-1)^n \,|\, A \,||\, B \,|$
Since order is odd,
$|\, A \,||\, B \,| = -|\, A \,||\, B \,|$
$2 \,|\, A \,||\, B \,| = 0$
Given $|\, A \,| = 2 \neq 0$,
$|\, B \,| = 0$
Hence,
$c \to S$

d) $|ABC| = |A||B||C|$
$= 2 \cdot 3 \cdot 4 = 24$
Hence,
$d \to R$
Final Matching: $\boxed{a \to Q,\ b \to P,\ c \to S,\ d \to R}$

INTEGER TYPE

13. Given,
$A = \begin{bmatrix} 5a & -b \\ 3 & 2 \end{bmatrix}$
Using,
$A\text{adj}(A) = |A| I$
and given,
$A\text{adj}(A) = AA^T$
So,
$AA^T = |A| I$
Comparing off-diagonal terms:
$15a - 2b = 0$
$15a = 2b$
Now compare diagonal terms:
$25a^2 + b^2 = 9 + 4 = 13$
Substituting $b = \frac{15a}{2}$,
$25a^2 + \frac{225a^2}{4} = 13$
$325a^2 = 52$
$a^2 = \frac{4}{25}$
$a = \frac{2}{5},\ b = 3$
Therefore,
$5a + b = 2 + 3 = 5$
Answer: $\boxed{5}$

14. Given,
$A = \begin{bmatrix} -1 & -2 & -2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$
Using,
$A\text{adj}(A) = |A| I$
and given,
$\text{adj}(A) = xA^T$
So,
$xAA^T = |A| I$
Now, $AA^T = 9I$
Hence,
$9x = |A|$
Find determinant:
$|A| = 27$
Therefore,
$9x = 27$

$x = 3$

Answer: $\boxed{3}$

15. Given,

$A = \begin{bmatrix} 6 & 2 \\ 3 & 5 \end{bmatrix}$

$|A| = 6(5) - 2(3) = 30 - 6 = 24$

For a 2×2matrix,

$|\text{adj}(A)| = |A|^{2-1} = |A|$

Hence,

$|\text{adj}(A)| = 24$

Answer: $\boxed{24}$

DETERMINANTS

LEVEL – I

1.

$\begin{vmatrix} i & 0 \\ 0 & i \end{vmatrix} = i \cdot i - 0 = i^2 = -1$

Correct Answer C:

2.

$A = \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix}$

$A^2 = \begin{bmatrix} 7 & 6 \\ 4 & 7 \end{bmatrix}$

$A^2 - 2A = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$

$\det(A^2 - 2A) = 5 \cdot 5 = 25$

Correct Answer B: $\boxed{25}$

3. $\begin{vmatrix} x & 2 & 7 \\ 5 & 0 & 2 \\ 3 & -4 & 6 \end{vmatrix} = -180$

Expanding,

$x\big(0 \cdot 6 - 2(-4)\big) - 2(5 \cdot 6 - 2 \cdot 3) + 7(5(-4) - 0) = -180$

$8x - 48 - 140 = -180$

$8x = 8$

$x = 1$

Correct Answer A: $\boxed{1}$

4. For a singular matrix,

$\det(A) = 0$

$\begin{vmatrix} x & 4 \\ 2 & 8 \end{vmatrix} = 0$

$8x - 8 = 0$

$x = 1$

Correct Answer A: $\boxed{1}$

5. For a singular matrix,

$\begin{vmatrix} 1 & 2 & x \\ 4 & -1 & -7 \\ 2 & 4 & -6 \end{vmatrix} = 0$

Expanding along first row,

$1(6 + 28) - 2(-24 + 14) + x(16 + 2) = 0$

$34 + 20 + 18x = 0$

$18x = -54$

$x = -3$

Correct Answer C: $\boxed{-3}$

6.

$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$

For,

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$\text{adj}(A) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Hence,

$\text{adj}(A) = \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix}$

Correct Answer B:

$\boxed{\begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix}}$

7.

$A = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix}$

$| A | = 1(-1) - 3(2) = -1 - 6 = -7$

$\text{adj}(A) = \begin{bmatrix} -1 & -3 \\ -2 & 1 \end{bmatrix}$

$A^{-1} = \frac{1}{-7}\begin{bmatrix} -1 & -3 \\ -2 & 1 \end{bmatrix}$

Correct Answer A:

$\boxed{-\frac{1}{7}\begin{bmatrix} -1 & -3 \\ -2 & 1 \end{bmatrix}}$

8. For element 1 at position $(1, 1)$,

$M_{11} = \begin{vmatrix} -1 & 4 \\ 7 & -6 \end{vmatrix} = 6 - 28 = -22$

For element -1 at position $(2, 2)$,

$M_{22} = \begin{vmatrix} 1 & 2 \\ -3 & -6 \end{vmatrix} = -6 + 6 = 0$

Correct Answer A: $\boxed{-22,\ 0}$

9. For element 7at position $(2, 3)$,

$C_{23} = (-1)^{2+3}\begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} = -(4 - 4) = 0$

For element 6 at position $(3, 3)$,

$C_{33} = (-1)^{3+3}\begin{vmatrix} 1 & 2 \\ 4 & -1 \end{vmatrix} = -1 - 8 = -9$

Correct Answer C: $\boxed{0,\ -9}$

10. Given,

$A = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

$| A | = 1$

$\text{adj}(A) = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$

Hence,

$A^{-1} = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$

Correct Answer D:

$$\boxed{\begin{bmatrix}1 & 3 & 3\\ 1 & 4 & 3\\ 1 & 3 & 4\end{bmatrix}}$$

11. Given,

$$A=\begin{bmatrix}2 & 2 & 1\\ -1 & 0 & 2\\ 4 & 1 & 0\end{bmatrix}$$

$|A|=11$

Using,

$$A^{-1}=\frac{1}{|A|}\text{adj}(A)$$

So,

$11A^{-1}=\text{adj}(A)$

Calculating adjoint,

$$11A^{-1}=\begin{bmatrix}-2 & 1 & 4\\ 8 & -4 & -5\\ -1 & 6 & 2\end{bmatrix}$$

Correct Answer D:

$$\boxed{\begin{bmatrix}-2 & 1 & 4\\ 8 & -4 & -5\\ -1 & 6 & 2\end{bmatrix}}$$

12. Given,

$$A=\begin{bmatrix}1 & -1 & 1\\ 1 & 2 & 0\\ 1 & 3 & 0\end{bmatrix}$$

$$|A|=1\begin{vmatrix}2 & 0\\ 3 & 0\end{vmatrix}-(-1)\begin{vmatrix}1 & 0\\ 1 & 0\end{vmatrix}+1\begin{vmatrix}1 & 2\\ 1 & 3\end{vmatrix}$$

$=0+0+(3-2)=1$

For a 3×3matrix,

$|\text{adj}(A)|=|A|^{3-1}$

$=1^2=1$

Correct Answer C: $\boxed{1}$

13. Find the value of the determinant:

$$\Delta=\begin{vmatrix}b+c & a & 1\\ c+a & b & 1\\ a+b & c & 1\end{vmatrix}$$

Apply,

$C_1\to C_1+C_2$

$$\Delta=\begin{vmatrix}a+b+c & a & 1\\ a+b+c & b & 1\\ a+b+c & c & 1\end{vmatrix}$$

Take $(a+b+c)$common from C_1:

$$\Delta=(a+b+c)\begin{vmatrix}1 & a & 1\\ 1 & b & 1\\ 1 & c & 1\end{vmatrix}$$

Since $C_1=C_3$,

$\Delta=0$

Correct Answer A: $\boxed{0}$

14. Find the equivalent form of the determinant:

$$\Delta=\begin{vmatrix}ax & by & cz\\ x^2 & y^2 & z^2\\ 1 & 1 & 1\end{vmatrix}$$

Take x,y,zcommon from columns:

$$\Delta = xyz\begin{vmatrix} a & b & c \\ x & y & z \\ \frac{1}{x} & \frac{1}{y} & \frac{1}{z} \end{vmatrix}$$

Now, $xyz\left(\frac{1}{x},\frac{1}{y},\frac{1}{z}\right) = (y, z, zxxy)$

Hence,

$$\Delta = \begin{vmatrix} a & b & c \\ x & y & z \\ yz & zx & xy \end{vmatrix}$$

Correct Answer A:

$$\boxed{\begin{vmatrix} a & b & c \\ x & y & z \\ yz & zx & xy \end{vmatrix}}$$

15. Find the value of the determinant:

$$\Delta = \begin{vmatrix} bc & b+c & 1 \\ ca & c+a & 1 \\ ab & a+b & 1 \end{vmatrix}$$

Apply,

$R_2 \to R_2 - R_1,\ R_3 \to R_3 - R_1$

$$\Delta = \begin{vmatrix} bc & b+c & 1 \\ c(a-b) & a-b & 0 \\ -b(c-a) & -(c-a) & 0 \end{vmatrix}$$

Take common factors:

$$\Delta = (a-b)(c-a)\begin{vmatrix} bc & b+c & 1 \\ c & 1 & 0 \\ -b & -1 & 0 \end{vmatrix}$$

Expanding along C_3:

$$\Delta = (a-b)(c-a)\begin{vmatrix} c & 1 \\ -b & -1 \end{vmatrix}$$

$$= (a-b)(c-a)(b-c)$$

$$= (a-b)(b-c)(c-a)$$

Correct Answer C: $\boxed{(a-b)(b-c)(c-a)}$

16. $$\Delta = \begin{vmatrix} a+b+2c & a & b \\ c & b+c+2a & b \\ c & a & c+a+2b \end{vmatrix}$$

Apply $C_1 + C_2 + C_3 \to C_1$:

$$\Delta = \begin{vmatrix} 2(a+b+c) & a & b \\ 2(a+b+c) & b+c+2a & b \\ 2(a+b+c) & a & c+a+2b \end{vmatrix}$$

Take $2(a+b+c)$common from C_1:

$$= 2(a+b+c)\begin{vmatrix} 1 & a & b \\ 1 & b+c+2a & b \\ 1 & a & c+a+2b \end{vmatrix}$$

Now apply $R_2 - R_1,\ R_3 - R_1$:

$$= 2(a+b+c)\begin{vmatrix} 1 & a & b \\ 0 & a+b+c & 0 \\ 0 & 0 & a+b+c \end{vmatrix}$$

$$= 2(a+b+c)(a+b+c)^2$$

$$\boxed{2(a+b+c)^3}$$

Answer: D

17. $\Delta = \begin{vmatrix} -a^2 & ab & ac \\ ba & -b^2 & bc \\ ca & cb & -c^2 \end{vmatrix}$

Take a, b, c common from rows:

$\Delta = abc \begin{vmatrix} -a & b & c \\ a & -b & c \\ a & b & -c \end{vmatrix}$

Again take a, b, c common from columns:

$= a^2b^2c^2 \begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$

Evaluate determinant:

$\begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix} = 4$

Hence,

$\Delta = 4a^2b^2c^2$

$\boxed{k = 4}$

Answer: C

18. $\begin{vmatrix} a+b & b+c & c+a \\ b+c & c+a & a+b \\ c+a & a+b & b+c \end{vmatrix} = k \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$

Let

$A = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$

Observe:

$a + b = (a) + (b)$

$b + c = (b) + (c)$

$c + a = (c) + (a)$

So left determinant becomes:

$\begin{vmatrix} a+b & b+c & c+a \\ b+c & c+a & a+b \\ c+a & a+b & b+c \end{vmatrix} = 2 \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$

Hence, $\boxed{k = 2}$

Answer: B

19. $\Delta = \begin{vmatrix} 1+a & b & c \\ a & 1+b & c \\ a & b & 1+c \end{vmatrix}$

Apply $R_1 \to R_1 - R_2$ and $R_2 \to R_2 - R_3$:

$\Delta = \begin{vmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ a & b & 1+c \end{vmatrix}$

Expand along R_1:

$\Delta = 1 \begin{vmatrix} 1 & -1 \\ b & 1+c \end{vmatrix} + 1 \begin{vmatrix} 0 & -1 \\ a & 1+c \end{vmatrix}$

$= (1 + c + b) + a$

$= 1 + a + b + c$

$\boxed{1 + a + b + c}$

Answer: C

20. $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$

Expand determinant:

$a(bc - a^2) - b(b^2 - ac) + c(ab - c^2) = 0$

$abc - a^3 - b^3 + abc + abc - c^3 = 0$

$3abc - (a^3 + b^3 + c^3) = 0$

$a^3 + b^3 + c^3 = 3abc$

Given $abc \neq 0$,

$\frac{a^3 + b^3 + c^3}{abc} = \frac{3abc}{abc} = 3$

Answer: B

LEVEL – II

SINGLE OPTION CORRECT TYPE

1. For an $n \times n$matrix,

$\det(\text{adj}\, A) = (\det A)^{n-1}$

Here $n = 3$ and $\det A = 5$.

$\det(\text{adj}\, A) = 5^{3-1} = 5^2 = 25$

$\boxed{25}$

Answer: B

2. $\begin{vmatrix} a & a^2 & 1 + a^3 \\ b & b^2 & 1 + b^3 \\ c & c^2 & 1 + c^3 \end{vmatrix} = 0$

Apply

$C_3 \to C_3 - C_1$

Then

$C_3 = \begin{bmatrix} 1 + a^3 - a \\ 1 + b^3 - b \\ 1 + c^3 - c \end{bmatrix}$

Using

$1 + x^3 = (x + 1)(x^2 - x + 1)$

Since a, b, c are distinct, determinant becomes a Vandermonde type determinant and for determinant to be zero,

$1 + x^3$ must be dependent on x, x^2

which happens when constant term vanishes:

$1 + abc = 0$

$abc = -1$

$\boxed{-1}$

Answer: D

3. $A = \begin{bmatrix} \alpha & 2 \\ 2 & \alpha \end{bmatrix}$

$|A| = \alpha^2 - 4$

Given

$|A^3| = 125$

Using

$|A^3| = |A|^3$

$(\alpha^2 - 4)^3 = 125$

$\alpha^2 - 4 = 5$

$\alpha^2 = 9$

$\alpha = \pm 3$

$\boxed{\pm 3}$

Answer: C

4. $A = \begin{bmatrix} 6 & 11 \\ 2 & 4 \end{bmatrix}$

$A^{2015} - 6A^{2014} = A^{2014}(A - 6I)$

Therefore,

$\det(A^{2015} - 6A^{2014}) = \det(A^{2014})\det(A - 6I)$

Now,

$|A| = 6 \cdot 4 - 11 \cdot 2 = 24 - 22 = 2$

So,

$|A^{2014}| = 2^{2014}$

Also,

$A - 6I = \begin{bmatrix} 0 & 11 \\ 2 & -2 \end{bmatrix}$

$|A - 6I| = 0(-2) - 22 = -22$

Hence,

$\det(A^{2015} - 6A^{2014}) = 2^{2014}(-22) = -11 \cdot 2^{2015}$

$\boxed{-11 \cdot 2^{2015}}$

Answer: B

5. $\Delta = \begin{vmatrix} 1 & w^n & w^{2n} \\ w^n & w^{2n} & 1 \\ w^{2n} & 1 & w^n \end{vmatrix}$

Add all rows:

Since $1 + w + w^2 = 0$and powers repeat modulo 3,

$1 + w^n + w^{2n} = 0$

Thus sum of entries in each row is zero.

Hence rows are linearly dependent.

$\boxed{0}$

Answer: D

6. Given

$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0$

$\Rightarrow ab + bc + ca = 0$

Now,

$\Delta = \begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix}$

Expanding,

$\Delta = (1+a)(1+b)(1+c) - (1+a) - (1+b) - (1+c) + 2$

$= 1 + a + b + c + ab + bc + ca + abc - 3 - a - b - c + 2$

$= ab + bc + ca + abc$

Using $ab + bc + ca = 0$,

$\Delta = abc$

$\boxed{abc}$

Answer: B

7. Let $\Delta = \begin{vmatrix} x+1 & x+2 & x+4 \\ x+3 & x+5 & x+8 \\ x+7 & x+10 & x+14 \end{vmatrix}$

Apply column operations:

$C_2 \to C_2 - C_1$, $C_3 \to C_3 - C_1$

Then,

$\Delta = \begin{vmatrix} x+1 & 1 & 3 \\ x+3 & 2 & 5 \\ x+7 & 3 & 7 \end{vmatrix}$

Now expand along the first column:

$\Delta = (x+1)\begin{vmatrix} 2 & 5 \\ 3 & 7 \end{vmatrix} - (x+3)\begin{vmatrix} 1 & 3 \\ 3 & 7 \end{vmatrix} + (x+7)\begin{vmatrix} 1 & 3 \\ 2 & 5 \end{vmatrix}$

$= (x+1)(14-15) - (x+3)(7-9) + (x+7)(5-6)$

$= -(x+1) + 2(x+3) - (x+7)$

$= -2$

Since determinant is always -2, given condition is true for every x.

Any real number

Answer: D) any real number

8. $\Delta = \begin{vmatrix} 0 & x-a & x-b \\ x+a & 0 & x-c \\ x+b & x+c & 0 \end{vmatrix} = 0$

Expand determinant:

$0 - (x-a)(x-c)(x+b) + (x-b)(x+a)(x+c) = 0$

$(x-b)(x+a)(x+c) = (x-a)(x-c)(x+b)$

Expanding both sides and simplifying:

$2x(a+c-b) = 2acx$

On simplification,

$x(a-b)(b-c)(c-a) = 0$

Since a, b, c are different,

$(a-b)(b-c)(c-a) \neq 0$

Hence,

$x = 0$

Answer: D

9. $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix}$

Apply column operations:

$C_2 \to C_2 - C_1$, $C_3 \to C_3 - C_1$

$\Delta = \begin{vmatrix} 1 & 0 & 0 \\ a & b-a & c-a \\ a^3 & b^3-a^3 & c^3-a^3 \end{vmatrix}$

$= \begin{vmatrix} b-a & c-a \\ b^3-a^3 & c^3-a^3 \end{vmatrix}$

Using

$b^3 - a^3 = (b-a)(a^2+ab+b^2)$

$c^3 - a^3 = (c-a)(a^2+ac+c^2)$

$\Delta = (b-a)(c-a)\begin{vmatrix} 1 & 1 \\ a^2+ab+b^2 & a^2+ac+c^2 \end{vmatrix}$

$= (b-a)(c-a)(ac+c^2-ab-b^2)$

$= (b-a)(c-a)(c-b)(a+b+c)$

$= (a-b)(b-c)(c-a)(a+b+c)$

$\boxed{(a-b)(b-c)(c-a)(a+b+c)}$

Answer: B

10. $D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+x & 1 \\ 1 & 1 & 1+y \end{vmatrix}$

Apply:

$R_2 \to R_2 - R_1,\ R_3 \to R_3 - R_1$

$D = \begin{vmatrix} 1 & 1 & 1 \\ 0 & x & 0 \\ 0 & 0 & y \end{vmatrix}$

$D = 1 \cdot x \cdot y = xy$

Hence D is divisible by both $x$and y.

$\boxed{\text{Divisible by both } x \text{ and } y}$

Answer: B

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. $A = \begin{bmatrix} \sin\alpha & -\cos\alpha & 0 \\ \cos\alpha & \sin\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Since

$AA^T = I$

matrix A is orthogonal.

Hence,

$|A| = 1$

Also,

$|A^T| = |A| = 1$

So A is true.

Further,

$|A^{-1}| = \frac{1}{|A|} = 1$

So B is true.

Since

$A^{-1} = \frac{\text{adj}\, A}{|A|}$

and $|A| = 1$,

$A^{-1} = \text{adj}\, A$

So C is true.

Also,

$|AA^T| = |I| = 1 \neq 10$

So D is false.

$\boxed{\boldsymbol{A, B, C}}$

12. Given

$\text{adj}(P) = \begin{bmatrix} 1 & 4 & 4 \\ 2 & 1 & 7 \\ 1 & 1 & 3 \end{bmatrix}$

Find determinant:

$$|\text{adj}(P)| = \begin{vmatrix} 1 & 4 & 4 \\ 2 & 1 & 7 \\ 1 & 1 & 3 \end{vmatrix}$$

$= 1(3-7) - 4(6-7) + 4(2-1)$

$= -4 + 4 + 4 = 4$

For a 3×3matrix,

$|\text{ adj}(P)| = |P|^2$

Thus,

$|P|^2 = 4$

$|P| = \pm 2$

$\boxed{-2, 2}$

Correct options: A, B

13. $A = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$

Compute A^2:

$$A^2 = \frac{1}{3}\begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}\begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$$

$$= \frac{1}{3}\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = I$$

Hence,

$A^{-1} = A$

So A is involutory.

Also,

$\text{adj}\, A = |A|\, A^{-1}$

Now,

$$|A| = \frac{1}{3}\begin{vmatrix} 1 & 1+i \\ 1-i & -1 \end{vmatrix}$$

$$= \frac{1}{3}[-1 - (1+i)(1-i)] = \frac{1}{3}(-1-2) = -1$$

Thus,

$\text{adj}\, A = (-1)A = -A \neq A$

Also $A^2 \neq A$, so not idempotent.

Correct answers $\boxed{B, C}$

14. A)

$|A| = |A^T|$

True.

B)

If $A = I$,

$|A| = |I| = 1$

True.

C)

If $A = 0$,

$|A| = 0$

True.

D)

For a skew-symmetric matrix, $A^T = -A$

Determinant is zero only for odd order skew-symmetric matrices, not always.

Hence false.

Correct answers: $\boxed{A, B, C}$

COMPREHENSION TYPE

Passage

15. $A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$

Characteristic equation:

$| A - \lambda I | = \begin{vmatrix} 2-\lambda & 3 \\ 1 & 2-\lambda \end{vmatrix}$

$= (2-\lambda)^2 - 3 = \lambda^2 - 4\lambda + 1$

By Cayley–Hamilton theorem,

$A^2 - 4A + I = 0$

$\boxed{A^2 - 4A + I = 0}$

Answer: A

16. Given

$\begin{vmatrix} 1 & -1 & x \\ 1 & x & 1 \\ x & -1 & 1 \end{vmatrix}$

For no inverse,

$| A | = 0$

Expand:

$= 1(x+1) - (-1)(1-x) + x(-1-x^2)$

$= x + 1 + 1 - x - x - x^3$

$= 2 - x - x^3$

Set equal to zero:

$x^3 + x - 2 = 0$

Checking options:

$x = 1 \Rightarrow 1 + 1 - 2 = 0$

$\boxed{x = 1}$

Answer: D

17. $A = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$

Characteristic equation:

$| A - \lambda I | = \begin{vmatrix} 3-\lambda & 2 \\ 1 & 1-\lambda \end{vmatrix}$

$= (3-\lambda)(1-\lambda) - 2$

$= \lambda^2 - 4\lambda + 1$

By Cayley–Hamilton theorem,

$A^2 - 4A + I = 0$

Compare with

$A^2 + aA + bI = 0$

So,

$a = -4,\ b = 1$

$| a + b | = | -4 + 1 | = 3$

$\boxed{3}$

Answer: C

MATRIX MATCHING TYPE

18. (a) $A = \begin{bmatrix} -1 & -2 & -2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$

$AA^T = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix} = 9I$

$A^{-1} = \frac{1}{9}A^T$

$\boxed{a \to P}$

(b)

$A = \begin{bmatrix} 1 & 0 \\ -1 & 7 \end{bmatrix}$

Characteristic equation:

$(1-\lambda)(7-\lambda) = 0$

$\lambda^2 - 8\lambda + 7 = 0$

By Cayley-Hamilton theorem,

$A^2 - 8A + 7I = 0$

$A^2 = 8A - 7I$

Comparing with $A^2 = 8A + KI$,

$K = -7$

$\boxed{b \to T}$

(c)

$$\begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix} = 0$$

$$\begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ y & y^2 & y^3 \\ z & z^2 & z^3 \end{vmatrix} = 0$$

$$(1+xyz)\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = 0$$

Since x, y, z are distinct,

$$\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} \neq 0$$

$1 + xyz = 0$

$xyz = -1$

$\boxed{c \to R}$

(d)

$A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix}$

$A^{-1} = A^T$

$\boxed{d \to Q}$

INTEGER TYPE

19. $\Delta = \begin{vmatrix} 0 & b & -c \\ -b & 0 & a \\ c & -a & 0 \end{vmatrix}$

Since $\Delta^T = -\Delta$, the determinant is skew-symmetric of odd order.

$\therefore \Delta = 0$

20. Given,

$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & \lambda \end{bmatrix}$

For singular matrix,

$|A| = 0$

$$\begin{vmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & \lambda \end{vmatrix} = 0$$

Expanding,

$8(7\lambda - 16) + 6(-6\lambda + 8) + 2(24 - 14) = 0$

$56\lambda - 128 - 36\lambda + 48 + 20 = 0$

$20\lambda - 60 = 0$

$\lambda = 3$

STUDENT PRACTISE CORNER

SINGLE CORRECT ANSWER TYPE

1. Evaluate the determinant

$$\begin{vmatrix} 4 & 3 \\ -1 & 2 \end{vmatrix}$$

Using,

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$= (4)(2) - (3)(-1)$

$= 8 + 3 = 11$

Correct Answer: B) 11

2. If $A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$

find $\det(A^2 - 3A)$

First find A^2:

$$A^2 = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}\begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 8 & 5 \\ 20 & 13 \end{bmatrix}$$

Now,

$$3A = \begin{bmatrix} 6 & 3 \\ 12 & 9 \end{bmatrix}$$

$$A^2 - 3A = \begin{bmatrix} 8-6 & 5-3 \\ 20-12 & 13-9 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 8 & 4 \end{bmatrix}$$

$\det(A^2 - 3A) = (2)(4) - (2)(8)$

$= 8 - 16 = -8$

Correct Answer: C) -8

3. Evaluate the determinant

$$\begin{vmatrix} 1 & 4 & 2 \\ 2 & -1 & 4 \\ -3 & 7 & -6 \end{vmatrix}$$

Expanding along first row:

$$= 1\begin{vmatrix} -1 & 4 \\ 7 & -6 \end{vmatrix} - 4\begin{vmatrix} 2 & 4 \\ -3 & -6 \end{vmatrix} + 2\begin{vmatrix} 2 & -1 \\ -3 & 7 \end{vmatrix}$$

$= 1[(6 - 28)] - 4[(-12 + 12)] + 2[(14 - 3)]$

$= -22 - 0 + 22$

$= 0$

Correct Answer: A) 0

4. Evaluate the determinant

$$\begin{vmatrix} a+b+c & a+b+c & a+b+c \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$$

Take $(a + b + c)$common from the first row:

$$= (a+b+c)\begin{vmatrix} 1 & 1 & 1 \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$$

Now apply:

$C_2 \to C_2 - C_1,\ C_3 \to C_3 - C_1$

$$= (a+b+c)\begin{vmatrix} 1 & 0 & 0 \\ 2b & -(a+b+c) & 0 \\ 2c & 0 & -(a+b+c) \end{vmatrix}$$

Determinant:

$= (a+b+c)\big[\big(-(a+b+c)\big)\big(-(a+b+c)\big)\big]$

$= (a+b+c)^3$

Correct Answer: A) $(a+b+c)^3$

5. Evaluate the determinant

$$\begin{vmatrix} y+z & x & x \\ y & z+x & y \\ z & z & x+y \end{vmatrix}$$

Apply:

$C_1 \to C_1 - C_2$

$$= \begin{vmatrix} y+z-x & x & x \\ -x+y-z & z+x & y \\ 0 & z & x+y \end{vmatrix}$$

Expanding and simplifying:

$= 2xyz + 2xyz$

$= 4xyz$

Correct Answer: D) $4xyz$

6. If $\begin{vmatrix} a & a & x \\ m & m & m \\ b & x & b \end{vmatrix} = 0$

find x.

Expand along the second row:

$= m\begin{vmatrix} a & x \\ x & b \end{vmatrix} - m\begin{vmatrix} a & x \\ b & b \end{vmatrix} + m\begin{vmatrix} a & a \\ b & x \end{vmatrix} = 0$

$= m(ab - x^2) - m(ab - bx) + m(ax - ab)$

$= mab - mx^2 - mab + mbx + max - mab$

$= -mx^2 + mbx + max - mab$

$= -m(x^2 - bx - ax + ab)$

$= -m(x-a)(x-b) = 0$

Hence, $x = a$ or $x = b$

Correct Answer: D) a(or) b

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

7. If $\begin{vmatrix} 3x-8 & 3 & 3 \\ 3 & 3x-8 & 3 \\ 3 & 3 & 3x-8 \end{vmatrix} = 0$

find x.

For a matrix of type

$$\begin{vmatrix} a & b & b \\ b & a & b \\ b & b & a \end{vmatrix}$$

determinant is

$(a-b)^2(a+2b)$

Here,

$a = 3x-8,\ b = 3$

So,

$[(3x-8)-3]^2[(3x-8)+6]=0$

$(3x-11)^2(3x-2)=0$

Hence,

$3x-11=0 \Rightarrow x=\frac{11}{3}$

or

$3x-2=0 \Rightarrow x=\frac{2}{3}$

Correct Answers: B) $\frac{2}{3}$ and C) $\frac{11}{3}$

8. Let $A=\begin{bmatrix}1 & 0 & 2\\ 3 & 1 & 4\\ 5 & 2 & 0\end{bmatrix}$

Check each statement.

A) Minor of a_{11}

Delete row 1 and column 1:

$M_{11}=\begin{vmatrix}1 & 4\\ 2 & 0\end{vmatrix}=(1)(0)-(4)(2)=-8$

So statement A is correct.

B) Cofactor of a_{11}

$C_{11}=(-1)^{1+1}M_{11}$

$=1(-8)=-8$

So statement B is correct.

C) Minor of a_{23}

Delete row 2 and column 3:

$M_{23}=\begin{vmatrix}1 & 0\\ 5 & 2\end{vmatrix}=(1)(2)-(0)(5)=2$

So statement C is correct.

D) Cofactor of a_{23}

$C_{23}=(-1)^{2+3}M_{23}$

$=-2$

So statement D is correct.

Correct Answers: A, B, C and D

9. Which of the following determinants are equal to zero?

A)

$\begin{vmatrix}1 & 2\\ 2 & 4\end{vmatrix}$

$=(1)(4)-(2)(2)=4-4=0$

Determinant is zero.

B)

$\begin{vmatrix}3 & 5\\ 5 & 3\end{vmatrix}$

$=(3)(3)-(5)(5)=9-25=-16\neq 0$

Not zero.

C)

$\begin{vmatrix}0 & 0 & 0\\ 1 & 2 & 3\\ 4 & 5 & 6\end{vmatrix}$

Since one row is completely zero,

$\det=0$

Determinant is zero.

D)

$\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$

$= (1)(1) - (0)(0) = 1$

Not zero.

Correct Answers: A and C

COMPREHENSION TYPE

Passage

Given,

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 5 & 6 & 8 \end{bmatrix}, B = \begin{bmatrix} 3 & 2 & 5 \\ 2 & 3 & 8 \\ 7 & 2 & 9 \end{bmatrix}$$

10. Find $\text{adj}(\text{adj}\, A)$

For an $n \times n$matrix,

$\text{adj}(\text{adj}\, A) = |A|^{n-2} A$

Here $n = 3$, so

$\text{adj}(\text{adj}\, A) = |A| A$

Now find $|A|$:

$|A| = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 5 & 6 & 8 \end{vmatrix}$

$= 1(24 - 24) - 2(16 - 20) + 3(12 - 15)$

$= 0 + 8 - 9 = -1$

Hence,

$\text{adj}(\text{adj}\, A) = (-1)A = -A$

Correct Answer: A) $-A$

11. Find $|\text{adj}(\text{adj}\, B)|$

For a 3×3matrix,

$\text{adj}(\text{adj}\, B) = |B| B$

Taking determinant on both sides,

$|\text{adj}(\text{adj}\, B)| = |\ |B| B|$

$= |B|^3 |B|$

$= |B|^4$

Correct Answer: D) $|B|^4$

12. Find $|\text{adj}(AB)|$

Use property:

$|\text{adj}(M)| = |M|^{n-1}$

For 3×3,

$|\text{adj}(AB)| = |AB|^2$

Also,

$|AB| = |A||B|$

Now,

$|A| = -1$

Find $|B|$:

$|B| = \begin{vmatrix} 3 & 2 & 5 \\ 2 & 3 & 8 \\ 7 & 2 & 9 \end{vmatrix}$

$= 3(27 - 16) - 2(18 - 56) + 5(4 - 21)$

$= 33 + 76 - 85 = 24$

Thus,

$|AB| = (-1)(24) = -24$

$|\text{adj}(AB)| = (-24)^2 = 24^2$

Correct Answer: B) 24^2

MATRIX MATCHING TYPE

13. $\begin{vmatrix} a^2+1 & ab & ac \\ ab & b^2+1 & bc \\ ac & bc & c^2+1 \end{vmatrix} =$

Write as $I + VV^T$, where $V = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$.

Using determinant property:

$|I + VV^T| = 1 + V^T V$

$= 1 + a^2 + b^2 + c^2$

$\boxed{(a) \to S}$

(b) $\begin{vmatrix} a-b-c & 2b & 2c \\ 2a & b-c-a & 2c \\ 2a & 2b & c-a-b \end{vmatrix}$

Apply:

$C_1 + C_2 + C_3 = \begin{bmatrix} a+b+c \\ a+b+c \\ a+b+c \end{bmatrix}$

Factor $(a+b+c)$:

$\Delta = (a+b+c)\begin{vmatrix} 1 & 2b & 2c \\ 1 & b-c-a & 2c \\ 1 & 2b & c-a-b \end{vmatrix}$

Evaluating gives:

$\Delta = (a+b+c)^3$

$\boxed{(b) \to R}$

(c) $\begin{vmatrix} a & b & c \\ a-b & b-c & c-a \\ b+c & c+a & a+b \end{vmatrix}$

Apply $R_3 \to R_1 + R_3$:

$$R_3 = (a, +, b+c\ a+b+c\ a+b+c)$$

Take $(a+b+c)$common:

$\Delta = (a+b+c)\begin{vmatrix} a & b & c \\ a-b & b-c & c-a \\ 1 & 1 & 1 \end{vmatrix}$

Now evaluate:

$\Delta = a^3 + b^3 + c^3 - 3abc$

$\boxed{(c) \to Q}$

(d) $\begin{vmatrix} a^2+2a & 2a+1 & 1 \\ 2a+1 & a+2 & 1 \\ 3 & 3 & 1 \end{vmatrix}$

Apply:

$R_1 - R_2 = (a^2, -, 1\ a-1\ 0)$

Factor $(a-1)$:

$$= (a-1)\begin{vmatrix} a+1 & 1 & 0 \\ 2a+1 & a+2 & 1 \\ 3 & 3 & 1 \end{vmatrix}$$

Evaluating:

$\Delta = (a-1)^3$

$\boxed{(d) \to P}$

INTEGER TYPE

14. Element 7 is at position $(2,3)$.

Its cofactor:

$$C_{23} = (-1)^{2+3}\begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix}$$

$= (-1)^5(1 \cdot 4 - 2 \cdot 2)$

$= -1(4-4) = 0$

$\boxed{0}$

15. $\begin{vmatrix} \sin\alpha & \cos\alpha \\ -\cos\alpha & \sin\alpha \end{vmatrix}$

$= \sin^2\alpha - (-\cos^2\alpha)$

$= \sin^2\alpha + \cos^2\alpha$

$= 1$

2. Functions

Domain, Range of Function and Composite Functions

Level – I

1. For $f(x) = \sqrt{(x-1)(x-2)}$

For square root,

$(x-1)(x-2) \geq 0$

Critical points: $x = 1,2$

Using sign analysis:

- $x \leq 1 \rightarrow$ positive
- $1 < x < 2 \rightarrow$ negative
- $x \geq 2 \rightarrow$ positive

So domain is:

$(-\infty, 1] \cup [2, \infty)$

Answer: C

2. For $f(x) = \frac{x}{x^2-3x+2}$

Denominator should not be zero:

$x^2 - 3x + 2 \neq 0$

Factorize:

$(x-1)(x-2) \neq 0$

So,

$x \neq 1,\ x \neq 2$

Domain:

$\mathbb{R} - \{1,2\}$

Answer: D

3. For $f(x) = \frac{1}{2-\cos 3x}$

Since

$-1 \leq \cos 3x \leq 1$

then

$1 \leq 2 - \cos 3x \leq 3$

Denominator is never zero.

So domain is all real numbers.

Answer: A

4. For

$$f(x) = \frac{1}{\log_{10}(1-x)} + \sqrt{x+2}$$

Conditions:

From square root: $x + 2 \geq 0$

$x \geq -2$

From logarithm:

$1 - x > 0$

$x < 1$

Also denominator cannot be zero: $\log_{10}(1-x) \neq 0$

$1 - x \neq 1$

$x \neq 0$

Combining: $[-2,1) - \{0\}$

Answer: C

5. For

$$f(x) = \sin\left\{\log\left(\sqrt{\frac{4-x^2}{1-x}}\right)\right\}$$

For logarithm:

$$\sqrt{\frac{4-x^2}{1-x}} > 0$$

So,

$$\frac{4-x^2}{1-x} > 0$$

Factorize:

$$\frac{(2-x)(2+x)}{1-x} > 0$$

Critical points:

$x = -2, 1, 2$

Sign analysis gives:

$(-2,1) \cup (2,\infty)$

But numerator becomes negative after 2, so only: $(-2,1)$

Answer: A

6. For

$$f(x) = \frac{x}{1-x}$$

Let

$$y = \frac{x}{1-x}$$

Solve for x:

$y(1-x) = x$

$y = x + xy$

$y = x(1+y)$

$$x = \frac{y}{1+y}$$

This exists for all real $y$except:

$1 + y \neq 0$

$y \neq -1$

Hence range: $\mathbb{R} - \{-1\}$

Answer: C

7. For

$$f(x) = \sqrt{9-x^2}$$

Since

$9 - x^2 \geq 0$

$-3 \leq x \leq 3$

Now,

$f(x) \geq 0$

Maximum value:
$\sqrt{9} = 3$
Minimum value: 0
Hence range: $[0, 3]$
Answer: D

8. For
$f(x) = \log_{10}(3x^2 - 4x + 5)$
Consider
$3x^2 - 4x + 5$
Minimum value occurs at
$x = \frac{-(-4)}{2(3)} = \frac{2}{3}$

Minimum value:
$3\left(\frac{2}{3}\right)^2 - 4\left(\frac{2}{3}\right) + 5 = \frac{11}{3}$

So,
$3x^2 - 4x + 5 \geq \frac{11}{3}$
Hence,
$f(x) \geq \log_{10}\left(\frac{11}{3}\right)$

Range:
$[\log_{10}\left(\frac{11}{3}\right), \infty)$
Answer: B

9. Given
$f(x) = x^2 - 2x$
$g(x) = -x + 6$
For equality:
$x^2 - 2x = -x + 6$
$x^2 - x - 6 = 0$
$(x - 3)(x + 2) = 0$
$x = 3, -2$
Hence domain where both are equal:
$\{-2,3\}$
Answer: B

10. Number of surjections from a 5-element set onto a 3-element set:
Using inclusion-exclusion:
$3^5 - \binom{3}{1}2^5 + \binom{3}{2}1^5$
$= 243 - 3(32) + 3(1)$
$= 243 - 96 + 3$
$= 150$
Answer: B

11. Number of many-one functions from
$A = \{1,2\}, B = \{3,4,5,6,7\}$
Total functions:
$5^2 = 25$
One-one functions:
${}^5P_2 = 5 \times 4 = 20$
Many-one functions:
$25 - 20 = 5$
Answer: A

12. Number of bijections from a set with 2018 elements to itself: 2018!
Answer: B

13. Number of non-surjective mappings from
$A = \{a, b, c, d\}, B = \{1,2,3,4,5\}$
Total functions:
$5^4 = 625$
Since $|A| < |B|$, no function can be surjective.
Hence all functions are non-surjective.
625
Answer: D

14. Given
$f = \{(4,5), (5,6), (6,-4)\}$
$g = \{(4,-4), (6,5), (8,5)\}$
Compute:
$2f = \{(4,10), (5,12), (6,-8)\}$
$4g = \{(4,-16), (6,20), (8,20)\}$
Now add common domain elements:
For $x = 4$:
$10 + (-16) = -6$
For $x = 6$:
$-8 + 20 = 12$
Hence,
$2f + 4g = \{(4,-6), (6,12)\}$
Answer: C

15. Given
$f(x) = 2x - 1, g(x) = x^2$
Find
$\left(\frac{\sqrt{f}}{g}\right)(4)$

First,
$f(4) = 2(4) - 1 = 7$
$g(4) = 4^2 = 16$

Therefore,
$\left(\frac{\sqrt{f}}{g}\right)(4) = \frac{\sqrt{f(4)}}{g(4)} = \frac{\sqrt{7}}{16}$

Answer: B

16. If $f \circ g$is one-one, then $g$must be one-one.
Because if $g$is not one-one, then distinct inputs can give same output, making $f \circ g$not one-one.
Answer: B

17. Given
$f(x) = 4x - 1$
$g(x) = x^2 + 2$

Find
$(g \circ f)\left(\frac{a+1}{4}\right)$
First evaluate f:
$f\left(\frac{a+1}{4}\right) = 4\left(\frac{a+1}{4}\right) - 1 = a + 1 - 1 = a$
Now apply g:
$g(a) = a^2 + 2$
Hence,
$(g \circ f)\left(\frac{a+1}{4}\right) = a^2 + 2$
Answer: B

18. Given
$f(x) = \begin{cases} 0 & \text{when } x \text{ is rational} \\ 1 & \text{when } x \text{ is irrational} \end{cases}$
$g(x) = \begin{cases} -1 & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$
Find:
$(f \circ g)(\pi) + (g \circ f)(e)$

Since $\pi$is irrational,
$g(\pi) = 0$
Now 0is rational, so
$f(0) = 0$
Thus,
$(f \circ g)(\pi) = 0$
Now $e$is irrational, so
$f(e) = 1$
Since 1is rational,
$g(1) = -1$
Thus,
$(g \circ f)(e) = -1$
Therefore,
$0 + (-1) = -1$
Answer: A

19. Given
$g(x) = 1 + x - [x]$
where $[x]$denotes greatest integer function.
Also,
$f(x) = \begin{cases} -1 & x < 0 \\ 0 & x = 0 \\ 1 & x > 0 \end{cases}$

Now,
$x - [x] \geq 0$

and
$0 \leq x - [x] < 1$
Hence,
$1 + x - [x] > 0$
So,
$g(x) > 0$

Therefore, $f[g(x)] = 1$
Answer: B

20. Given
$f(x) = \sqrt{x^3 - 1}$
$g(x) = \sqrt[3]{x^2 + 1}$

Find
$(f \circ g)(x)$
$f(g(x)) = \sqrt{\left(\sqrt[3]{x^2 + 1}\right)^3 - 1}$
$= \sqrt{x^2 + 1 - 1}$
$= \sqrt{x^2}$
$= x$
Correct Answer: A) x

LEVEL – II

SINGLE OPTION CORRECT TYPE

1. Given
$f(x) = \begin{cases} 2x & 0 \leq x \leq 2 \\ 3x & 2 \leq x \leq 5 \end{cases}$
At $x = 2$,
From first part:
$f(2) = 2(2) = 4$
From second part:
$f(2) = 3(2) = 6$
Since two different values occur at the same input, $f(2)$is not defined.
Answer: D

2. Given
$f(x) = \dfrac{\alpha x^2 + 6x - 8}{\alpha + 6x - 8x^2}$

For onto function $f: \mathbb{R} \to \mathbb{R}$, let
$y = \dfrac{\alpha x^2 + 6x - 8}{\alpha + 6x - 8x^2}$

Cross multiply:
$y(\alpha + 6x - 8x^2) = \alpha x^2 + 6x - 8$
$(-8y - \alpha)x^2 + (6y - 6)x + (\alpha y + 8) = 0$

For real x, discriminant must be non-negative for all y.
After simplification:
$\Delta = 4(y+1)(\alpha - 8)(y-1)$
For all real y,
$\alpha - 8 \leq 0$
and checking boundary conditions gives
$0 \leq \alpha \leq 15$
Answer: C

3. Let
$f(x) = ax + b$
Given
$f(f(0)) = 0$
Since
$f(0) = b$
$f(b) = ab + b = b(a+1) = 0$

So either $b = 0$or $a = -1$.
Also,
$f(f(f(4))) = 9$
Testing possibilities gives:
If $a = -1$,
$f(x) = -x + b$
Then
$f(f(x)) = x$
$f(f(f(x))) = f(x)$

Thus,
$f(4) = 9$
$-4 + b = 9$
$b = 13$
Hence,
$f(x) = -x + 13$
Now compute:
$f(10) = -10 + 13 = 3$
$f(f(10)) = f(3) = 10$
$f(f(f(10))) = f(10) = 3$
$f(f(f(f(10)))) = f(3) = 10$
Answer: D

4. Given

$$f(n) = \begin{cases} \frac{n+1}{2} & n \text{ odd} \\ \frac{n}{2} & n \text{ even} \end{cases}$$

Observe:
$f(1) = 1, f(2) = 1$
Different inputs give same output.
So not injective.
For every $m \in N$,

$f(2m) = m$
Hence every natural number has a preimage.
So function is surjective.
Answer: B

5. Given
$[x^2] + x - a = 0$
where $[x^2]$denotes greatest integer function.
$a = [x^2] + x$
Checking possible integer values for $a \leq 20$, equation has solutions for:
$a = 1,2,3,4, \ldots$
Distinct valid values among options count to: 6
Answer: D

6. Given
$f(x) = x^2, g(x) = 2^x$
Find solution set of
$(f \circ g)(x) = (g \circ f)(x)$

Now,
$(f \circ g)(x) = f(2^x) = (2^x)^2 = 2^{2x}$
$(g \circ f)(x) = g(x^2) = 2^{x^2}$

Equating:
$2^{2x} = 2^{x^2}$
Thus,
$2x = x^2$
$x(x-2) = 0$
$x = 0,2$
Solution set:
$\{0, 2\}$
Answer: C

7. Given
$g(x) = 1 + \sqrt{x}$
and
$f(g(x)) = 3 + 2\sqrt{x} + x$
Let
$t = g(x) = 1 + \sqrt{x}$
Then
$\sqrt{x} = t - 1$
So, $x = (t-1)^2$
Now,
$f(t) = 3 + 2(t-1) + (t-1)^2$
$= 3 + 2t - 2 + t^2 - 2t + 1$
$= t^2 + 2$
Hence,
$f(x) = x^2 + 2$
Answer: B

8. Given

$f(x) = \frac{2x-2}{x^2-2x+3}$

Find range.

Let

$y = \frac{2x-2}{x^2-2x+3}$

Cross multiply:

$yx^2 - 2yx + 3y = 2x - 2$

$yx^2 + (-2y-2)x + (3y+2) = 0$

For real x, discriminant must satisfy:

$(-2y-2)^2 - 4y(3y+2) \geq 0$

$4(y+1)^2 - 4y(3y+2) \geq 0$

$(y+1)^2 - y(3y+2) \geq 0$

$y^2 + 2y + 1 - 3y^2 - 2y \geq 0$

$1 - 2y^2 \geq 0$

$y^2 \leq \frac{1}{2}$

$-\frac{1}{\sqrt{2}} \leq y \leq \frac{1}{\sqrt{2}}$

Range:

$\left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$

Answer: D

9. Given

$f(x) = \sqrt[3]{x+8}$

$g(x) = x^3 - 8$

Find

$(f \circ g)(x)$

$f(g(x)) = \sqrt[3]{(x^3-8)+8}$

$= \sqrt[3]{x^3}$

$= x$

Answer: A

ONE OR MORE THAN ONE OPTION CORRECT TYPE

10. Area of an equilateral triangle with side $s$is

$\frac{\sqrt{3}}{4}s^2$

Given area:

$\frac{\sqrt{3}}{4}$

So,

$s^2 = 1$

Distance between $(0,0)$and $(x, g(x))$is:

$\sqrt{x^2 + g(x)^2}$

Hence,
$x^2 + g(x)^2 = 1$
$g(x)^2 = 1 - x^2$
$g(x) = \pm\sqrt{1 - x^2}$

Thus possible functions are:
$g(x) = \sqrt{1 - x^2}$

And $g(x) = -\sqrt{1 - x^2}$
Correct options: B, D

11. Given
$g(f(x)) = |\sin x|$
and
$f(g(x)) = (\sin\sqrt{x})^2$

Check options.
Option A
$f(x) = \sin^2 x, g(x) = \sqrt{x}$

Then
$g(f(x)) = \sqrt{\sin^2 x} = |\sin x|$

Correct.
Also,
$f(g(x)) = \sin^2(\sqrt{x}) = (\sin\sqrt{x})^2$
Correct.
So A is correct.

Option B
$f(x) = \sin x, g(x) = |x|$
Then
$g(f(x)) = |\sin x|$
Correct.
But
$f(g(x)) = \sin|x| \neq (\sin\sqrt{x})^2$
Not correct.

Option C
$f(x) = x^2, g(x) = \sin\sqrt{x}$
Then
$g(f(x)) = \sin\sqrt{x^2} = \sin|x| \neq |\sin x|$

Not correct.
Hence only option A is correct.
Correct option: A

12. Given
$f(x) = \cos([\pi^2]x) + \cos([-\pi^2]x)$
Since
$\pi^2 \approx 9.86$
$[\pi^2] = 9$
and
$[-\pi^2] = -10$
Therefore,
$f(x) = \cos(9x) + \cos(-10x)$

Using
$\cos(-\theta) = \cos\theta$
we get
$f(x) = \cos(9x) + \cos(10x)$
Option A
$f\left(\frac{\pi}{2}\right) = \cos\frac{9\pi}{2} + \cos 5\pi$
$= 0 + (-1) = -1$
Correct.

Option B
$f(\pi) = \cos 9\pi + \cos 10\pi$
$= -1 + 1 = 0$
Not correct.

Option C
$f(-\pi) = \cos(-9\pi) + \cos(-10\pi)$
$= \cos 9\pi + \cos 10\pi$
$= -1 + 1 = 0$
Correct.

Option D
$f\left(\frac{\pi}{4}\right) = \cos\frac{9\pi}{4} + \cos\frac{10\pi}{4}$
$= \cos\frac{\pi}{4} + \cos\frac{5\pi}{2}$
$= \frac{\sqrt{2}}{2} + 0 \neq 1$

Not correct.
Correct options: A, C

13. Solve
$|x-1| + |x-2| + |x-3| \geq 6$
Break into intervals.
Case 1: $x \leq 1$
$(1-x) + (2-x) + (3-x) \geq 6$
$6 - 3x \geq 6$
$x \leq 0$
So valid:

$x \leq 0$

Case 2: $1 < x < 2$

$(x-1)+(2-x)+(3-x) \geq 6$

$4-x \geq 6$

$x \leq -2$

No solution in this interval.

Case 3: $2 < x < 3$

$(x-1)+(x-2)+(3-x) \geq 6$

$x \geq 6$

No solution.

Case 4: $x \geq 3$

$(x-1)+(x-2)+(x-3) \geq 6$

$3x-6 \geq 6$

$x \geq 4$

So valid:

$x \geq 4$

Hence solution set: $(-\infty, 0] \cup [4, \infty)$

Correct options: A, D

COMPREHENSION TYPE

14. Correct Answer: B) -1

Short Solution:

$$f(x) = \frac{kx}{x+1}$$

$$(f \circ f)(x) = x \Rightarrow \frac{k^2 x}{(k+1)x+1} = x$$

Comparing coefficients:

$k + 1 = 0 \Rightarrow k = -1.$

15. **Correct Answer**: **D)** $(-\infty, 1)$

Short Solution:

$f(x) = \sqrt{1-x}, g(x) = \ln x$

$(g \circ f)(x) = \ln\left(\sqrt{1-x}\right)$

For ln, input must be > 0:

$\sqrt{1-x} > 0 \Rightarrow 1 - x > 0 \Rightarrow x < 1.$

16. Correct Answer: B) r = 1, sarbitrary

Short Solution:

Given p = 1, q = 2:

$f(x) = x + 2, g(x) = rx + s$

$(f \circ g)(x) = rx + s + 2$

$(g \circ f)(x) = r(x+2) + s = rx + 2r + s$

Equating:

$s + 2 = 2r + s \Rightarrow r = 1.$

Hence, sis arbitrary.

MATRIX MATCHING TYPE

Functions

a) $f_1(x) = | x - 1 | - 1$

- Shift $y = | x |$ right by 1 and down by 1
- Vertex: $(1, -1)$
- Opens upward

So it matches **Q**

b) $f_2(x) = || x - 1 | - 1 |$

- First graph $| x - 1 | - 1$
- Then take modulus again, reflecting negative part upward
- Produces a "W" shaped graph with points $(0,0), (1,1), (2,0)$

So it matches **R**

c) $f_3(x) = | x | - 1$

- Standard absolute value shifted down by 1
- Vertex: $(0, -1)$

So it matches **P**

d) $f_4(x) = 1 - | x |$

- Inverted absolute value graph
- Vertex at $(0, 1)$

So it matches **S**

Final Matching: a – Q, b – R, c – P, d – S

INTEGER TYPE

18. Range of the function

$$f(x) = \frac{x^2 + x + c}{x^2 + 2x + c}$$

Let

$$y = \frac{x^2 + x + c}{x^2 + 2x + c}$$

Then

$$y(x^2 + 2x + c) = x^2 + x + c$$

$$(y - 1)x^2 + (2y - 1)x + c(y - 1) = 0$$

For real x, discriminant ≥ 0:

$$(2y - 1)^2 - 4(y - 1)^2 c \geq 0$$

Given range is

$$\left[\frac{5}{6}, \frac{3}{2}\right]$$

The boundary values occur when discriminant = 0:

$$(2y - 1)^2 - 4c(y - 1)^2 = 0$$

$$\frac{2y - 1}{y - 1} = \pm 2\sqrt{c}$$

Substitute $y = \frac{5}{6}$:

$$\frac{\frac{10}{6} - 1}{\frac{5}{6} - 1} = \frac{\frac{4}{6}}{-\frac{1}{6}} = -4$$

Hence

$2\sqrt{c} = 4$

$\sqrt{c} = 2$

$\boxed{c = 4}$

19. Number of solutions of

$\text{sgn}(x) = | 1 - x |$

Signum function:

$$\text{sgn}(x) = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases}$$

Check cases:

Case 1: $x > 0$

$1 = | 1 - x |$

$x = 0$ or 2

Only $x = 2$valid.

Case 2: $x = 0$

$0 = | 1 | = 1$

No solution.

Case 3: $x < 0$

$-1 = | 1 - x |$

Impossible since modulus is nonnegative.

Hence number of solutions: $\boxed{1}$

20. Integer in the range of

$f(x) = \sqrt{9 - x^2} + \sqrt{x^2 - 4}$

Domain:

$4 \le x^2 \le 9$

Let

$t = x^2$

Then

$f(t) = \sqrt{9 - t} + \sqrt{t - 4}, 4 \le t \le 9$

Using Cauchy or checking endpoints:

At $t = 4$:

$f = \sqrt{5}$

At $t = 9$:

$f = \sqrt{5}$

Maximum occurs at midpoint $t = \frac{13}{2}$:

$$f_{\max} = 2\sqrt{\frac{5}{2}} = \sqrt{10}$$

Hence range:

$[\sqrt{5}, \sqrt{10}]$

Since

$\sqrt{5} \approx 2.236, \sqrt{10} \approx 3.162$

The only integer in the range is $\boxed{3}$

Student practice Corner

SINGLE OPTION CORRECT TYPE

1. A relation is a function if every $x \in \{1,2,3,4\}$has exactly one image.

D) $x + y = 5$

$(1,4), (2,3), (3,2), (4,1)$

Each $x$has exactly one image. $\boxed{D}$

2. **Range of**

$f(x) = \frac{1}{x-5}$

Denominator cannot be zero, so $x \neq 5$.

Also, $\frac{1}{x-5} \neq 0$

Hence range is $\boxed{\mathbb{R} - \{0\}}$

Answer: $\boxed{A}$

3. **Number of functions from**

$A = \{p, q, r, s\}, |\ A\ | = 4$

to

$B = \{x, y, z\}, |\ B\ | = 3$

Number of functions:

$3^4 = 81$

$\boxed{81}$

Answer: $\boxed{C}$

4. **Number of relations from $A$to Bwhich are not functions**

Total relations: 2^{mn}

Total functions: n^m

Therefore non-functions:

$\boxed{2^{mn} - n^m}$

Answer: $\boxed{D}$

5. **One-one functions from set with 3 elements to set with 10 elements**

Number of injective functions:

$^{10}P_3 = \frac{10!}{7!} = 10 \cdot 9 \cdot 8$

$= 720$

$\boxed{720}$

Answer: $\boxed{A}$

6. **Given**

$f(x) = x^2 - 1, g(x) = 4x - 3$

Then

$(f + g + 8)(x) = f(x) + g(x) + 8$

$= x^2 - 1 + 4x - 3 + 8$

$= x^2 + 4x + 4$

$= (x + 2)^2$

$\boxed{(x + 2)^2}$

Answer: $\boxed{B}$

7. If f and g are functions such that $f \circ g$ is onto, then

If $f \circ g$ is onto, every element in the codomain is achieved by $f(g(x))$

This is only possible if f itself is onto.

g need not be onto.

Hence $\boxed{\text{A) } f \text{ is onto}}$

8. Given

$f(x) = 2x^2 + 3, g(x) = 3x - 2$

Find

$[g \circ (f \circ g)](3)$

First compute:

$g(3) = 3(3) - 2 = 7$

Then

$f(g(3)) = f(7) = 2(7)^2 + 3 = 98 + 3 = 101$

Now

$g(101) = 3(101) - 2 = 303 - 2 = 301$

$\boxed{301}$

9. Let

$f(x) = x + 1, g(x) = 2x - 3$

A)

$(f + g)(x) = x + 1 + 2x - 3 = 3x - 2$

Correct.

B)

$(f - g)(x) = x + 1 - (2x - 3) = 4 - x$

$(f - g)(2) = 4 - 2 = 2$

Not 6. False.

C)

$\left(\frac{f}{g}\right)(x) = \frac{x+1}{2x-3}$

At $x = 2$,

$\frac{2+1}{4-3} = 3$

Correct.

D)

$(f - g)(x) = 4 - x$

Correct.

Correct options: $\boxed{A,\ C,\ D}$

10. Given

$f(x) = x^2 + 2x + 3, g(x) = x + 1$

Find x such that

$g(f(x)) = f(g(x))$

First,

$g(f(x)) = f(x) + 1 = x^2 + 2x + 4$

Now,

$f(g(x)) = f(x + 1)$

$= (x+1)^2 + 2(x+1) + 3$
$= x^2 + 2x + 1 + 2x + 2 + 3$
$= x^2 + 4x + 6$
Equating:
$x^2 + 2x + 4 = x^2 + 4x + 6$
$2x + 4 = 4x + 6$
$-2 = 2x$
$x = -1$
$\boxed{-1}$
Answer: $\boxed{A}$

11. Domain of

$$\sqrt{\frac{4 - x^2}{[x] + 2}}$$

For square root,
$\frac{4 - x^2}{[x] + 2} \geq 0$
and denominator $\neq 0$.
Also,
$4 - x^2 \geq 0 \Rightarrow -2 \leq x \leq 2$
Now check intervals:
For $-2 \leq x < -1$ $[x] = -2$
Denominator = 0, not allowed.
For $-1 \leq x < 0$
$[x] = -1$

Denominator positive, numerator positive.
Allowed.
For $0 \leq x < 1$
$[x] = 0$
Allowed.
For $1 \leq x < 2$
$[x] = 1$
Allowed.
At $x = 2$,
$4 - x^2 = 0$
Allowed.
Hence domain: $\boxed{[-1,2]}$
Answer: $\boxed{B}$

12. If $a^2 + b^2 + c^2 = 1$
Find range of
$ab + bc + ca$

Using
$(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$
$(a + b + c)^2 = 1 + 2(ab + bc + ca)$
Since square is nonnegative,

$1 + 2(ab + bc + ca) \geq 0$

$ab + bc + ca \geq -\frac{1}{2}$

Maximum occurs when

$a = b = c = \frac{1}{\sqrt{3}}$

Then

$ab + bc + ca = 3\left(\frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}}\right) = 1$

Hence range is

$\boxed{\left[-\frac{1}{2}, 1\right]}$

Answer: $\boxed{D}$

MATRIX MATCHING TYPE

13. Given

$f(x) = |x - 1| + |x - 2| + |x - 3|$

Match for different intervals.

a) $-\infty < x < 1$

All terms negative inside modulus:

$f(x) = -(x - 1) - (x - 2) - (x - 3)$

$= -x + 1 - x + 2 - x + 3$

$= 6 - 3x$

Matches: $\boxed{R}$

b) $1 \leq x \leq 2$

$|x - 1| = x - 1, |x - 2| = -(x - 2), |x - 3| = -(x - 3)$

$f(x) = x - 1 - x + 2 - x + 3$

$= 4 - x$

Matches: $\boxed{S}$

c) $2 < x < 3$

$|x - 1| = x - 1, |x - 2| = x - 2, |x - 3| = -(x - 3)$

$f(x) = x - 1 + x - 2 - x + 3$

$= x$

Matches: $\boxed{Q}$

d) $3 \leq x < \infty$

All positive:

$f(x) = x - 1 + x - 2 + x - 3$

$= 3x - 6$

Matches: $\boxed{P}$

Final Matching $\boxed{a \to R, b \to S, c \to Q, d \to P}$

INTEGER TYPE

14. Minimum value of

$f(x) = (x - 2)(x - 4)(x - 6)(x - 8) + 16$

Group symmetrically:

$= (x - 2)(x - 8)(x - 4)(x - 6) + 16$

Let
$y = (x-5)^2$
Then
$(x-2)(x-8) = y-9$
$(x-4)(x-6) = y-1$
So
$f(x) = (y-9)(y-1)+16$
$= y^2 - 10y + 25$
$= (y-5)^2$
Hence minimum value is $\boxed{0}$
So $k = 0$.
Therefore,
$k + 5 = 5$
$\boxed{5}$

15. Integer in the range of
$f(x) = \sqrt{4-x^2} + \sqrt{x^2-1}$

Domain: $1 \le x^2 \le 4$
Let $t = x^2$
Then
$f(t) = \sqrt{4-t} + \sqrt{t-1}$
where
$1 \le t \le 4$
At endpoints:
$f(1) = \sqrt{3}, f(4) = \sqrt{3}$
Maximum occurs at midpoint: $t = \frac{5}{2}$

$$f_{\max} = 2\sqrt{\frac{3}{2}} = \sqrt{6}$$

Hence range: $[\sqrt{3}, \sqrt{6}]$
Since $\sqrt{3} \approx 1.732, \sqrt{6} \approx 2.449$
The only integer in the range is $\boxed{2}$

ODD, EVEN AND PERIODIC FUNCTIONS

LEVEL – I

1. If f and g are odd, then $f \circ g$ is
Given:
$f(-x) = -f(x), g(-x) = -g(x)$
Now,
$(f \circ g)(-x) = f(g(-x))$
$= f(-g(x))$

Since f is odd,
$f(-g(x)) = -f(g(x))$
$= -(f \circ g)(x)$
Hence $f \circ g$ is odd.
$\boxed{\text{A) Odd}}$

2. If f is even and g is odd, then $f \circ g$ is

Given:

$f(-x) = f(x), g(-x) = -g(x)$

Then

$(f \circ g)(-x) = f(g(-x))$

$= f(-g(x))$

Since f is even,

$f(-g(x)) = f(g(x))$

$= (f \circ g)(x)$

Hence $f \circ g$ is even.

B) Even

3. If f is odd and g is even, then $f \circ g$ is

Given:

$g(-x) = g(x)$

Then

$(f \circ g)(-x) = f(g(-x))$

$= f(g(x))$

$= (f \circ g)(x)$

Hence $f \circ g$ is even.

B) Even

4. Given

$f(x) = \log\left(x+\sqrt{x^2+1}\right)$

Check $f(-x)$:

$f(-x) = \log\left(-x+\sqrt{x^2+1}\right)$

Using

$\left(x+\sqrt{x^2+1}\right)\left(-x+\sqrt{x^2+1}\right) = 1$

So,

$f(-x) = \log\left(\frac{1}{x+\sqrt{x^2+1}}\right)$

$= -\log\left(x+\sqrt{x^2+1}\right)$

$= -f(x)$

Hence f is odd.

A) Odd

5. Given

$f(x) = x\left(\frac{a^x-1}{a^x+1}\right)$

Find nature of f.

Compute $f(-x)$:

$f(-x) = (-x)\left(\frac{a^{-x}-1}{a^{-x}+1}\right)$

Multiply numerator and denominator by a^x:

$= -x\left(\frac{1-a^x}{1+a^x}\right)$

$= x\left(\frac{a^x - 1}{a^x + 1}\right)$

$= f(x)$

Hence $f$is even.

B) Even

6. Given

$f(x + k) + f(x) = 0$

So,

$f(x + k) = -f(x)$

Now replace $x$by $x + k$:

$f(x + 2k) = -f(x + k)$

Using $f(x + k) = -f(x)$,

$f(x + 2k) = f(x)$

Hence the period is $2k$

Answer: *B*

7. Period of the function

$f(x) = x[x]$

where $[x]$denotes greatest integer function.

Check:

$f(x + T) = (x + T)[x + T]$

This cannot equal $x[x]$for all x, since the function grows without bound and changes structure on every interval.

Hence it is not periodic.

Non periodic

Answer: *D*

8. If $f(g(x))$is periodic and $g$is polynomial

Since $f(g(x))$is periodic, values of $g(x)$repeat after equal intervals.

Given:

$g(2) = 3, g(4) = 7$

For a polynomial compatible with periodic composition, $g$must be linear.

Slope:

$\frac{7 - 3}{4 - 2} = 2$

So,

$g(x) = 2x - 1$

Therefore,

$g(6) = 2(6) - 1 = 11$

11

Answer: *A*

9. For

$f(x) = |\sin 4x| + \cos 4x$

Period of $|\sin 4x|$is $\frac{\pi}{4}$

Period of $\cos 4x$is $\frac{\pi}{2}$

LCM of $\frac{\pi}{4}$and $\frac{\pi}{2}$

Answer: C) $\frac{\pi}{8}$

10. For

$f(x) = 5\sin 3x - 7\sin 8x$

Period of $\sin 3x = \frac{2\pi}{3}$

Period of $\sin 8x = \frac{\pi}{4}$

Find LCM:

$T = \text{LCM}\left(\frac{2\pi}{3}, \frac{\pi}{4}\right) = 2\pi$

Answer: B) 2π

11. For

$f(x) = \cos x + \{x\}$

$\cos x$has period 2π

- Fractional part function $\{x\}$has period 1

A common period must satisfy both:

$T = 2\pi m = n$

for integers m, n.

Since $2\pi$is irrational, no such nonzero common period exists.

Answer: D) Does not exist

12. For

$$f(x) = \frac{1}{2}\left[\frac{|\sin x|}{\cos x} + \frac{\sin x}{|\cos x|}\right]$$

Simplify in different quadrants:

- Quadrant I:

$f(x) = \tan x$

Quadrant II:

$f(x) = -\tan x$

- Quadrant III:

$f(x) = \tan x$

- Quadrant IV:

$f(x) = -\tan x$

Now,

$f(x + \pi) = f(x)$

and also no smaller positive value satisfies the condition everywhere.

Hence the least period is 2π

Answer: **D)** 2π

13. If

$f(x) = x^2$

then

$f^{-1}(9) = \{x : x^2 = 9\}$

$x = \pm 3$

Hence,

$f^{-1}(9) = \{-3, 3\}$

Answer: C) $\{-3, 3\}$

14. If
$f(x) = x^2 + 5x + 9$
then $f^{-1}(9) = \{x: x^2 + 5x + 9 = 9\}$
$x^2 + 5x = 0$
$x(x + 5) = 0$
$x = 0, -5$
Hence,
$f^{-1}(9) = \{-5,0\}$
Answer: C) $\{-5, 0\}$

15. Given
$f(x) = 3x + 5$
Let
$y = 3x + 5$
Interchange $x$and y:
$x = 3y + 5$

Solve for y:
$3y = x - 5$
$y = \frac{x-5}{3}$
Hence,
$f^{-1}(x) = \frac{x-5}{3}$
Answer: A) $\frac{x-5}{3}$

16. Given
$$f(n) = \begin{cases} \frac{n}{2}, & n \text{ even} \\ -\frac{n-1}{2}, & n \text{ odd} \end{cases}$$
We need:
$f^{-1}(-100)$

Use odd case:
$-\frac{n-1}{2} = -100$
$\frac{n-1}{2} = 100$
$n - 1 = 200$
$n = 201$
Answer: B) 201

17. Given
$f(x) = x - [x]$
But
$x - [x] = \{x\}$
(the fractional part function).
Its range is [0,1), so many values of $x$give the same output.
Hence $f$is not one-one, therefore inverse is not defined.
Answer: D) Not defined

18. Given
$f(x) = \log_5 x$
Let
$y = \log_5 x$
Then
$x = 5^y$
Interchanging variables:
$f^{-1}(x) = 5^x$
Answer: C) 5^x

19. Given
$f(x) = \sqrt{1-x}$
Let
$y = \sqrt{1-x}$
Squaring:
$y^2 = 1 - x$
$x = 1 - y^2$
Hence,
$f^{-1}(x) = 1 - x^2$
Also,
$g(x) = \log_e x$
so
$g^{-1}(x) = e^x$

Now,
$(g^{-1} \circ f^{-1})(x) = g^{-1}(f^{-1}(x))$
$= e^{\,1-x^2}$
Answer: C) e^{1-x^2}

20. Given
$f(x) = 2^{x(x-1)}$
Let
$y = 2^{x(x-1)}$
Taking $\log_2$:
$\log_2 y = x(x-1)$
$x^2 - x - \log_2 y = 0$
Using quadratic formula:
$x = \dfrac{1 \pm \sqrt{1 + 4\log_2 y}}{2}$
Replacing $y$by x:
$f^{-1}(x) = \dfrac{1}{2}\left(1 \pm \sqrt{1 + 4\log_2 x}\right)$

Answer: A) $\frac{1}{2}\left(1 \pm \sqrt{1 + 4\log_2 x}\right)$

LEVEL – II

SINGLE OPTION CORRECT TYPE

1. Since $f$is an even function,
$f(a) = f(b) \Rightarrow a = \pm b$
Given

$f(x) = f\left(\frac{x+1}{x+2}\right)$

So,

$x = \pm\frac{x+1}{x+2}$

Case 1:

$x = \frac{x+1}{x+2}$

$x(x+2) = x+1$

$x^2 + x - 1 = 0$

Sum of roots $= -1$.

Case 2:

$x = -\frac{x+1}{x+2}$

$x(x+2) = -(x+1)$

$x^2 + 3x + 1 = 0$

Sum of roots $= -3$.

Total sum:

$-1 + (-3) = -4$

Answer: C) -4

2. Given

$f(x) = \cot(\sin x) + \left[\frac{x^2}{|\ a\ |}\right]$

$\cot(\sin x)$is an odd function.

For $f(x)$to be odd,

$\left[\frac{x^2}{|\ a\ |}\right]$

must be identically zero on $[-4,4]$.

Since

$0 \le x^2 \le 16$

we need

$0 \le \frac{x^2}{|\ a\ |} < 1$

Maximum occurs at $x^2 = 16$:

$\frac{16}{|\ a\ |} < 1$

$|\ a\ | > 16$

Thus

$a \in (-\infty, -16) \cup (16, \infty)$

Answer: B) $(-\infty, -16) \cup (16, \infty)$

3. Given

$p(x) = x - [x + a] - b$

Using
$[x+a+1]=[x+a]+1$

Then
$p(x+1)=x+1-[x+a+1]-b$
$=x+1-([x+a]+1)-b$
$=x-[x+a]-b$
$=p(x)$
Hence least period is 1
Answer: D) 1

4. Given
$f(x)=2\cos\left(\frac{x-\pi}{3}\right)+4\sin\left(\frac{x-\pi}{3}\right)$
Both sine and cosine have argument
$\frac{x-\pi}{3}$
For
$\sin\left(\frac{x-\pi}{3}\right)$
period is obtained from
$\frac{T}{3}=2\pi$
$T=6\pi$
Same for cosine.
Hence period of $f(x)$is 6π
Answer: D) 6π

5. Given
$f(x)=\frac{\cos x}{\left[\frac{2x}{\pi}\right]+\frac{1}{2}}$

Check $f(-x)$:
$f(-x)=\frac{\cos(-x)}{\left[\frac{-2x}{\pi}\right]+\frac{1}{2}}$

Since $\cos(-x)=\cos x$and
$[-t]=-[t]-1$(when $t\notin\mathbb{Z}$)

let
$n=\left[\frac{2x}{\pi}\right]$

Then
$\left[\frac{-2x}{\pi}\right]=-n-1$

So denominator becomes
$-n-1+\frac{1}{2}=-(n+\frac{1}{2})$

Hence

$$f(-x) = \frac{\cos x}{-(n+\frac{1}{2})} = -f(x)$$

Therefore $f(x)$is odd.

Answer: A) An odd function

6. Given periods:

$f(x)$: 7

$g(x)$: 11

Then

$g\left(\frac{x}{5}\right)$

has period 55, and similarly

$f\left(\frac{x}{5}\right)$

has period 35.

Hence

$F(x) = f(x)g\left(\frac{x}{5}\right) - g(x)f\left(\frac{x}{5}\right)$

has period

LCM(7,11,35,55)

= 385

Answer: C) 385

7. Given

$f(x) = 1 + x + x^2 + x^3 + \cdots$

For $|\ x\ | < 1$,

$$f(x) = \frac{1}{1-x}$$

Let

$$y = \frac{1}{1-x}$$

Then

$$1 - x = \frac{1}{y}$$

$$x = 1 - \frac{1}{y} = \frac{y-1}{y}$$

Hence

$$f^{-1}(x) = \frac{x-1}{x}$$

Answer: C) $\frac{x-1}{x}$

8. Given

$f(x) = x^2 - 4x + 3$

Complete square:

$f(x) = (x-2)^2 - 1$

Let
$y = (x-2)^2 - 1$
$y + 1 = (x-2)^2$
Since domain is $[2, \infty)$,
$x - 2 \geq 0$
So
$x - 2 = \sqrt{y+1}$
$x = 2 + \sqrt{y+1}$
Hence
$f^{-1}(x) = 2 + \sqrt{x+1}$
Answer: B) $\mathbf{2 + \sqrt{x+1}}$

9. Given
$f(x) = k$
(constant function)
Then
$f(-x) = k = f(x)$
Hence $f(x)$is even.
Answer: B) Even

ONE OR MORE THAN ONE OPTION CORRECT TYPE

10. Given
$f(x) = 3x - 5$
Let
$y = 3x - 5$
Then
$x = \dfrac{y+5}{3}$
Hence
$f^{-1}(x) = \dfrac{x+5}{3}$
So only option **B** is correct.
Answer: B

11. Given
$f(x) = \sin x + \cos x$
$g(x) = x^2 - 1$
Then
$g(f(x)) = (\sin x + \cos x)^2 - 1$

Using
$(\sin x + \cos x)^2 = \sin^2 x + \cos^2 x + 2\sin x \cos x$
$= 1 + \sin 2x$
Thus
$g(f(x)) = \sin 2x$
For invertibility, $\sin 2x$must be one-one on the interval.

- On $\left[0, \frac{\pi}{2}\right]$, $\sin 2x$increases then decreases ⇒ not one-one.
- On $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$, $2x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, where sine is one-one.
- On $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, not one-one.
- On $[0, \pi]$, not one-one.

Answer: B

COMPREHENSION TYPE

12. Given
$f(x) = x^3 + 3$
Let
$y = x^3 + 3$
Then
$x^3 = y - 3$
$x = (y-3)^{1/3}$
Replacing $y$by x,
$f^{-1}(x) = (x-3)^{1/3}$
Answer: B) $\boldsymbol{(x-3)^{1/3}}$

13. Given
$f(x) = \log_a\left(x+\sqrt{x^2+1}\right)$
Let
$y = \log_a\left(x+\sqrt{x^2+1}\right)$
Then
$a^y = x + \sqrt{x^2+1}$
Also,
$\sqrt{x^2+1} = a^y - x$

Squaring:
$x^2 + 1 = a^{2y} - 2a^y x + x^2$
$1 = a^{2y} - 2a^y x$
$2a^y x = a^{2y} - 1$
$x = \dfrac{a^{2y}-1}{2a^y}$
$x = \dfrac{a^y - a^{-y}}{2}$

Hence
$f^{-1}(x) = \dfrac{a^x - a^{-x}}{2}$

Answer: D) $\boldsymbol{\frac{a^x - a^{-x}}{2}}$

14. Given
$f(x) = (a - x^n)^{1/n}$

Find
$(f \circ f)(x)$
$f(f(x)) = \left(a - \left[(a-x^n)^{1/n}\right]^n\right)^{1/n}$
$= (a - (a - x^n))^{1/n}$
$= (x^n)^{1/n} = x$
Thus
$(f \circ f)(x) = x$

Hence its inverse is also x

Answer: A) x

MATRIX MATCHING TYPE

Match the functions with their periods:

- $\sin 2x \rightarrow \pi$
- $\cos 3x \rightarrow \frac{2\pi}{3}$
- $\tan 4x \rightarrow \frac{\pi}{4}$
- $\sin x \rightarrow 2\pi$

Correct matching: a → P, b → R, c → S, d → Q

Useful formulas:

For sine/cosine:$T = \frac{2\pi}{|a|}$

For tangent:$T = \frac{\pi}{|a|}$

INTEGER TYPE

16. Given

$f\left(\frac{2x-3}{x-2}\right) = 5x - 2$

Let

$\frac{2x-3}{x-2} = 1$

Then,

$2x - 3 = x - 2$

$x = 1$

So,

$f(1) = 5(1) - 2 = 3$

Therefore,

$f^{-1}(3) = 1$

Hence,

$f^{-1}(13)$

Take

$5x - 2 = 13$

$5x = 15$

$x = 3$

Now,

$f\left(\frac{2(3)-3}{3-2}\right) = 13$

$f(3) = 13$

Therefore,

$\boxed{f^{-1}(13) = 3}$

17.

$f(\theta) = 4 + 4\sin^3\theta - 3\sin\theta$

Using identity:

$4\sin^3\theta - 3\sin\theta = -\sin 3\theta$

So,

$f(\theta) = 4 - \sin 3\theta$

Period of $\sin 3\theta$is:

$\frac{2\pi}{3}$

Given period $= \frac{k\pi}{3}$,

$\frac{k\pi}{3} = \frac{2\pi}{3}$

$k = 2$

Useful identity:

$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$

18. If $f(x) = \tan 3x$

Fundamental period:

$\frac{\pi}{3}$

$\boxed{\frac{\pi}{3}}$

Using:

$T = \frac{\pi}{|a|}$

19. Functions given:

$x^2 + 1 \rightarrow$ even

$x^3 + x \rightarrow$ odd

$|x| \rightarrow$ even

$\sin x \rightarrow$ odd

Number of even functions = 2

20.

$f(x) = x^5 - 3x^3 + 2x$

$f(2) = 32 - 24 + 4 = 12$

$f(-2) = -32 + 24 - 4 = -12$

$f(2) + f(-2) = 12 + (-12) = 0$

$\boxed{0}$

Student Practice Corner

SINGLE OPTION CORRECT TYPE

1. is even and $g$is odd.

$(f \circ g)(-x) = f(g(-x))$

Since $g$is odd,

$g(-x) = -g(x)$

Since $f$is even,

$f(-g(x)) = f(g(x))$

Therefore, $f \circ g$is even.

B) Even

2.

$f(x) = x^2 - |x|$
$f(-x) = (-x)^2 - |-x|$
$= x^2 - |x| = f(x)$
Hence f is even.
B) Even

3.

$f(x) = x - [x]$
This is the fractional part function.
$f(x+1) = x + 1 - [x+1]$
$= x + 1 - ([x] + 1)$
$= x - [x] = f(x)$
Hence period = 1.
A) 1

4. $f(x)$ is odd and periodic with period 2.
$f(4) = f(0)$
Since f is odd,
$f(0) = -f(0)$
$2f(0) = 0$
$f(0) = 0$

Therefore,
$f(4) = 0$
D) 0

5.

Now,
$\sin 2x$ has period π
$\cos 2x$ has period π
But because of symmetry,
$f\left(x+\frac{\pi}{2}\right) = \cos(\cos(2x+\pi)) + \cos(\sin(2x+\pi))$
$= \cos(-\cos 2x) + \cos(-\sin 2x)$
Since cosine is even,
$= \cos(\cos 2x) + \cos(\sin 2x)$
Hence the least period is $\frac{\pi}{2}$
Relevant function transformation:
$f\left(x+\frac{\pi}{2}\right) = f(x)$
$\frac{\pi}{2}$

6.

$f(x) = \sin x + \cos x$
$f(-x) = \sin(-x) + \cos(-x)$
$= -\sin x + \cos x$
This is neither equal to $f(x)$ nor $-f(x)$.
Hence the function is neither even nor odd.
D) Neither even nor odd

7.

$f(x) = \sin\left(\log_{10}\left(x+\sqrt{x^2+1}\right)\right)$

Using identity:
$\log\left(x+\sqrt{x^2+1}\right) = \sinh^{-1} x$

Now,
$\sinh^{-1}(-x) = -\sinh^{-1} x$
Hence,
$f(-x) = \sin(-\sinh^{-1} x)$
$= -\sin(\sinh^{-1} x)$
$= -f(x)$
Therefore $f(x)$is odd.
A) Odd
Useful identity:
$\log\left(x+\sqrt{x^2+1}\right) = \sinh^{-1} x$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

8.

$f(x) = \cos([\pi^2]x) + \cos([-\pi^2]x)$
Since
$[\pi^2] = 9, [-\pi^2] = -10$
So,
$f(x) = \cos 9x + \cos 10x$
A)
$f\left(\frac{\pi}{2}\right) = \cos\frac{9\pi}{2} + \cos 5\pi$
$= 0 + (-1) = -1$
Correct
B)
$f(\pi) = \cos 9\pi + \cos 10\pi$
$= -1 + 1 = 0$
False
C)
$f(-\pi) = \cos(-9\pi) + \cos(-10\pi)$
$= \cos 9\pi + \cos 10\pi = 0$
Correct
D)
$f\left(\frac{\pi}{4}\right) = \cos\frac{9\pi}{4} + \cos\frac{10\pi}{4}$
$= \frac{\sqrt{2}}{2} + 0 \neq 1$
False
Correct options: A and C

9.

$f(x) = (\log(\sec x + \tan x))^3$

Using identity:
$\log(\sec(-x) + \tan(-x)) = \log(\sec x - \tan x)$
Also,
$(\sec x + \tan x)(\sec x - \tan x) = 1$
Hence,

$\log(\sec x - \tan x) = -\log(\sec x + \tan x)$

Therefore,
$f(-x) = (-\log(\sec x + \tan x))^3$
$= -f(x)$

So $f(x)$is odd.
A correct
Since $\log(\sec x + \tan x)$is strictly increasing on
$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

its cube is also strictly increasing.
Hence one-one.
B correct
Range:
As $x \to -\frac{\pi}{2}$,
$f(x) \to -\infty$

As $x \to \frac{\pi}{2}$,
$f(x) \to \infty$
Hence range is $\mathbb{R}$.
C correct
Correct options: **A, B, C**

COMPREHENSION TYPE

10.

$f(x) = \sin\frac{2\pi x}{3} + \cos\frac{\pi x}{2}$
Period of
$\sin\frac{2\pi x}{3}$
is obtained from
$\frac{2\pi}{3}T = 2\pi \Rightarrow T = 3$
Period of
$\cos\frac{\pi x}{2}$
is
$\frac{\pi}{2}T = 2\pi \Rightarrow T = 4$

LCM of 3and 4is 12.
C) 12
Relevant formula:

$T = \frac{2\pi}{|a|}$

11.

$f(x) = \sqrt[3]{(1-x)^2} + \sqrt[3]{(1+x)^2}$

Now,
$f(-x) = \sqrt[3]{(1+x)^2} + \sqrt[3]{(1-x)^2}$
$= f(x)$
Hence $f(x)$is even.
A) even

12.

$f(x) = 1 - x^3 - x^4 - 2x^5$
Check:
$f(-x) = 1 + x^3 - x^4 + 2x^5$
This is neither equal to $f(x)$nor $-f(x)$.
Hence the function is neither even nor odd.
D) neither even nor odd

MATRIX MATCHING TYPE

13. Correct matching:
a) $\sin 3x \rightarrow$ Q) $\frac{2\pi}{3}$
b) $\tan 2x \rightarrow$ R) $\frac{\pi}{2}$
c) $\cos x \rightarrow$ S) 2π
d) $\sin^2 x \rightarrow$ P) π
Using periods:

- Period of $\sin(ax) = \frac{2\pi}{a}$
- Period of $\tan(ax) = \frac{\pi}{a}$
- Period of $\sin^2 x = \pi$

$T_{\sin(ax)} = \frac{2\pi}{a}, T_{\tan(ax)} = \frac{\pi}{a}$

INTEGER TYPE

14. Given,
$f\left(\frac{2x-5}{x+1}\right) = 3x - 4$

To find $f^{-1}(11)$,
set
$3x - 4 = 11$
$3x = 15$
$x = 5$
Now,
$\frac{2(5)-5}{5+1} = \frac{5}{6}$

Hence,
$f\left(\frac{5}{6}\right) = 11$

Therefore,

$$\boxed{f^{-1}(11) = \frac{5}{6}}$$

15.

$f(\theta) = 7 - 6\sin^3\theta + \sin\theta$

Using identity:

$3\sin\theta - 4\sin^3\theta = \sin 3\theta$

Rewrite:

$-6\sin^3\theta + \sin\theta = -\frac{3}{2}(4\sin^3\theta) - ?$

Better simplification:

$6\sin^3\theta - \sin\theta = (4\sin^3\theta - 3\sin\theta) + 2\sin^3\theta + 2\sin\theta$

Instead of simplifying, directly observe all terms involve $\sin\theta$, so period remains 2π.

Hence,

$\frac{k\pi}{2} = 2\pi$

$k = 4$

3. Permutations and Combinations

Permutations

Level-1

1. Given:

$^{12}P_r = 1320$

Using permutation formula:

$^{n}P_r = \frac{n!}{(n-r)!}$

$^{12}P_2 = 12 \times 11 = 132$

$^{12}P_3 = 12 \times 11 \times 10 = 1320$

Hence, $r = 3$

Answer: B) 3

2. Given: $^{n}P_r = 5040$

$^{7}P_6 = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 5040$

So, $r = 6$

Answer: D) 6

3. Given: $^{9}P_5 + 5 \cdot {}^{9}P_4 = {}^{10}P_r$

Calculate:

$^{9}P_5 = 9 \times 8 \times 7 \times 6 \times 5 = 15120$

$^{9}P_4 = 9 \times 8 \times 7 \times 6 = 3024$

$5 \times 3024 = 15120$

Therefore, $15120 + 15120 = 30240$

Now, $^{10}P_5 = 10 \times 9 \times 8 \times 7 \times 6 = 30240$

Hence, $r = 5$

Answer: D) 5

4. Given: $\frac{(n-1)_{P_3}}{(n+1)_{P_3}} = \frac{5}{12}$

Using formula:

$^{(n-1)}P_3 = (n-1)(n-2)(n-3)$

$^{(n+1)}P_3 = (n+1)n(n-1)$

So, $\frac{(n-1)(n-2)(n-3)}{(n+1)n(n-1)} = \frac{5}{12}$

Cancel $(n-1)$: $\frac{(n-2)(n-3)}{n(n+1)} = \frac{5}{12}$

Check options:

For $n = 8$, $\frac{6\times5}{8\times9} = \frac{30}{72} = \frac{5}{12}$

Hence, $n = 8$

Answer: C) 8

5. There are:

3 routes from Warangal to Hyderabad

4 routes from Hyderabad to Karimnagar

By multiplication principle: $3 \times 4 = 12$

Hence total ways: 12

Answer: A) 12

6. A man has 3 sons and 6 schools.

No two sons should go to the same school.

So, arrange 3 different schools out of 6 for the 3 sons.

$${}^6P_3 = \frac{6!}{(6-3)!}$$

$$= \frac{6!}{3!} = 6 \times 5 \times 4 = 120$$

Hence,

$${}^6P_3 = 120$$

Answer: C) ${}^6P_3$

7. There are 50 stations on a railway line.

A ticket is needed from one station to another station.

Number of ways:

$${}^{50}P_2 = 50 \times 49$$

$= 2450$

Hence, total kinds of tickets: 2450

Answer: A) 2450

8. Word: "EQUATION"

Total letters = 8

Vowels: $E, U, A, I, O \rightarrow$ 5 vowels

Consonants: $Q, T, N \rightarrow$ 3 consonants

Beginning and ending must be vowels.

First place can be filled in 5ways.

Last place can be filled in 4ways.

Remaining 6 letters can be arranged in:

6!

Therefore,

$5 \times 4 \times 6!$

$= 20 \times 720$

$= 14400$

Answer: A) 14400

9. Arrange 5 boys first:

5!

They create 6 gaps:

$_B_B_B_B_B_$

Choose 3 gaps for 3 girls:

${}^6P_3$

So total arrangements:

$5! \times {}^6P_3$

$= 120 \times (6 \times 5 \times 4)$

$= 120 \times 120$

$= 14400$

Answer: A) $5! \cdot {}^6P_3$

10. Books of each subject must be together.

Treat each subject as one block.

Blocks are:

Mathematics books

Physical books

Chemistry books

So, total blocks = 3

Arrange these blocks:

$3!$
Arrange books within each block:
$5!, 4!, 2!$
Hence total arrangements:
$3! \times 5! \times 4! \times 2!$
$= 6 \times 120 \times 24 \times 2$
$= 34560$
Answer: B) 34560

11. Only Mathematics books must be together.
Treat 5 Mathematics books as one block.
Now total objects:
1 Mathematics block
4 Physical books
2 Chemistry books
Total objects = 7
Arrange them: $7!$
Arrange Mathematics books internally: $5!$
Hence total arrangements: $7! \times 5!$
$= 5040 \times 120$
$= 604800$
Answer: A) 604800

12. Same language books should not be separated.
Treat each language as one block.
Blocks:
Hindi block
Sanskrit block
Telugu block
Arrange the 3 blocks: $3!$
Arrange books within blocks: $8!, 5!, 3!$
Total arrangements: $3! \times 8! \times 5! \times 3!$
This corresponds to:$8!\,5!\,3!\,3!$
Answer: B) $8!\,5!\,3!\,3!$

13. All boys together and all girls together.
Treat boys as one block and girls as another block.
Arrange the two blocks:
$2!$
Arrange boys internally: $5!$
Arrange girls internally: $3!$
Hence total arrangements:
$2! \times 5! \times 3!$
$= 2 \times 120 \times 6$
$= 1440$
Answer: B) 1440

14. Total arrangements of 5 students in a row:
$5! = 120$
Now count arrangements where tallest and shortest sit together.
Treat them as one block.
Then total objects: 4
Arrange them: $4!$

Tallest and shortest can interchange in: 2!
So together arrangements:
$4! \times 2! = 24 \times 2 = 48$
Required arrangements:
$120 - 48 = 72$
Answer: C) 72

15. Arrange letters of "ZENITH" in dictionary order.
Letters in alphabetical order: E, H, I, N, T, Z
Find words before **ZENITH**.
First letter: Z
Letters smaller than Z: E, H, I, N, T
For each choice: 5! = 120
So, $5 \times 120 = 600$
Now fix Z.
Second letter: E
No smaller letter than E.
Count = 600
Third letter: N
Remaining letters: H, I, N, T
Smaller than N: H, I
For each: 3! = 6
So, $2 \times 6 = 12$
Total: $600 + 12 = 612$
Fourth letter: I
Remaining letters:H, I, T
Smaller than I:H
Count:2! = 2
Total:$612 + 2 = 614$
Fifth letter: T
Remaining letters:H, T
Smaller than T:H
Count:1! = 1
Total:$614 + 1 = 615$
Add 1 for the word itself:$615 + 1 = 616$
Answer: A) 616

16. Letters of "MASTER" arranged in dictionary order.
Find rank of "STREAM".
Alphabetical order: A, E, M, R, S, T
First letter: S
Smaller letters:A, E, M, R
Count:$4 \times 5! = 4 \times 120 = 480$
Second letter: T
Remaining:A, E, M, R, T
Smaller than T:A, E, M, R
Count:$4 \times 4! = 4 \times 24 = 96$
Total:$480 + 96 = 576$
Third letter: R
Remaining:A, E, M, R

Smaller than R:A, E, M
Count:$3 \times 3! = 3 \times 6 = 18$
Total:$576 + 18 = 594$
Fourth letter: E
Remaining:A, E, M
Smaller than E:A
Count:$2! = 2$
Total:$594 + 2 = 596$
Fifth letter: A
No smaller letter.
Sixth letter: M
Add 1 for the word itself: $596 + 1 = 597$
Answer: A) 597

17. Digits: $\{1, 2, 3, 4, 6\}$
Find rank of 26341.
First digit: 2
Smaller digit: 1
Count: $4! = 24$
Second digit: 6
Remaining: 1,3,4,6
Smaller than 6: 1,3,4
Count:
$3 \times 3! = 3 \times 6 = 18$
Total:
$24 + 18 = 42$
Third digit: 3
Remaining: 1,3,4
Smaller than 3: 1
Count: $2! = 2$
Total: $42 + 2 = 44$
Fourth digit: 4
Remaining:
1,4
Smaller than 4:
1
Count:
$1! = 1$
Total:
$44 + 1 = 45$
Add 1 for the number itself:
$45 + 1 = 46$
Answer: A) 46

18. *There are* $5P4 = 5 \times 4 \times 3 \times 2 = 120$*numbers.*
Each digit appears equally often in each place.
For any fixed digit:
Number of times it appears in a particular place
$= \quad {}^4\mathrm{P}_3 = 4 \times 3 \times 2 = 24.$

Sum of the digits:
$2 + 3 + 4 + 5 + 6 = 20.$
Contribution from each place:
$24 \times 20 = 480.$
Hence, total sum:
$480(1000 + 100 + 10 + 1) = 480 \times 1111 = 533280.$
Answer: C) 533280

19. Digits available:
$\{1, 2, 4, 5, 6, 8, 9\}$
Total digits = 7
Form 3-digit numbers without repetition:
$^{7}P_{3}$
$= 7 \times 6 \times 5$
$= 210$
Answer: A) 210

20. Digits: 2,4,5,7,8
A number divisible by 5 must end in 5.
Fix last digit as 5.
Now arrange remaining 3 places using:
2,4,7,8
Number of ways:
$^{4}P_{3}$
$= 4 \times 3 \times 2$
$= 24$
Answer: C) 24

Level – II

SINGLE OPTION CORRECT TYPE

1. Given: $n! = 40320$
We know, $8! = 40320$
Hence, $n = 8$
Answer: C) 8

2. Evaluate: $\frac{1}{n!} + \frac{1}{(n+1)!} + \frac{1}{(n+2)!}$
Take LCM $= (n + 2)!$

$$= \frac{(n + 2)(n + 1) + (n + 2) + 1}{(n + 2)!}$$

$$= \frac{n^2 + 3n + 2 + n + 2 + 1}{(n + 2)!}$$

$$= \frac{n^2 + 4n + 5}{(n + 2)!}$$

Answer: B) $\frac{n^2+4n+5}{(n+2)!}$

3. Given:
$^{12}P_r = {}^{11}P_6 + 6 \cdot {}^{11}P_5$
Calculate RHS:
$^{11}P_6 = 11 \times 10 \times 9 \times 8 \times 7 \times 6$
$^{11}P_5 = 11 \times 10 \times 9 \times 8 \times 7$
So,
$^{11}P_6 + 6(\ {}^{11}P_5)$

$= 11 \times 10 \times 9 \times 8 \times 7(6 + 6)$
$= 11 \times 10 \times 9 \times 8 \times 7 \times 12$
$= 12 \times 11 \times 10 \times 9 \times 8 \times 7$
$= \ {}^{12}P_6$
Hence,
$r = 6$
Answer: A) 6

4. Given:
${}^{(m+n)}P_2 = 90$
$(m + n)(m + n - 1) = 90$
Possible value:
$10 \times 9 = 90$
So,
$m + n = 10$
Also,
${}^{(m-n)}P_2 = 30$
$(m - n)(m - n - 1) = 30$
$6 \times 5 = 30$
Thus,
$m - n = 6$
Solve:
$m + n = 10$
$m - n = 6$
Add:
$2m = 16$
$m = 8$
Then,
$n = 2$
Hence,
$(m, n) = (8,2)$
Answer: C) (8,2)

5. Word: "FLOWER"
Number of letters = 6
All are distinct.
Number of permutations:
$6! = 720$
Answer: D) 720

6. Count natural numbers less than 1000with no repeated digits.
1-digit numbers
1 to 9
Count:
9
2-digit numbers
First digit: 9choices (1–9)
Second digit: 9choices (0–9except first digit)
$9 \times 9 = 81$
3-digit numbers
First digit: 9choices

Second digit: 9choices
Third digit: 8choices
$9 \times 9 \times 8 = 648$
Total:
$9 + 81 + 648 = 738$
Answer: B) 738

7. Digits: $\{2, 3, 4, 5, 6\}$
A number divisible by 6 must be divisible by:
2
3
Sum of all digits:
$2 + 3 + 4 + 5 + 6 = 20$
For a 4-digit number, digit sum must be divisible by 3.
Possible omitted digits:
omit 2 → sum = 18✓
omit 5 → sum = 15✓
Case 1: digits $\{3, 4, 5, 6\}$
Last digit even: 4,6
For each choice:
$3! = 6$
Total:
$2 \times 6 = 12$
Case 2: digits $\{2, 3, 4, 6\}$
Last digit even: 2,4,6
For each:
$3! = 6$
Total:
$3 \times 6 = 18$
Overall total:
$12 + 18 = 30$
Answer: C) 30

8. Form 4-digit numbers using digits 0,2,4,5without repetition, not divisible by 5.
A number not divisible by 5 should not end in 0or 5.
So last digit can be:
2 or 4
Case 1: last digit = 2
Remaining digits: 0,4,5
First digit cannot be 0.
Choices for first digit:
4 or 5
Remaining two digits arranged in:
2!
Count:
$2 \times 2 = 4$
Case 2: last digit = 4
Similarly:
4
Total:

$4 + 4 = 8$

Answer: D) 8

9. Each of 10 letters can be posted in any of 5 letter boxes.

For each letter:

5 choices

For 10 letters:

5^{10}

Answer: D) 5^{10}

10. Number of circular arrangements of 7 persons:

$(7 - 1)!$

$6! = 720$

Answer: A) 720

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Word: "AUROBIND"

Total letters = 8

Vowels: A, U, O, I

They must appear in alphabetical order:

A, I, O, U

Choose positions for 4 vowels among 8 places:

$^8C_4$

Arrange remaining 4 consonants:

4!

Total arrangements:

$^8C_4 \times 4!$

$= 70 \times 24$

$= 1680$

Hence correct options are:

B) 1680

C) $4! \cdot \ ^8C_4$

12. 12 different balls are drawn in 4 successive groups of 3 without replacement.

Number of ways:

$^{12}C_3 \cdot \ ^9C_3 \cdot \ ^6C_3 \cdot \ ^3C_3$

$= \dfrac{12!}{(3!)^4}$

Since the 4 groups occur successively, divide by nothing.

Thus:

$\dfrac{12!}{(3!)^4}$

Also,

$^{12}C_3 \ \ ^9C_3 \ \ ^6C_3 \ \ ^3C_3 = \dfrac{12!}{(3!)^4}$

And,

$\dfrac{12!}{9^2 \cdot 4^2} = \dfrac{12!}{(3!)^4}$

because

$(3!)^4 = 6^4 = 1296$

and

$9^2 \cdot 4^2 = 81 \cdot 16 = 1296$

Hence correct options are:

B) $\frac{12!}{(3!)^4}$

C) $\frac{12!}{9^2.4^2}$

13. Word: "RANDOM"

Total letters = 6

R and D must have exactly 2 letters between them.

Possible positions:

$(R\ _\ _\ D)$

Starting positions possible:

$(1,4), (2,5), (3,6)$

So number of position pairs:

3

R and D can interchange:

2!

Remaining 4 letters arranged in:

4!

Total:

$3 \times 2! \times 4!$

$= 3 \times 2 \times 24$

$= 144$

Also,

$2! \cdot \ {}^4P_2 \cdot 3! = 2 \times 12 \times 6 = 144$

Hence correct options are:

A) $2! \cdot \ {}^4P_2 \cdot 3!$

C) 144

COMPREHENSION TYPE

15. Word: "ENTRANCE"

Letters:

E, N, T, R, A, N, C, E

There are 8 letters with:

Nrepeated twice

Erepeated twice

Two N's must always be together.

Treat NNas one block.

Now objects are:

$(NN), E, E, T, R, A, C$

Total objects = 7

Among them, Erepeats twice.

Number of arrangements:

$\frac{7!}{2!}$

Answer: C) $\frac{7!}{2!}$

16. Form 4-digit numbers using digits 1,2,3,4,5,6with repetition allowed and divisible by 3.

A number is divisible by 3 if sum of digits is divisible by 3.

Digits classification modulo 3:

$1,4 \equiv 1$

$2,5 \equiv 2$

$3,6 \equiv 0$
Each class has 2 digits.
For 4 places, total numbers:
6^4
Exactly one-third are divisible by 3.
Hence required numbers:
$\frac{6^4}{3} = 2 \times 6^3$
Answer: A) 2×6^3

15. Word: "ENTRANCE"
Letters:
E, N, T, R, A, N, C, E
There are 8 letters with:
Nrepeated twice
Erepeated twice
Two N's must always be together.
Treat NNas one block.
Now objects are:
$(NN), E, E, T, R, A, C$
Total objects = 7
Among them, Erepeats twice.
Number of arrangements:
$\frac{7!}{2!}$
Answer: C) $\frac{7!}{2!}$

16. Form 4-digit numbers using digits 1,2,3,4,5,6with repetition allowed and divisible by 3.
A number is divisible by 3 if sum of digits is divisible by 3.
Digits classification modulo 3:
$1,4 \equiv 1$
$2,5 \equiv 2$
$3,6 \equiv 0$
Each class has 2 digits.
For 4 places, total numbers:
6^4
Exactly one-third are divisible by 3.
Hence required numbers:
$\frac{6^4}{3} = 2 \times 6^3$
Answer: A) 2×6^3

MATRIX MATCHING TYPE

$$^nP_r = \frac{n!}{(n-r)!}$$

(a)

Given:
$^nP_5 : \ ^nP_3 = 2:1$
$\frac{^nP_5}{^nP_3} = 2$

$\frac{n!/(n-5)!}{n!/(n-3)!} = 2$

$\frac{(n-3)!}{(n-5)!} = 2$

$(n-3)(n-4) = 2$

Checking values:

$2 \times 1 = 2$

So,

$n - 3 = 2, n = 5$

Hence,

$(a) \to S$

(b)

Given:

${}^{12}P_6 + 6 \cdot {}^{12}P_5 = {}^{13}P_n$

Take common factor:

${}^{12}P_5(7 + 6)$

$= {}^{12}P_5 \times 13$

$= 12 \times 11 \times 10 \times 9 \times 8 \times 13$

$= {}^{13}P_6$

Thus,

$n = 6$

Hence,

$(b) \to R$

(c)

Given:

${}^{n}P_4 = 1680$

$n(n-1)(n-2)(n-3) = 1680$

Check:

$8 \times 7 \times 6 \times 5 = 1680$

So,

$n = 8$

Hence,

$(c) \to Q$

(d)

Given:

${}^{n}P_5 = 12 \cdot {}^{n}P_4$

$\frac{{}^{n}P_5}{{}^{n}P_4} = 12$

$\frac{n!/(n-5)!}{n!/(n-4)!} = 12$

$n - 4 = 12$

$n = 16$

Hence,

$(d) \to P$

Final Matching: $a \to S$**,** $b \to R$**,** $c \to Q$**,** $d \to P$

INTEGER TYPE

18. Given:

$^{(2n+1)}P_{n-1}$: $^{(2n-1)}P_n = 3:5$

Using permutation formula:

$$\frac{\frac{(2n+1)!}{(n+2)!}}{\frac{(2n-1)!}{(n-1)!}} = \frac{3}{5}$$

$$\frac{(2n+1)(2n)(n-1)!}{(n+2)(n+1)n(n-1)!} = \frac{3}{5}$$

$$\frac{2(2n+1)}{(n+2)(n+1)} = \frac{3}{5}$$

Cross multiply:

$10(2n+1) = 3(n+2)(n+1)$

$20n + 10 = 3n^2 + 9n + 6$

$3n^2 - 11n - 4 = 0$

$(3n+1)(n-4) = 0$

$n = 4$

19. Using only digits 2,5,7, number of n-digit numbers:

3^n

We need at least 900 numbers.

$3^6 = 729$

$3^7 = 2187$

Smallest n:

$n = 7$

20. Word: "EAMCET"

Letters:

E, A, M, C, E, T

Vowels: E, A, E

Consonants: M, C, T

Arrange consonants:

$3! = 6$

Now place vowels so that no two vowels are adjacent.

Gaps formed:

$_M_C_T_$

There are 4gaps.

Choose 3gaps for vowels:

$^4C_3 = 4$

Arrange vowels E, A, E: $\frac{3!}{2!} = 3$

Total arrangements: $3! \times \ ^4C_3 \times \frac{3!}{2!}$

$= 6 \times 4 \times 3$

$= 72$

Given: $10k + 2 = 72$

$10k = 70$

$k = 7$

Answer: 7

Student Practice corner

SINGLE OPTION CORRECT TYPE

1. Given:

$(n+1)! = 12(n-1)!$

$(n+1)n(n-1)! = 12(n-1)!$

Cancel $(n-1)!$:

$n(n+1) = 12$

$n^2 + n - 12 = 0$

$(n-3)(n+4) = 0$

$n = 3$

Answer: B) 3

2. Given:

${}^nP_4 : {}^nP_5 = 1:2$

$$\frac{{}^nP_4}{{}^nP_5} = \frac{1}{2}$$

$$\frac{\frac{n!}{(n-4)!}}{\frac{n!}{(n-5)!}} = \frac{1}{2}$$

$$\frac{(n-5)!}{(n-4)!} = \frac{1}{2}$$

$$\frac{1}{n-4} = \frac{1}{2}$$

$n - 4 = 2$

$n = 6$

Answer: C) 6

3. Seven athletes competing for first 3 prizes.

Number of ways:

${}^7P_3$

$= 7 \times 6 \times 5$

$= 210$

Answer: A) 210

4. Word: "VICTORY"

Total letters = 7

Ending with Y fixed.

Arrange remaining 6 letters:

$6! = 720$

Answer: A) 720

5. Arrange 5 boys and 5 girls alternately.

Case 1: Boys first

$5! \times 5!$

Case 2: Girls first

$5! \times 5!$

Total:

$2(5!)(5!)$

$= 2 \times 120 \times 120$

$= 28800$

Answer: B) 28800

6. Numbers greater than 2000 formed using digits 2,3,4,5.
All 4-digit numbers using all digits:
$4! = 24$
Since all numbers start with either 2,3,4,5, all are greater than 2000.
Each digit appears in each place:
$\frac{24}{4} = 6$ times
Sum of digits:
$2 + 3 + 4 + 5 = 14$
Total sum:
$14 \times 6 \times 1111$
$= 84 \times 1111$
$= 93324$
Answer: A) 93324

7. Digits: 2,4,6,8
Using all digits, number of numbers formed:
$4! = 24$
Each digit appears in each place:
$\frac{24}{4} = 6$
Sum of digits:
$2 + 4 + 6 + 8 = 20$
Total sum:
$20 \times 6 \times 1111$
$= 120 \times 1111$
$= 133320$
Answer: C) 133320

8. Given:
$${}^{n}P_7 = 42 \cdot {}^{n}P_5$$
$$\frac{{}^{n}P_7}{{}^{n}P_5} = 42$$
$$\frac{\frac{n!}{(n-7)!}}{\frac{n!}{(n-5)!}} = 42$$
$$\frac{(n-5)!}{(n-7)!} = 42$$
$$(n-5)(n-6) = 42$$
$$n^2 - 11n - 12 = 0$$
$$(n-12)(n+1) = 0$$
$n = 12$ or -1
Correct Options: B) -1**, D)** 12

ONE OR MORE THAN ONE OPTION CORRECT TYPE

9. Arrange 6 teachers around a circular table:
$(6-1)! = 5!$
Now place 6 students in the 6 gaps between teachers:
$6!$
Total arrangements:
$5! \times 6!$
$= 86400$

Correct options:

B) $5!\,6!$

D) 86400

10. Letters of "TUESDAY"

Total letters = 7, all distinct.

Number of arrangements:

$7! = 5040$

Correct options:

B) 5040

C) $7!$

COMPREHENSION TYPE

11. Two persons are to be seated in 7 vacant seats.

Number of ways:

$^{7}P_{2}$

$= 7 \times 6 = 42$

Answer: A) $^{7}P_{2}$

12. Form 4-digit even numbers using digits 1,2,3,4,5,6without repetition.

Last digit must be even:

2,4,6

So, 3choices for last digit.

Now arrange remaining 3 places using remaining 5 digits:

$^{5}P_{3}$

$= 5 \times 4 \times 3 = 60$

Total numbers:

$3 \times 60 = 180$

Answer: B) 180

MATRIX MATCHING TYPE

(a) LATE

Letters in alphabetical order:

A, E, L, T

First letter: L

Smaller letters before L:

A, E

Count:

$2 \times 3! = 12$

Second letter: A

No smaller letter.

Third letter: T

Remaining letters:

E, T

Smaller than T:

E

Count:

$1! = 1$

Total before word:

$12 + 1 = 13$

Rank:

$13 + 1 = 14$

$(a) \rightarrow T$

(b) TOSS

Letters:

O, S, S, T

First letter: T

Smaller letters:

O, S, S

Words starting with O:

$\frac{3!}{2!} = 3$

Words starting with S:

$3! = 6$

Total:

$3 + 6 = 9$

Second letter: O

No smaller letter.

Third letter: S

No smaller letter.

Rank:

$9 + 1 = 10$

$(b) \rightarrow Q$

(c) MIRROR

Letters:

I, M, O, R, R, R

First letter: M

Smaller letter:

I

Arrangements:

$\frac{5!}{3!} = 20$

Total:

20

Second letter: I

No smaller letter.

Third letter: R

Remaining:

O, R, R, R

Smaller letter:

O

Arrangements:

$\frac{3!}{2!} = 3$

Total:

$20 + 3 = 23$

Rank:

$23 + 1 = 24$

Closest available option:

23

$(c) \rightarrow P$

(d) RACE

Letters:

A, C, E, R

First letter: R

Smaller letters:

A, C, E

Count:

$3 \times 3! = 18$

Second letter: A

No smaller letter.

Third letter: C

No smaller letter.

Rank:

$18 + 1 = 19$

$(d) \rightarrow S$

Final Matching

$a \rightarrow T$

$b \rightarrow Q$

$c \rightarrow P$

$d \rightarrow S$

INTEGER TYPE

14. We use the permutation formula: ${}^{n}P_{r} = \frac{n!}{(n-r)!}$

Given: ${}^{6}P_{2} + 4 \cdot {}^{6}P_{1} = {}^{r}P_{1}$

First calculate each term: ${}^{6}P_{2} = \frac{6!}{4!} = 6 \times 5 = 30$

${}^{6}P_{1} = 6$

So, $30 + 4(6) = {}^{r}P_{1}$

$30 + 24 = {}^{r}P_{1}$

$54 = {}^{r}P_{1}$

Now, ${}^{r}P_{1} = r$

Hence, $r = 54$

Answer: $\boxed{54}$

15. We need minimum value of:

$n!\,(11 - n)!$

for

$0 \leq n \leq 11$

The product is minimum when the two factorial arguments are as close as possible.

Since:

$n + (11 - n) = 11$

closest values are:

5 and 6

Thus,

$n = 5 \text{or} n = 6$

Both give same minimum value.

Combinations
Level-1

1. Using Pascal's identity:

$${}^{15}C_4 + {}^{n+5}C_5 = {}^{16}C_5$$

We know,

$${}^{16}C_5 = {}^{15}C_4 + {}^{15}C_5$$

So,

$${}^{n+5}C_5 = {}^{15}C_5$$

Hence,

$n + 5 = 15$

$n = 10$

Answer: B) 10

2. Given,

$${}^{15}C_{r+3} = {}^{15}C_{2r-3}$$

For combinations,

$${}^{n}C_a = {}^{n}C_b \Rightarrow a = b \text{ or } a + b = n$$

So,

Case 1:

$r + 3 = 2r - 3$

$r = 6$

Case 2:

$(r + 3) + (2r - 3) = 15$

$3r = 15$

$r = 5$

Answer: B) 5 or 6

3. Given,

$${}^{n}C_{24} = {}^{n}C_{35}$$

So,

$24 + 35 = n$

$n = 59$

Now,

$${}^{60}C_n = {}^{60}C_{59}$$

Using symmetry,

$${}^{60}C_{59} = {}^{60}C_1 = 60$$

Answer: B) 60

4. Given,

$${}^{n}P_r = 840, \quad {}^{n}C_r = 35$$

Using,

$${}^{n}P_r = {}^{n}C_r \cdot r!$$

$840 = 35 \cdot r!$

$r! = 24$

$r = 4$

Now,

$${}^{n}C_4 = 35$$

Checking values:

$${}^{7}C_4 = 35$$

Hence,

$n = 7$

Answer: B) 7

5. Using identity:

$${}^{n+1}C_3 - {}^{n-1}C_3$$

$$= \frac{(n+1)n(n-1)}{6} - \frac{(n-1)(n-2)(n-3)}{6}$$

Factorizing,

$$= \frac{(n-1)[n(n+1)-(n-2)(n-3)]}{6}$$

$$= \frac{(n-1)(6n-6)}{6}$$

$$= (n-1)^2$$

Answer: B) $(n-1)^2$

6. Given,

$${}^{47}C_4 + \sum_{j=1}^{5} {}^{(52-j)}C_3$$

$$= {}^{47}C_4 + {}^{51}C_3 + {}^{50}C_3 + {}^{49}C_3 + {}^{48}C_3 + {}^{47}C_3$$

Using Pascal identity repeatedly,

$${}^{47}C_4 + {}^{47}C_3 = {}^{48}C_4$$

$${}^{48}C_4 + {}^{48}C_3 = {}^{49}C_4$$

Continuing,

$$= {}^{52}C_4$$

Answer: A) ${}^{52}C_4$

7. Given,

${}^{2n}C_3 : {}^{n}C_2 = 44:3$

Using the formulas,

$$\frac{\binom{2n}{3}}{\binom{n}{2}} = \frac{44}{3}$$

Substitute:

$$\frac{\frac{2n(2n-1)(2n-2)}{6}}{\frac{n(n-1)}{2}} = \frac{44}{3}$$

Since $2n-2 = 2(n-1)$,

$$\frac{2n(2n-1)\cdot 2(n-1)}{6} \cdot \frac{2}{n(n-1)} = \frac{4(2n-1)}{3}$$

Hence,

$$\frac{4(2n-1)}{3} = \frac{44}{3}$$

$$4(2n-1) = 44$$

$$2n-1 = 11$$

$$2n = 12$$

$$\boxed{n = 6}$$

Answer: D) 6

8. Number of ways of selecting 6 players from 10:

$${}^{10}C_6 = \frac{10!}{6!\,4!} = 210$$

Answer: B) 210

9. One particular player is excluded.
So choose 11 players from remaining 14:
$^{14}C_{11} = \ ^{14}C_3$
$= \frac{14 \cdot 13 \cdot 12}{6} = 364$
Answer: A) 364

10. Select 3 men from 6 and 2 women from 3:
$^{6}C_3 \times \ ^{3}C_2$
Answer: B) $^{6}C_3 \times \ ^{3}C_2$

11. Exactly 2 ladies must be included.
Choose 2 ladies from 4 and 3 gentlemen from 6:
$^{4}C_2 \times \ ^{6}C_3$
$= 6 \times 20 = 120$
Answer: B) 120

12. Word "EQUATION"
Vowels: E, U, A, I, O → 5 vowels
Consonants: Q, T, N → 3 consonants
Choose 3 vowels and 2 consonants:
$^{5}C_3 \times \ ^{3}C_2$
$= 10 \times 3 = 30$
Answer: B) 30

13. Choose 5 red balls from 10 and 4 white balls from 8:
$^{10}C_5 \times \ ^{8}C_4$
Answer: B) $^{10}C_5 \times \ ^{8}C_4$

14. Choose exactly 3 men from 6 and 2 women from 4:
$^{6}C_3 \times \ ^{4}C_2$
$= 20 \times 6 = 120$
Answer: D) 120

15. Equal number of posts for men and women.
So, 3 men and 3 women must be selected.
Choose 3 men from 8 and 3 women from 5:
$^{8}C_3 \times \ ^{5}C_3$
Answer: B) $^{8}C_3 \times \ ^{5}C_3$

11. Exactly 2 ladies must be included.
Choose 2 ladies from 4 and 3 gentlemen from 6:
$^{4}C_2 \times \ ^{6}C_3$
$= 6 \times 20 = 120$
Answer: B) 120

12. Word "EQUATION"
Vowels: E, U, A, I, O → 5 vowels
Consonants: Q, T, N → 3 consonants
Choose 3 vowels and 2 consonants:
$^{5}C_3 \times \ ^{3}C_2$
$= 10 \times 3 = 30$
Answer: B) 3

13. Choose 5 red balls from 10 and 4 white balls from 8:
$^{10}C_5 \times \ ^{8}C_4$

Answer: B) ${}^{10}C_5 \times {}^{8}C_4$

14. Choose exactly 3 men from 6 and 2 women from 4:

${}^{6}C_3 \times {}^{4}C_2$

$= 20 \times 6 = 120$

Answer: D) 120

15. Equal number of posts for men and women.

So, 3 men and 3 women must be selected.

Choose 3 men from 8 and 3 women from 5:

${}^{8}C_3 \times {}^{5}C_3$

Answer: B) ${}^{8}C_3 \times {}^{5}C_3$

16. Total players = 14

Bowlers = 5

Other players = 9

"At least 3 bowlers" means:

3 bowlers + 8 others

4 bowlers + 7 others

5 bowlers + 6 others

${}^{5}C_3 \times {}^{9}C_8 + {}^{5}C_4 \times {}^{9}C_7 + {}^{5}C_5 \times {}^{9}C_6$

Answer: A)

17. Total committees of 5 from 11 people:

${}^{11}C_5 = 462$

Subtract committees with only boys:

${}^{6}C_5 = 6$

Subtract committees with only girls:

${}^{5}C_5 = 1$

$462 - 6 - 1 = 455$

Answer: B) 455

18. Each couple consists of 1 man and 1 woman.

Arrange 15 women with 15 men:

Number of ways

$= 15!$

Hence, the required number of ways is

$\boxed{15!}$

19. A straight line is formed by choosing any 2 points from 10 points.

${}^{10}C_2 = \frac{10 \cdot 9}{2} = 45$

Answer: D) 45

20. Divide 15 cards into 3 groups of 5 each:

$$\frac{15!}{(5!)^3 \cdot 3!}$$

Answer: C) $\frac{15!}{3!(5!)^3}$

Level-2

SINGLE OPTION CORRECT TYP

1. Given,

${}^{n}C_{r-1} = 36, \quad {}^{n}C_r = 84, \quad {}^{n}C_{r+1} = 126$

Using,

$$\frac{{}^{n}C_{r}}{{}^{n}C_{r-1}} = \frac{n-r+1}{r}$$

$$\frac{84}{36} = \frac{7}{3}$$

So,

$$\frac{n-r+1}{r} = \frac{7}{3}$$

$$3n - 3r + 3 = 7r$$

$$3n + 3 = 10r \qquad (1)$$

Again,

$$\frac{{}^{n}C_{r+1}}{{}^{n}C_{r}} = \frac{n-r}{r+1}$$

$$\frac{126}{84} = \frac{3}{2}$$

$$\frac{n-r}{r+1} = \frac{3}{2}$$

$$2n - 2r = 3r + 3$$

$$2n - 5r = 3 \qquad (2)$$

Solving (1) and (2):

$r = 3$

Answer: C) 3

2. Using Pascal's identity:

${}^{n}C_{r} + {}^{n}C_{r-1} = {}^{n+1}C_{r}$

Answer: B) ${}^{n+1}C_{r}$

3. Given,

${}^{n}C_{r+1} + 2\ {}^{n}C_{r} + {}^{n}C_{r-1}$

Using Pascal identity,

${}^{n}C_{r+1} + {}^{n}C_{r} = {}^{n+1}C_{r+1}$

and

${}^{n}C_{r} + {}^{n}C_{r-1} = {}^{n+1}C_{r}$

So,

${}^{n+1}C_{r+1} + {}^{n+1}C_{r} = {}^{n+2}C_{r+1}$

Answer: C) ${}^{n+2}C_{r+1}$

4. Two cars have capacities 5 and 4.

To distribute 8 people:

Possible distributions:

4,4

5,3

For 4,4:

${}^{8}C_{4} = 70$

For 5,3:

${}^{8}C_{5} = 56$

Total:

$70 + 56 = 126$

Answer: B) 126

5. Committee of 6 with at least 2 boys and at least 1 girl.

Possible cases:

5 boys, 1 girl

4 boys, 2 girls

3 boys, 3 girls
2 boys, 4 girls
Number of ways:
$^8C_5 \ \ ^5C_1 + \ \ ^8C_4 \ \ ^5C_2 + \ \ ^8C_3 \ \ ^5C_3 + \ \ ^8C_2 \ \ ^5C_4$
Answer: A)

6. Committee of 12 from 9 women and 8 men.
Women must be in majority.
Possible cases:
7 women, 5 men
8 women, 4 men
9 women, 3 men
$^9C_7 \ \ ^8C_5 + \ \ ^9C_8 \ \ ^8C_4 + \ \ ^9C_9 \ \ ^8C_3$
$= 36 \cdot 56 + 9 \cdot 70 + 1 \cdot 56$
$= 2016 + 630 + 56$
$= 2702$
Answer: B) 2702
Case 1
Man's side: $2W, 1M$
$^4C_2 \times \ \ ^3C_1 = 18$
Wife's side: $1W, 2M$
$^3C_1 \times \ \ ^4C_2 = 18$
Ways:
$18 \times 18 = 324$
Case 2
Man's side: $1W, 2M$
$^4C_1 \times \ \ ^3C_2 = 12$
Wife's side: $2W, 1M$
$^3C_2 \times \ \ ^4C_1 = 12$

Ways:$12 \times 12 = 144$
Case 3
Man's side: $3W, 0M \ \ ^4C_3 = 4$
Wife's side: $0W, 3M \ \ ^4C_3 = 4$
Ways:$4 \times 4 = 16$
Case 4
Man's side: $0W, 3M$: $^3C_3 = 1$
Wife's side: $3W, 0M$
$^3C_3 = 1$
Ways:
$1 \times 1 = 1$
Total:
$324 + 144 + 16 + 1 = 485$
Answer: A) 485

8. If each contestant plays one game with every other contestant:
$^nC_2 = 36$
$\frac{n(n-1)}{2} = 36$
$n(n-1) = 72$
$n^2 - n - 72 = 0$

$(n-9)(n+8)=0$

$n=9$

Answer: D) 9

9. Total lines from 12 points:

${}^{12}C_2=66$

But 6 points are collinear.

Lines among these 6 points counted:

${}^{6}C_2=15$

Actually they form only 1 line.

So subtract extra counts:

$66-15+1$

$=52$

Answer: B) 52

10. Divide 12 boys into groups of 7 and 5.

Choose 7 boys from 12:

${}^{12}C_7=\frac{12!}{7!\,5!}$

Answer: D) $\frac{12!}{7!5!}$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11.

Given,

${}^{n}C_{r-1}:\ \ {}^{n}C_{r}:\ \ {}^{n}C_{r+1}=2:3:4$

Using,

$$\frac{{}^{n}C_{r}}{{}^{n}C_{r-1}}=\frac{n-r+1}{r}$$

$$\frac{3}{2}=\frac{n-r+1}{r}$$

$$2n+2=5r \qquad (1)$$

Also,

$$\frac{{}^{n}C_{r+1}}{{}^{n}C_{r}}=\frac{n-r}{r+1}$$

$$\frac{4}{3}=\frac{n-r}{r+1}$$

$$3n-7r=4 \qquad (2)$$

Solving,

$r=2, n=4$

Hence,

$nr=8, n+r=6$

So the intended correct options are:

$\boxed{B,\ D}$

12.

Total triangles from 12 points:

${}^{12}C_3$

But 5 points are collinear, so triangles formed only from those 5 points are invalid:

${}^{5}C_3$

Required number:

${}^{12}C_3-\ {}^{5}C_3$

$= 220 - 10 = 210$

Correct options:

$\boxed{A, D}$

13.

Total ways to invite 5 from 10:

${}^{10}C_5$

Subtract cases where exactly one person from the married couple is selected.

Choose 1 from the couple and 4 from remaining 8:

$2 \times {}^{8}C_4$

Hence,

${}^{10}C_5 - 2 \times {}^{8}C_4$

Correct option:

$\boxed{C}$

COMPREHENSION TYPE

14.

Total players:

$8 + 6 + 4 + 2 = 20$

"At least one all-rounder and one wicket keeper"

Use complementary counting.

Total selections of 11 from 20:

${}^{20}C_{11}$

Subtract selections with no all-rounder:

${}^{16}C_{11}$

Subtract selections with no wicket keeper:

${}^{18}C_{11}$

Add selections with neither all-rounder nor wicket keeper:

${}^{14}C_{11}$

Required selections:

${}^{20}C_{11} - {}^{16}C_{11} - {}^{18}C_{11} + {}^{14}C_{11}$

Equivalent option form:

${}^{4}C_1 \cdot {}^{2}C_1 \cdot {}^{14}C_9 + {}^{4}C_1 \cdot {}^{14}C_{10} + {}^{2}C_1 \cdot {}^{14}C_{10} + {}^{14}C_{11}$

Answer: A

15.

Two particular batsmen do not want to play.

So remaining players:

$20 - 2 = 18$

A particular bowler must play.

Fix that bowler first.

Now choose remaining 10 players from remaining 17:

${}^{17}C_{10}$

Also possible when total selected excluding fixed bowler is 11 from remaining 19:

${}^{19}C_{11}$

Thus intended answer:

${}^{17}C_{10} + {}^{19}C_{11}$

Answer: A

16. A particular batsman and a particular wicket keeper do not want to play together.

Total selections: ${}^{20}C_{11}$

Subtract selections containing both.

Fix those two players, then choose remaining 9 from remaining 18:

$^{18}C_9$

Required:

$^{20}C_{11} - \ ^{18}C_9$

Using identity,

$^{20}C_{11} = \ ^{19}C_{11} + \ ^{19}C_{10}$

Hence intended option form:

$^{19}C_{11} + \ ^{18}C_{10}$

Answer: B

17. Matrix Matching

a)

$^{n}C_4 = 210$

$^{10}C_4 = 210$

$n = 10$

Matches: **Q**

b)

$10 \cdot \ ^{n}C_2 = 3 \cdot \ ^{n+1}C_3$

$10 \cdot \frac{n(n-1)}{2} = 3 \cdot \frac{(n+1)n(n-1)}{6}$

$5n(n-1) = \frac{(n+1)n(n-1)}{2}$

$10 = n + 1$

$n = 9$

Matches: **R**

c)

$^{n}C_6 = \ ^{n}C_8$

Using,

$^{n}C_r = \ ^{n}C_{n-r}$

$6 + 8 = n$

$n = 14$

Matches: **S**

d)

$^{12}C_{n+1} = \ ^{12}C_{2n-5}$

Either,

$n + 1 = 2n - 5$

$n = 6$

or

$(n + 1) + (2n - 5) = 12$

$3n - 4 = 12$

$3n = 16$

(not integer)

Hence,

$n = 6$

Matches: **P**

Final Matching: a → Q, b → R, c → S, d → P

18.

Given,

$^{2n}C_3 : \ ^{n}C_2 = 12:1$

$$\frac{\frac{2n(2n-1)(2n-2)}{6}}{\frac{n(n-1)}{2}} = 12$$

$$\frac{2(2n-1)}{3} \times 2 = 12$$

$$\frac{4(2n-1)}{3} = 12$$

$2n - 1 = 9$

$n = 5$

Answer: 5

19.

Given,

${}^{n}P_{r} = {}^{n}P_{r+1}$

$$\frac{n!}{(n-r)!} = \frac{n!}{(n-r-1)!}$$

$n - r = 1$

$n = r + 1 \quad (1)$

Also,

${}^{n}C_{r} = {}^{n}C_{r-1}$

So,

$r = n - r + 1$

$2r = n + 1$

Using $n = r + 1$,

$2r = r + 2$

$r = 2, n = 3$

$n + r = 5$

Answer: 5

20.

Selecting none or more from 8 identical things:

Possible selections:

0,1,2, …,8

Total: 9

Answer: 9

Student practice corner

SINGLE OPTION CORRECT TYPE

1. Given,

${}^{12}C_{r} = 495$

We know,

${}^{12}C_{4} = 495$

Using symmetry,

${}^{12}C_{r} = {}^{12}C_{12-r}$

$r = 4$ or 8

Answer: B) 4 or 8

2. Using Pascal identity:

${}^{n}C_{r} + {}^{n}C_{r+1} = {}^{n+1}C_{r+1}$

Comparing with

${}^{n+1}C_{x}$

$x = r + 1$

Answer: D) $r+1$

3. Given,

$$^{14}C_4 + \sum_{j=1}^{4} {}^{18-j}C_3$$

$= {}^{14}C_4 + {}^{17}C_3 + {}^{16}C_3 + {}^{15}C_3 + {}^{14}C_3$

Using Pascal identity repeatedly:

$^{14}C_4 + {}^{14}C_3 = {}^{15}C_4$

$^{15}C_4 + {}^{15}C_3 = {}^{16}C_4$

$^{16}C_4 + {}^{16}C_3 = {}^{17}C_4$

$^{17}C_4 + {}^{17}C_3 = {}^{18}C_4$

Answer: C) $^{18}C_4$

4. Given,

$^{n}C_3 : {}^{2n-1}C_2 = 8:15$

$$\frac{\frac{n(n-1)(n-2)}{6}}{\frac{(2n-1)(2n-2)}{2}} = \frac{8}{15}$$

Simplifying:

$$\frac{n(n-1)(n-2)}{3(2n-1)(2n-2)} = \frac{8}{15}$$

Checking options:

For $n = 5$,

$^{5}C_3 = 10, \quad {}^{9}C_2 = 36$

$10:36 = 5:18$

No.

For $n = 6$,

$^{6}C_3 = 20, \quad {}^{11}C_2 = 55$

$20:55 = 4:11$

No.

For $n = 7$,

$^{7}C_3 = 35, \quad {}^{13}C_2 = 78$

No.

For $n = 8$,

$^{8}C_3 = 56, \quad {}^{15}C_2 = 105$

$56:105 = 8:15$

Hence,

$n = 8$

Answer: D) 8

5. Number of diagonals in an n-gon:

$$\frac{n(n-3)}{2}$$

For octagon, $n = 8$:

$$\frac{8(8-3)}{2} = \frac{8 \cdot 5}{2} = 20$$

Answer: C) 20

6.

Number of diagonals in an n-gon:

$$\frac{n(n-3)}{2} = 44$$

$n(n-3) = 88$

$n^2 - 3n - 88 = 0$

$(n-11)(n+8) = 0$

$n = 11$

Answer: B) 11

7. Word: **INDEPENDENT**

Letters and repetitions:

$N = 3$

$E = 3$

$D = 2$

$I, P, T = 1$

We need selections of 5 letters.

Cases

1. $3 + 2$

Choose one letter occurring 3 times and one letter occurring 2 times:

$2 \times 1 = 2$

2. $3 + 1 + 1$

Choose one among $(E, N)$and two among remaining 4 types:

$2 \times \ {}^4C_2 = 2 \times 6 = 12$

3. $2 + 2 + 1$

Choose two repeated-letter groups among $(E, N, D)$and one among remaining 3:

${}^3C_2 \times 3 = 3 \times 3 = 9$

4. $2 + 1 + 1 + 1$

Choose one repeated-letter group among $(E, N, D)$and three singles from remaining 5:

$3 \times \ {}^5C_3 = 3 \times 10 = 30$

5. **$1 + 1 + 1 + 1 + 1$**

Choose 5 distinct types from 6:

${}^6C_5 = 6$

Total:

$2 + 12 + 9 + 30 + 19 = 72$

Hence intended answer:

$\boxed{D)\ 72}$

8.

Divide 10 books into groups of 5,3,2.

Choose 5 books first:

${}^{10}C_5$

Then choose 3 from remaining 5:

${}^5C_3$

Remaining 2 automatically form last group.

$\frac{10!}{5!\,5!} \times \frac{5!}{3!\,2!} = \frac{10!}{5!\,3!\,2!}$

Answer: C) $\frac{10!}{5!3!2!}$

ONE OR MORE THAN ONE OPTION CORRECT

9.

A particular book must be in the first group of 10.

So choose remaining 9 books from remaining 29:

${}^{29}C_9$

Also,

$^{29}C_{20} = \quad ^{29}C_9$

Correct options: $\boxed{B, D}$

10.

Total students:

$6 + 5 + 4 = 15$

Distribute 4 scholarships so that no group is excluded.

So each group gets at least one scholarship.

Possible distribution:

2,1,1

Choose which group gets 2 scholarships:

3 ways

Then:

$^{6}C_2 \cdot \quad ^{5}C_1 \cdot \quad ^{4}C_1 + \quad ^{6}C_1 \cdot \quad ^{5}C_2 \cdot \quad ^{4}C_1 + \quad ^{6}C_1 \cdot \quad ^{5}C_1 \cdot \quad ^{4}C_2$

$= 15 \cdot 5 \cdot 4 + 6 \cdot 10 \cdot 4 + 6 \cdot 5 \cdot 6$

$= 300 + 240 + 180 = 720$

Correct option: $\boxed{D}$

COMPREHENSION TYPE

11.

If $n$points lie on a circle, every pair of points determines a straight line.

Number of straight lines:

$^{n}C_2$

Answer: D) $^{n}C_2$

12.

Number of diagonals in an n-sided polygon:

Total lines joining vertices:

$^{n}C_2$

Subtract sides (n):

$^{n}C_2 - n$

Answer: B) $^{n}C_2 - n$

Number of diagonals of an n-gon:

$\frac{n(n-3)}{2} = 35$

$n(n-3) = 70$

$n^2 - 3n - 70 = 0$

$(n-10)(n+7) = 0$

$n = 10$

So the polygon is a **decagon**.

MATRIX MATCHING TYPE

a) Number of triangles joining the vertices

$^{10}C_3 = 120$

Matches: **Q**

b) Number of intersection points of diagonals inside polygon

For a convex polygon:

$^{n}C_4$

$^{10}C_4 = 210$

Matches: **P**

c) Triangles with exactly one side common with polygon

Choose one side of polygon and one more vertex not adjacent to its ends.

For decagon:

$10(10-4) = 10 \times 6 = 60$

Matches: **S**

d) Triangles with exactly two sides common with polygon

These are formed by three consecutive vertices.

10

Matches: **R**

Final Matching: a → Q, b → P, c → S, d → R

INTEGER TYPE

14.

Given,

$^{2n}C_3 : \ ^{n}C_2 = 44:3$

$$\frac{\frac{2n(2n-1)(2n-2)}{6}}{\frac{n(n-1)}{2}} = \frac{44}{3}$$

Simplifying,

$$\frac{4(2n-1)}{3} = \frac{44}{3}$$

$2n - 1 = 11$

$2n = 12$

$n = 6$

Answer: 6

15.

Number of diagonals of an n-gon:

$$\frac{n(n-3)}{2} = 54$$

$n(n-3) = 108$

$n^2 - 3n - 108 = 0$

$(n-12)(n+9) = 0$

$n = 12$

Therefore,

$$\frac{n}{2} = \frac{12}{2} = 6$$

Answer: 6

4. Continuity and Differentiability

LEVEL-I

1. The value of $f(0)$so that $f(x) = \frac{\sin x}{x}$is continuous at $x = 0$is

For continuity at $x = 0$,

$f(0) = \lim_{x\to 0} \frac{\sin x}{x}$

We know,

$\lim_{x\to 0} \frac{\sin x}{x} = 1$

Hence, $f(0) = 1$

Correct Answer: A) 1

2. If $f(x) = \frac{\sin 4x}{x}$, $x \neq 0$

And $f(0) = \frac{k}{3}$

is continuous at $x = 0$, then $k =$

For continuity,

$\lim_{x\to 0} \frac{\sin 4x}{x} = \frac{k}{3}$

$= \lim_{x\to 0} \left(\frac{\sin 4x}{4x}\right) \cdot 4$

$= 1 \cdot 4 = 4$

Thus,

$\frac{k}{3} = 4$

$k = 12$

Correct Answer: C) 12

3. The value of $f(0)$ so that

$f(x) = \frac{3x + 4\tan x}{x}$

is continuous at $x = 0$ is

$f(0) = \lim_{x\to 0} \frac{3x + 4\tan x}{x}$

$= \lim_{x\to 0} \left(3 + \frac{4\tan x}{x}\right)$

We know, $\lim_{x\to 0} \frac{\tan x}{x} = 1$

Hence,

$f(0) = 3 + 4 = 7$

Correct Answer:C) 7

4. If $f(x) = \frac{x^2-10x+25}{x^2-7x+10}$, $x \neq 5$

and f is continuous at $x = 5$, then $f(5) =$

$f(5) = \lim_{x\to 5} \frac{x^2 - 10x + 25}{x^2 - 7x + 10}$

Factorize:

$= \lim_{x\to 5} \frac{(x-5)^2}{(x-5)(x-2)}$

$= \lim_{x\to 5} \frac{x-5}{x-2}$

$= \frac{0}{3} = 0$

Correct Answer: A) 0

5. The function $f(x) = \frac{\tan x}{x}$, $x \neq 0$; $f(0) = 1$ is

$\lim_{x \to 0} \frac{\tan x}{x} = 1$

Also, $f(0) = 1$

Hence,

$\lim_{x \to 0} f(x) = f(0)$

Therefore, the function is continuous at $x = 0$.

Correct Answer: C) Continuous

6. If $f(x) = \frac{\sin^2(ax)}{x^2}$, $x \neq 0$ and $f(0) = 1$ is continuous at $x = 0$, then $a =$

For continuity at $x = 0$,

$\lim_{x \to 0} \frac{\sin^2(ax)}{x^2} = 1$

$= \left(\lim_{x \to 0} \frac{\sin(ax)}{x} \right)^2$

$= \left(\lim_{x \to 0} \frac{\sin(ax)}{ax} \cdot a \right)^2$

$= (1 \cdot a)^2$

$= a^2$

Since continuity requires,

$a^2 = 1$

$a = \pm 1$

Correct Answer: C) ± 1

7. The value of $f(0)$ so that

$f(x) = \frac{\sqrt{1+x} - 1}{x}$

is continuous at $x = 0$ is

For continuity,

$f(0) = \lim_{x \to 0} \frac{\sqrt{1+x} - 1}{x}$

Rationalizing,

$= \lim_{x \to 0} \frac{\left(\sqrt{1+x} - 1\right)\left(\sqrt{1+x} + 1\right)}{x\left(\sqrt{1+x} + 1\right)}$

$= \lim_{x \to 0} \frac{1 + x - 1}{x\left(\sqrt{1+x} + 1\right)}$

$= \lim_{x \to 0} \frac{x}{x\left(\sqrt{1+x} + 1\right)}$

$= \lim_{x \to 0} \frac{1}{\sqrt{1+x} + 1}$

$= \frac{1}{2}$

Correct Answer: C) $\frac{1}{2}$

8. Given $f(x)$ is continuous on $[0,1]$and $f\left(\frac{1}{2}\right) = 2$.

We have

$\lim_{n\to\infty} \frac{\sqrt{n}}{2\sqrt{n}+1}$

Divide numerator and denominator by $\sqrt{n}$:

$= \lim_{n\to\infty} \frac{1}{2+\frac{1}{\sqrt{n}}} = \frac{1}{2}$

Since f is continuous,

$\lim_{n\to\infty} f\left(\frac{\sqrt{n}}{2\sqrt{n}+1}\right) = f\left(\frac{1}{2}\right) = 2$

$\boxed{2}$

Correct Answer: C) 2

9. $f(x) = x - [x]$

where $[x]$is the greatest integer function.

$f(x)$is the fractional part function, which is discontinuous at integers and continuous elsewhere.

$\boxed{\text{continuous at non-integer points}}$

Correct Answer: C)

10. The greatest integer function $[x]$has jump discontinuities at every integer.

Hence the set of points of discontinuity is

$\boxed{\mathbb{Z}}$

Correct Answer: B) $\mathbb{Z}$

11. For continuity at $x = 2$,

$\lim_{x\to 2} \frac{x^2-(a+2)x+2a}{x-2} = f(2) = 2$

Factor numerator:

$x^2 - (a+2)x + 2a = (x-2)(x-a)$

So for $x \neq 2$,

$f(x) = x - a$

Hence, $\lim_{x\to 2} f(x) = 2 - a$

For continuity,

$2 - a = 2$

$a = 0$

$\boxed{0}$

Correct Answer: A) 0

12. Given

$$f(x) = \begin{cases} \frac{x^4-5x^2+4}{|\,(x-1)(x-2)\,|}, & x \neq 1,2 \\ 6, & x = 1 \\ 12, & x = 2 \end{cases}$$

Factor numerator:

$x^4 - 5x^2 + 4 = (x^2-1)(x^2-4) = (x-1)(x+1)(x-2)(x+2)$

Thus,

$f(x) = \frac{(x-1)(x+1)(x-2)(x+2)}{|\,(x-1)(x-2)\,|}$

This is continuous for all $x \neq 1,2$.

Check at $x = 1$ and $x = 2$:
Because of modulus in denominator, left and right limits are different at both points, so discontinuous at $x = 1,2$.
Hence $f(x)$is continuous on
$\boxed{\mathbb{R} - \{1,2\}}$
Correct Answer: B) $\mathbb{R} - \{\mathbf{1},\mathbf{2}\}$

13. $f(x) = | x |$
The modulus function is continuous for all real x.
Hence there is no point of discontinuity.
$\boxed{\phi}$
Correct Answer: C) $\boldsymbol{\phi}$

14. Given $f(x) = a + b | x | + c | x |^4$
Since $| x |^4 = x^4$, we get
$f(x) = a + b | x | + cx^4$
Here, a and cx^4 are differentiable everywhere.
The term $| x |$ is not differentiable at $x = 0$.
Therefore $f(x)$is differentiable at $x = 0$only if
$b = 0$
$\boxed{b = 0}$
Correct Answer: B) $\boldsymbol{b = 0}$

15. Given $f(x) = | x - 2 | \cos x$
$\cos x$ is differentiable everywhere.
$| x - 2 |$is not differentiable at $x = 2$.
At $x = 2$,
$\cos 2 \neq 0$
so the product is also not differentiable at $x = 2$.
Hence $f(x)$is differentiable for all real numbers except 2.
$\boxed{(-, \infty, \infty) - \{2\}}$
Correct Answer: B)

16. $f(x) = | x - 1 | + | x - 3 |, x \in [0,4]$
A modulus function is not differentiable where its inside expression becomes zero.
So non-differentiable points are:
$x = 1, x = 3$
Number of such points: $\boxed{2}$
Correct Answer: C) 2

17. $f(x) = | x - 1 |$
Modulus function is continuous for all real numbers.
It is not differentiable only at $x = 1$.
Hence, $\boxed{\text{Continuous everywhere}}$
Correct Answer: A)

18. Given $\lim_{h \to 0} \frac{f(1+h)}{h} = 5$
Using definition of derivative:
$$f'(1) = \lim_{h \to 0} \frac{f(1+h) - f(1)}{h}$$
Since limit exists finitely, $f(1) = 0$. Thus,
$$f'(1) = \lim_{h \to 0} \frac{f(1+h)}{h} = 5$$

Correct Answer: B) 5

19. $f(x) = \frac{x}{1+|x|}$

For $x \geq 0$,

$f(x) = \frac{x}{1+x}$

For $x < 0$,

$f(x) = \frac{x}{1-x}$

Both parts are differentiable in their domains.

Check at $x = 0$:

$f'_{-}(0) = \left(\frac{x}{1-x}\right)'_{x=0} = 1$

$f'_{+}(0) = \left(\frac{x}{1+x}\right)'_{x=0} = 1$

Hence differentiable at $x = 0$ also.

Therefore $f(x)$is differentiable for all real numbers.

$\boxed{(-,\infty,\infty)}$

Correct Answer: B)

20. Given

$f(x) = \begin{cases} cx+1, & x \leq 3 \\ cx^2-1, & x > 3 \end{cases}$

For continuity at $x = 3$,

$3c + 1 = 9c - 1$

$2 = 6c$

$c = \frac{1}{3}$

$\boxed{\frac{1}{3}}$

Correct Answer: A) $\frac{1}{3}$

LEVEL – II

SINGLE OPTION CORRECT TYPE

1. Given

$f(x) = x\sin\frac{1}{x}$

At $x = 0$,

$\lim_{x\to 0} x\sin\frac{1}{x} = 0$

so $f(x)$ is continuous at $x = 0$(if $f(0) = 0$).

Now,

$f'(0) = \lim_{h\to 0} \frac{h\sin\frac{1}{h} - 0}{h} = \lim_{h\to 0} \sin\frac{1}{h}$

This limit does not exist.

Hence $f(x)$is continuous but not differentiable at $x = 0$.

Correct answer $\boxed{\textbf{A)}}$

2. $f(x) = \sin 2x - 1$

Since sine function is continuous everywhere, $f(x)$is continuous for all real x.

Therefore continuous at both $x = 0$ and $x = \pi$.

Correct answer $\boxed{\textbf{A) Continuous at } x = 0, \pi}$

3. Given $f(x) = \max\{x, x^3\}$

Compare x and x^3:

$x^3 > x \iff x(x-1)(x+1) > 0$

Hence,

$$f(x) = \begin{cases} x^3, & -1 < x < 0 \text{ or } x > 1 \\ x, & x < -1 \text{ or } 0 < x < 1 \end{cases}$$

Possible non-differentiable points are where $x = x^3$:

$x(x-1)(x+1) = 0$

So points are $x = -1, 0, 1$.

Check derivatives:

At $x = -1$:

Left derivative = 1, right derivative = $3x^2 = 3$

Not equal ⇒not differentiable.

At $x = 0$:

Left derivative = $3x^2 = 0$, right derivative = 1

Not equal ⇒not differentiable.

At $x = 1$:

Left derivative = 1, right derivative = 3

Not equal ⇒not differentiable.

Therefore, the set is

$\{-, 1, 01\}$

Correct Answer: D) $\{-, \mathbf{1}, \mathbf{01}\}$

4. $y = ||x| - 1|$

Write in piecewise form:

$$y = \begin{cases} |x| - 1, & |x| \geq 1 \\ 1 - |x|, & |x| < 1 \end{cases}$$

Non-differentiability occurs where the modulus changes sign:

$|x| - 1 = 0 \Rightarrow |x| = 1 \Rightarrow x = \pm 1$

Also $|x|$ itself is non-differentiable at $x = 0$, but near 0,

$y = 1 - |x|$

which still has cusp at $x = 0$.

Hence non-differentiable points are

$-1, 0, 1$

Correct Answer: A) $\mathbf{-1, 0, 1}$

5. Given: $f(1) = -2$, $f'(x) \geq 2$ for $x \in [1, 6]$

By Mean Value Theorem,

$f(6) - f(1) \geq 2(6 - 1) = 10$

Since $f(1) = -2$,

$f(6) + 2 \geq 10$

$f(6) \geq 8$

Correct Answer: D) $\boldsymbol{f(6) \geq 8}$

6. Given $f(x) = \begin{cases} -x, & x < 0 \\ x^2, & 0 \leq x \leq 1 \\ x^2 - x + 1, & x > 1 \end{cases}$

Check differentiability at $x = 0$ and $x = 1$.

At $x = 0$

Continuity:

$\lim_{x \to 0^-} f(x) = 0$, $\lim_{x \to 0^+} f(x) = 0$, $f(0) = 0$

Continuous.

Derivatives:
LHD:
$\frac{d}{dx}(-x) = -1$
RHD:
$\frac{d}{dx}(x^2) = 2x \Rightarrow 0$
Since $-1 \neq 0$, not differentiable at $x = 0$.
At $x = 1$
Continuity:
$f(1) = 1$
$\lim_{x\to 1^+}(x^2 - x + 1) = 1$
Continuous.
Derivatives:
Left derivative:
$\frac{d}{dx}(x^2) = 2x \Rightarrow 2$
Right derivative:
$\frac{d}{dx}(x^2 - x + 1) = 2x - 1 \Rightarrow 1$
Since $2 \neq 1$, not differentiable at $x = 1$.
Therefore $f(x)$is not differentiable at both $x = 0$and $x = 1$.
Correct Answer: D)

7. Given $f(x) = \begin{cases} \frac{1-\sqrt{2}\sin x}{\pi - 4x}, & x \neq \frac{\pi}{4} \\ a, & x = \frac{\pi}{4} \end{cases}$
For continuity at $x = \frac{\pi}{4}$,
$a = \lim_{x\to \pi/4} \frac{1 - \sqrt{2}\sin x}{\pi - 4x}$
Substituting gives $0/0$, so use L'Hospital Rule:
$a = \lim_{x\to \pi/4} \frac{-\sqrt{2}\cos x}{-4}$
$= \frac{\sqrt{2}\cos(\pi/4)}{4}$
$= \frac{\sqrt{2} \cdot \frac{\sqrt{2}}{2}}{4} = \frac{1}{4}$
Correct Answer: D) $\frac{1}{4}$

8. Given
$f(x) = \begin{cases} ax^2 + b, & 0 \leq x < 1 \\ 4, & x = 1 \\ x + 3, & 1 < x \leq 2 \end{cases}$
For continuity at $x = 1$,
$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^+} f(x) = f(1) = 4$
Left limit:
$a + b$
Right limit:
$1 + 3 = 4$
Hence,

$a + b = 4$

Check options:

A) $(2, 2) \Rightarrow 2 + 2 = 4$

B) $(3, 1) \Rightarrow 3 + 1 = 4$

C) $(4, 0) \Rightarrow 4 + 0 = 4$

D) $(5, 12) \Rightarrow 5 + 12 = 17 \neq 4$

Therefore continuity is not possible for $(5, 12)$.

Correct Answer: D) $(5, 12)$

9. Given

$$f(x) = \begin{cases} \frac{1}{5}(2x^2 + 3), & x \leq 1 \\ 6 - 5x, & 1 < x < 3 \\ x - 3, & x \geq 3 \end{cases}$$

Check possible discontinuity points $x = 1$and $x = 3$.

At $x = 1$

Left value:

$f(1) = \frac{1}{5}(2 + 3) = 1$

Right limit:

$\lim_{x \to 1^+}(6 - 5x) = 1$

Hence continuous at $x = 1$.

At $x = 3$

Left limit:

$\lim_{x \to 3^-}(6 - 5x) = 6 - 15 = -9$

Right limit and value:

$f(3) = 3 - 3 = 0$

Since $-9 \neq 0$, discontinuous at $x = 3$.

Correct Answer: B) $x = 3$

10. Given

$$f(x) = \begin{cases} \frac{x^2 - 1}{x - 1}, & x \neq 1 \\ 2, & x = 1 \end{cases}$$

Simplify for $x \neq 1$:

$\frac{x^2 - 1}{x - 1} = \frac{(x - 1)(x + 1)}{x - 1} = x + 1$

So,

$\lim_{x \to 1} f(x) = \lim_{x \to 1}(x + 1) = 2$

and

$f(1) = 2$

Since limit equals function value, $f(x)$is continuous at $x = 1$.

Correct Answer: B) continuous at $x = 1$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Given

$$f(x) = \begin{cases} |x - 3|, & x \geq 1 \\ \frac{x^2}{4} - \frac{3x}{2} + \frac{13}{4}, & x < 1 \end{cases}$$

At $x = 1$

Left limit:

$\lim_{x\to 1^-} f(x) = \frac{1}{4} - \frac{3}{2} + \frac{13}{4} = \frac{1-6+13}{4} = 2$

Right value:

$f(1) = |\ 1-3\ | = 2$

Hence continuous at $x = 1$.

Now check differentiability.

Left derivative:

$\frac{d}{dx}\left(\frac{x^2}{4} - \frac{3x}{2} + \frac{13}{4}\right) = \frac{x}{2} - \frac{3}{2}$

At $x = 1$:

$\frac{1}{2} - \frac{3}{2} = -1$

Right derivative:

For $1 \le x < 3$,

$|\ x-3\ | = 3-x$

so derivative $= -1$.

Thus LHD = RHD $= -1$.

Hence differentiable at $x = 1$.

At $x = 3$

$f(x) = |\ x-3\ |$

Absolute value function is continuous at $x = 3$, but not differentiable there.

So continuous at $x = 3$, not differentiable at $x = 3$.

Correct Options: A, B, C

12. Given $f(x) = \begin{cases} ax^2+1, & x \le 1 \\ x^2+ax+b, & x > 1 \end{cases}$

For differentiability at $x = 1$:

Continuity condition

$a(1)^2 + 1 = 1 + a + b$

$a + 1 = a + b + 1$

$b = 0$

Derivative condition

Left derivative:

$\frac{d}{dx}(ax^2+1) = 2ax$

At $x = 1$:

$2a$

Right derivative:

$\frac{d}{dx}(x^2+ax+b) = 2x + a$

+

At $x = 1$: $2 + a$

Equate:

$2a = 2 + a$

$a = 2$

Thus,

$a = 2,\ b = 0$

Correct Options: A, B

13. Given

$$f(x)=\begin{cases}-2\sin x, & x\le -\frac{\pi}{2}\\ p\sin x+q, & -\frac{\pi}{2}<x<\frac{\pi}{2}\\ \cos x, & x\ge \frac{\pi}{2}\end{cases}$$

For continuity everywhere, check at boundary points.

At $x=-\frac{\pi}{2}$

Left value:

$-2\sin\left(-\frac{\pi}{2}\right)=2$

Middle limit:

$p\sin\left(-\frac{\pi}{2}\right)+q=-p+q$

So,

$-p+q=2$ (1)

At $x=\frac{\pi}{2}$

Right value:

$\cos\left(\frac{\pi}{2}\right)=0$

Middle limit:

$p\sin\left(\frac{\pi}{2}\right)+q=p+q$

So,

$p+q=0$ (2)

Solving (1) and (2):

$-p+q=2$

$p+q=0$

Add:

$2q=2\Rightarrow q=1$

Then

$p=-1$

Correct Options: C, D

14. Given

$$f(x)=\begin{cases}\frac{\sin(p+1)x+\sin x}{x}, & x<0\\ q, & x=0\\ \frac{\sqrt{x+x^2}-\sqrt{x}}{x^{3/2}}, & x>0\end{cases}$$

For continuity at $x=0$,

$\lim\limits_{x\to 0^-}f(x)=q=\lim\limits_{x\to 0^+}f(x)$

Left hand limit

$\lim\limits_{x\to 0^-}\frac{\sin(p+1)x+\sin x}{x}$

Using

$\lim\limits_{x\to 0}\frac{\sin(ax)}{x}=a$

we get

$(p+1)+1=p+2$

Hence,

LHL $= p + 2$

Right hand limit

$$\lim_{x\to 0^+} \frac{\sqrt{x+x^2}-\sqrt{x}}{x^{3/2}}$$

Factor $\sqrt{x}$:

$\sqrt{x+x^2} = \sqrt{x(1+x)} = \sqrt{x}\sqrt{1+x}$

So,

$$= \lim_{x\to 0^+} \frac{\sqrt{x}\left(\sqrt{1+x}-1\right)}{x^{3/2}}$$

$$= \lim_{x\to 0^+} \frac{\sqrt{1+x}-1}{x}$$

Rationalize:

$$= \lim_{x\to 0^+} \frac{(1+x)-1}{x\left(\sqrt{1+x}+1\right)}$$

$$= \lim_{x\to 0^+} \frac{1}{\sqrt{1+x}+1} = \frac{1}{2}$$

Thus, $q = \frac{1}{2}$

and

$$p + 2 = \frac{1}{2}$$

$$p = -\frac{3}{2}$$

Correct Options: A, B

COMPREHENSION TYPE

PASSAGE

15. Find $f'(0)$

Given

$$f(x) = \frac{x}{1+|x|}$$

At $x = 0$,

$f(0) = 0$

Using definition of derivative:

$$f'(0) = \lim_{x\to 0} \frac{f(x)-f(0)}{x-0} = \lim_{x\to 0} \frac{\frac{x}{1+|x|}}{x} = \lim_{x\to 0} \frac{1}{1+|x|} = 1$$

∴ Correct answer: B) 1

16. Check differentiability at $x = 0$ and $x = 1$

Given

$$f(x) = \begin{cases} -x, & x < 0 \\ x^2, & 0 \le x \le 1 \\ 3x^2 - x + 1, & x > 1 \end{cases}$$

At $x = 0$

Left derivative:

$$f'_-(0) = \frac{d}{dx}(-x) = -1$$

Right derivative:

$$f'_+(0) = \frac{d}{dx}(x^2)\,|_{x=0} = 0$$

Since LHD ≠RHD,

$f'(0)$ does not exist

At $x = 1$

Left derivative:

$f'_-(1) = \frac{d}{dx}(x^2)\ |_{x=1} = 2$

Right derivative:

$f'_+(1) = \frac{d}{dx}(3x^2 - x + 1)\ |_{x=1} = 6(1) - 1 = 5$

Since LHD ≠RHD,

$f'(1)$ does not exist

∴ Correct answer: D) $f'(0), f'(1)$does not exist

17. Find $g'(x)$

Given

$f(x) = |\ x - 2\ |$

And $g(x) = f\big(f(x)\big)$

For $2 < x < 4$,

$f(x) = x - 2$

Hence

$g(x) = f(x - 2) = |\ (x - 2) - 2\ | = |\ x - 4\ |$

Since $2 < x < 4$,

$x - 4 < 0$

So

$g(x) = 4 - x$

Therefore,

$g'(x) = -1$

Correct answer: A) -1

MATRIX MATCHING TYPE

18. Given

$$f(x) = \begin{cases} ax(x-1) + b, & x < 1 \\ x - 1, & 1 \le x \le 3 \\ px^2 + qx + 2, & x > 3 \end{cases}$$

Conditions:

1. $f(x)$is continuous for all x
2. $f(x)$is not differentiable at $x = 1$
3. $f'(x)$is continuous at $x = 3$

Step 1: Continuity at $x = 1$

For continuity,

$\lim_{x \to 1^-} f(x) = f(1)$

Since middle function gives

$f(1) = 1 - 1 = 0$

Left limit:

$a(1)(1 - 1) + b = b$

Hence

$b = 0$

So, $\boxed{b = 0}$

Step 2: Not differentiable at $x = 1$

Left derivative:

$\frac{d}{dx}[ax(x - 1) + b] = a(2x - 1)$

At $x = 1$,

$f'_-(1) = a$

Right derivative of $x - 1$ is

$f'_+(1) = 1$

Not differentiable means

$a \neq 1$

So, $\boxed{a \neq 1}$

Step 3: Continuity at $x = 3$

Middle function:

$f(3) = 3 - 1 = 2$

Right-side value:

$9p + 3q + 2$

Continuity gives

$9p + 3q + 2 = 2$

$9p + 3q = 0$

$3p + q = 0 \quad (1)$

Step 4: $f'(x)$continuous at $x = 3$

Left derivative:

$\frac{d}{dx}(x - 1) = 1$

Right derivative:

$\frac{d}{dx}(px^2 + qx + 2) = 2px + q$

At $x = 3$,

$6p + q = 1 \quad (2)$

From (1),

$q = -3p$

Substitute into (2):

$6p - 3p = 1$

$3p = 1$

$p = \frac{1}{3}$

Then $q = -1$

Matching

a) $a \neq 1 \rightarrow$ S

b) $b = 0 \rightarrow$ R

c) $p = \frac{1}{3} \rightarrow$ P

d) $q = -1 \rightarrow$ Q

INTEGER TYPE

19. Number of discontinuities of $\tan x$ in $(-\frac{\pi}{2}, \frac{\pi}{2})$

$\tan x$ is discontinuous where

$\cos x = 0$

i.e. $x = \frac{\pi}{2} + n\pi$

But the interval is open:

$(-\frac{\pi}{2}, \frac{\pi}{2})$

The points $\pm\frac{\pi}{2}$are not included.

Hence there is no discontinuity inside the interval.

$\boxed{0}$

20. Find $(a+b)$

Given

$$f(x)=\begin{cases}\frac{1}{|x|}, & |x|\geq 1\\ ax^2+b, & |x|<1\end{cases}$$

For continuity and differentiability, check at $x=\pm 1$.

Continuity at $x=1$

From right:

$f(1)=\frac{1}{|1|}=1$

From left:

$a+b$

So,

$a+b=1 \qquad (1)$

Differentiability at $x=1$

For $x>0$,

$\frac{1}{|x|}=\frac{1}{x}$

Derivative:

$-\frac{1}{x^2}$

At $x=1$,

$f'_+(1)=-1$

For ax^2+b,

$f'_-(1)=2a$

Equating,

$2a=-1$

$a=-\frac{1}{2}$

Using (1),

$-\frac{1}{2}+b=1$

$b=\frac{3}{2}$

Therefore,

$a+b=1$

$\boxed{1}$

STUDENT PRACTISE CORNER

SINGLE CORRECT ANSWER TYPE

1. $f(x)=\frac{1}{x-3}$ is discontinuous at

A rational function is discontinuous where the denominator becomes zero.

$x-3=0\Rightarrow x=3$

Correct Answer: C) $x=3$

2. $f(x)=\sqrt{x-5}$ is continuous on

For square root functions, the expression inside the root must be non-negative.

$x-5\geq 0\Rightarrow x\geq 5$

So the function is defined and continuous on:

$[5,\infty)$

Correct Answer: B) $[5,\infty)$

3. $f(x) = \begin{cases} x^2, & x \geq 1 \\ 2x - 1, & x < 1 \end{cases}$

Check continuity and differentiability at $x = 1$.

LHL:

$\lim_{x \to 1^-} (2x - 1) = 2(1) - 1 = 1$

RHL:

$\lim_{x \to 1^+} x^2 = 1$

Also,

$f(1) = 1^2 = 1$

Hence continuous.

Now derivatives:

Left derivative:

$\frac{d}{dx}(2x - 1) = 2$

Right derivative:

$\frac{d}{dx}(x^2) = 2x$

At $x = 1$,

$2x = 2$

Both derivatives are equal, so function is differentiable.

Correct Answer: A) differentiable

4. Find k so that

$f(x) = \begin{cases} x^2 + k, & x \leq 1 \\ 2x + 1, & x > 1 \end{cases}$

is continuous at $x = 1$.

For continuity:

LHL = RHL = $f(1)$

From left side:

$1^2 + k = 1 + k$

From right side:

$2(1) + 1 = 3$

Equating:

$1 + k = 3$

$k = 2$

Correct Answer: B) 2

5. $f(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$

Check continuity at $x = 0$.

$\lim_{x \to 0} \frac{\sin x}{x} = 1$

and

$f(0) = 1$

Since limit =function value, the function is continuous at $x = 0$.

Correct Answer: B) continuous at $x = 0$

6. The function $f(x) = x^2$ is differentiable at

Polynomial functions are differentiable for every real number.

$f'(x) = 2x$

exists for all $x \in \mathbb{R}$.

Correct Answer: C) all real numbers

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

7. $f(x) = 1+|\sin x|$

$|\sin x|$ is continuous for all x, so $f(x)$ is continuous everywhere.

At $x = 0$,

$f(x) = 1+|\sin x|$

Near 0, $\sin x \approx x$, so behavior is like $1+|x|$, which is not differentiable at $x = 0$.

Hence:

continuous everywhere

not differentiable at $x = 0$

Correct Answers: B, D

8. Continuous but not differentiable at $x = 0$

A) $|x|$

Continuous at 0, but left and right derivatives are different.

Correct

B) $x^{2/3}$

Continuous at 0, but derivative becomes infinite at 0.

Correct

C) x^2

Differentiable at 0.

Incorrect

D) $\sin x$

Differentiable everywhere.

Incorrect

Correct Answers: A, B

9. Functions differentiable for all real numbers

A) x^3

Polynomial ⇒ differentiable everywhere.

Correct

B) $\sin x$

Differentiable everywhere.

Correct

C) $|x|$

Not differentiable at $x = 0$.

Incorrect

D) e^x

Differentiable everywhere.

Correct

Correct Answers: A, B, D

COMPREHENSION TYPE

PASSAGE

Given $f(x) = x^2 + ax + 3$, $g(x) = x + b$

and

$$F(x) = \lim_{n\to\infty} \frac{f(x) + x^{2n} g(x)}{1 + x^{2n}}$$

As $n \to \infty$:

If $|x| < 1$, then $x^{2n} \to 0$, so

$F(x) = f(x)$

If $|x| > 1$, then $x^{2n} \to \infty$, so

$F(x) = g(x)$
For continuity at $x = \pm 1$, we need
$f(\pm 1) = g(\pm 1)$

10. Continuity at $x = 1$
$f(1) = 1 + a + 3 = a + 4$
$g(1) = 1 + b$
For continuity:
$a + 4 = 1 + b$
$b = a + 3$
Also,
$a = b - 3$
Hence only option A is correct.
Correct Answer: A) $b = a + 3$

11. Continuity at $x = -1$
$f(-1) = 1 - a + 3 = 4 - a$
$g(-1) = -1 + b = b - 1$
For continuity:
$4 - a = b - 1$
$a + b = 5$
Correct Answer: C) $a + b = 5$

12. Continuity at $x = \pm 1$
From Q10:
$b = a + 3$
From Q11:
$a + b = 5$
Substitute:
$a + (a + 3) = 5$
$2a = 2 \Rightarrow a = 1$
$b = 4$
Now,
$f(x) = x^2 + x + 3$
$g(x) = x + 4$
Equation $f(x) = g(x)$:
$x^2 + x + 3 = x + 4$
$x^2 - 1 = 0$
$(x - 1)(x + 1) = 0$
Roots are 1and -1, which are of opposite signs.
Correct Answer: D) roots of opposite signs

MATRIX MATCHING TYPE

13. (A) Progressive derivative / Right hand derivative → (P)
The right hand derivative is calculated by approaching a from the right side $(h \to 0^+)$.
$f'_+(a) = \lim_{h \to 0^+} \frac{f(a+h) - f(a)}{h}$
Hence, (A) matches with (P).
(B) Regressive derivative / Left hand derivative → (Q)
The left hand derivative is calculated by approaching $a$from the left side $(h \to 0^-)$.
$f'_-(a) = \lim_{h \to 0^-} \frac{f(a+h) - f(a)}{h}$

Hence, (B) matches with (Q).

(C) Function is differentiable at $a\rightarrow$ (R)

A function is differentiable at a point when both one-sided derivatives exist and are equal.

$LHD = RHD$

Hence, (C) matches with (R).

(D) Function is not differentiable at $a\rightarrow$ (S)

If the left hand derivative and right hand derivative are not equal, then derivative does not exist at that point.

$LHD \neq RHD$

Hence, (D) matches with (S).

INTEGER TYPE

14. Number of points where $f(x) = |\,|\,x\,| - 1\,|$ is not differentiable

Given,

$f(x) = |\,|\,x\,| - 1\,|$

First write it in piecewise form.

$|\,x\,| - 1 = \begin{cases} x - 1, & x \geq 0 \\ -x - 1, & x < 0 \end{cases}$

Now the outer modulus changes sign when

$|\,x\,| - 1 = 0 \Rightarrow |\,x\,| = 1 \Rightarrow x = \pm 1$

So critical points are:

$x = -1, 0, 1$

At these points, the graph has sharp corners/cusps, hence function is not differentiable.

Therefore, the number of non-differentiable points is $\boxed{3}$

15. Continuity at $x = 1$

Given,

$f(x) = \begin{cases} x^2 + k, & x < 1 \\ 2x, & x \geq 1 \end{cases}$

For continuity at $x = 1$,

LHL = RHL = $f(1)$

Left hand limit:

$\lim\limits_{x \to 1^-} (x^2 + k) = 1 + k$

Right hand limit:

$\lim\limits_{x \to 1^+} 2x = 2$

Equating,

$1 + k = 2$

$k = 1$

Hence, $\boxed{k = 1}$

5. Methods of Differentiation

LEVEL – I

1. Differentiate $3\sqrt[3]{x}$

$y = 3x^{1/3}$

Using power rule,

$\frac{dy}{dx} = 3 \cdot \frac{1}{3}x^{-2/3} = x^{-2/3}$

$= \frac{1}{x^{2/3}}$

Correct Answer: A) $x^{-2/3}$

2. Differentiability at $x = 1$

Given

$$f(x) = \begin{cases} x^2 + bx + c, & x < 1 \\ x, & x \geq 1 \end{cases}$$

For differentiability, continuity and derivatives must be equal.

Continuity at $x = 1$

$1 + b + c = 1$

$b + c = 0 \ldots (1)$

Equality of derivatives

Left derivative:

$\frac{d}{dx}(x^2 + bx + c) = 2x + b$

At $x = 1$:

$2 + b$

Right derivative:

$\frac{d}{dx}(x) = 1$

So,

$2 + b = 1$

$b = -1$

From (1),

$-1 + c = 0 \Rightarrow c = 1$

Therefore,

$b - c = -1 - 1 = -2$

Correct Answer: B) -2

3. Find $f'(1)$

$$f(x) = \begin{cases} \frac{x-1}{2x^2 - 7x + 5}, & x \neq 1 \\ -\frac{1}{3}, & x = 1 \end{cases}$$

Factor denominator:

$2x^2 - 7x + 5 = (2x - 5)(x - 1)$

So for $x \neq 1$,

$f(x) = \frac{1}{2x - 5}$

Differentiate:

$f'(x) = \frac{-2}{(2x - 5)^2}$

At $x = 1$,

$f'(1) = \frac{-2}{(-3)^2} = -\frac{2}{9}$

Correct Answer: D) $-\frac{2}{9}$

4. **Find $f'(x)$**

$f(x) = x - x^2 + x^3 - x^4 + \cdots$

This is a GP with first term $x$and common ratio $-x$.

$f(x) = \frac{x}{1+x}, \ |x| < 1$

Differentiate:

$f'(x) = \frac{(1+x) - x}{(1+x)^2}$

$f'(x) = \frac{1}{(1+x)^2}$

Correct Answer: B) $\frac{1}{(1+x)^2}$

5. Evaluate:

$\frac{d}{dx}\left(\log\left(x + \sqrt{x^2+1}\right)\right)$

Step-by-step Solution:

Let $y = \log\left(x + \sqrt{x^2+1}\right)$

Using chain rule,

$\frac{dy}{dx} = \frac{1}{x + \sqrt{x^2+1}} \cdot \frac{d}{dx}\left(x + \sqrt{x^2+1}\right)$

Differentiate inside term:

$\frac{d}{dx}\left(x + \sqrt{x^2+1}\right) = 1 + \frac{1}{2\sqrt{x^2+1}} \cdot 2x$

$= 1 + \frac{x}{\sqrt{x^2+1}}$

Taking LCM,

$= \frac{\sqrt{x^2+1} + x}{\sqrt{x^2+1}}$

Substitute back:

$\frac{dy}{dx} = \frac{1}{x + \sqrt{x^2+1}} \cdot \frac{\sqrt{x^2+1} + x}{\sqrt{x^2+1}}$

Cancel common terms:

$\frac{dy}{dx} = \frac{1}{\sqrt{x^2+1}}$

$= \frac{1}{\sqrt{1+x^2}}$

Correct Answer: $\boxed{\text{B) } \frac{1}{\sqrt{1+x^2}}}$

6. Given that $y = 3^{\log_9(1+\tan^2 x)}$.find $\frac{dy}{dx}$.

Step-by-step Solution:

Using identity,

$1 + \tan^2 x = \sec^2 x$

So,

$y = 3^{\log_9(\sec^2 x)}$

Since,

$9 = 3^2$

Use change of base:

$\log_9(\sec^2 x) = \frac{\log_3(\sec^2 x)}{\log_3 9} = \frac{\log_3(\sec^2 x)}{2}$

Hence,

$y = 3^{\frac{1}{2}\log_3(\sec^2 x)}$

Using logarithm property,

$\frac{1}{2}\log_3(\sec^2 x) = \log_3(\sec x)$

Therefore,

$y = 3^{\log_3(\sec x)}$

Using

$a^{\log_a b} = b$

we get $y = \sec x$

Differentiate:

$\frac{dy}{dx} = \sec x \tan x$

Correct Answer: C) $\sec x \tan x$

7. **If $f(x) = \sqrt{ax} + \frac{a^2}{\sqrt{ax}}$**

Write in powers:

$f(x) = a^{1/2}x^{1/2} + a^{3/2}x^{-1/2}$

Differentiate:

$f'(x) = \frac{a^{1/2}}{2\sqrt{x}} - \frac{a^{3/2}}{2x^{3/2}}$

At $x = a$,

$f'(a) = \frac{\sqrt{a}}{2\sqrt{a}} - \frac{a^{3/2}}{2a^{3/2}}$

$= \frac{1}{2} - \frac{1}{2} = 0$

Correct Answer: B) 0

8. Evaluate: $\frac{d}{dx}(\cos x^\circ)$

Step-by-step Solution:

Since $x^\circ$is in degrees,

$x^\circ = \frac{\pi x}{180}$ radians

So,

$\cos x^\circ = \cos\left(\frac{\pi x}{180}\right)$

Differentiate using chain rule:

$\frac{d}{dx}\left[\cos\left(\frac{\pi x}{180}\right)\right] = -\sin\left(\frac{\pi x}{180}\right) \cdot \frac{\pi}{180}$

Therefore,

$\frac{d}{dx}(\cos x^\circ) = -\frac{\pi}{180}\sin x^\circ$

Correct Answer: B) $-\frac{\pi}{180}\sin x^\circ$

9. If
$e^x + e^y = e^{x+y}$
find $\frac{dy}{dx}$at $(1, 1)$.
Step-by-step Solution:
Differentiate implicitly:
$e^x + e^y \frac{dy}{dx} = e^{x+y}\left(1 + \frac{dy}{dx}\right)$
At $(1, 1)$,
$e + e\frac{dy}{dx} = e^2\left(1 + \frac{dy}{dx}\right)$
Divide by e:
$1 + \frac{dy}{dx} = e\left(1 + \frac{dy}{dx}\right)$
Since $e \neq 1$,
$1 + \frac{dy}{dx} = 0$
$\frac{dy}{dx} = -1$
Correct Answer: B) -1

10. If $\sqrt{x} + \sqrt{y} = 4$. find $\frac{dx}{dy}$at $y = 1$.
Step-by-step Solution:
Differentiate implicitly with respect to y:
$\frac{1}{2\sqrt{x}}\frac{dx}{dy} + \frac{1}{2\sqrt{y}} = 0$
Multiply by 2:
$\frac{1}{\sqrt{x}}\frac{dx}{dy} + \frac{1}{\sqrt{y}} = 0$
$\frac{dx}{dy} = -\frac{\sqrt{x}}{\sqrt{y}}$
At $y = 1$,
$\sqrt{x} + 1 = 4$
$\sqrt{x} = 3$
Hence,
$\frac{dx}{dy} = -\frac{3}{1} = -3$
Correct Answer: B) -3

11. If
$Q(x) = \log_5(\log_3 x)$
find $Q'(e)$.
Step-by-step Solution:
Using change of base,
$Q(x) = \frac{\ln(\log_3 x)}{\ln 5}$
Differentiate:
$Q'(x) = \frac{1}{\ln 5} \cdot \frac{1}{\log_3 x} \cdot \frac{d}{dx}(\log_3 x)$
Now,

$\frac{d}{dx}(\log_3 x) = \frac{1}{x \ln 3}$
Also,
$\log_3 x = \frac{\ln x}{\ln 3}$
Substitute:
$Q'(x) = \frac{1}{\ln 5} \cdot \frac{\ln 3}{\ln x} \cdot \frac{1}{x \ln 3}$
$Q'(x) = \frac{1}{x \ln x \ln 5}$
At $x = e$,
$\ln e = 1$
Therefore,
$Q'(e) = \frac{1}{e \ln 5}$
Correct Answer:
C) $\frac{1}{e \log 5}$

12. If $x = \sin\theta + \theta\cos\theta$, $y = \cos\theta - \theta\sin\theta$.find $\left(\frac{dy}{dx}\right)_{\theta=\frac{\pi}{2}}$

Step-by-step Solution:
For parametric equations,
$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$
Differentiate x:
$\frac{dx}{d\theta} = \cos\theta + (\cos\theta - \theta\sin\theta)$
$= 2\cos\theta - \theta\sin\theta$
Differentiate y:
$\frac{dy}{d\theta} = -\sin\theta - (\sin\theta + \theta\cos\theta)$
$= -2\sin\theta - \theta\cos\theta$
At $\theta = \frac{\pi}{2}$,
$\sin\frac{\pi}{2} = 1,\ \cos\frac{\pi}{2} = 0$
So, $\frac{dx}{d\theta} = -\frac{\pi}{2}$
and
$\frac{dy}{d\theta} = -2$
Hence,
$\frac{dy}{dx} = \frac{-2}{-\pi/2} = \frac{4}{\pi}$
Correct Answer: D) $\frac{4}{\pi}$

13. If $y = x^x$.find $\frac{dy}{dx}$.
Step-by-step Solution:
Take logarithm on both sides:
$\ln y = \ln(x^x)$

$\ln y = x \ln x$
Differentiate implicitly:
$\frac{1}{y}\frac{dy}{dx} = \ln x + x \cdot \frac{1}{x}$
$= \ln x + 1$
Multiply by $y = x^x$:
$\frac{dy}{dx} = x^x(1 + \ln x)$
Correct Answer:
B) $x^x(1 + \log x)$

14. If $y = 2^{2^x}$.find $\frac{dy}{dx}$.
Step-by-step Solution:
Using derivative of a^u,
$\frac{d}{dx}(a^u) = a^u \ln(a) \cdot \frac{du}{dx}$

Here, $y = 2^{2^x}$
So,
$\frac{dy}{dx} = 2^{2^x} \ln 2 \cdot \frac{d}{dx}(2^x)$
Now,
$\frac{d}{dx}(2^x) = 2^x \ln 2$
Therefore,
$\frac{dy}{dx} = 2^{2^x} \ln 2 \cdot 2^x \ln 2$
$= y\, 2^x (\ln 2)^2$
Correct Answer: C) $y\, 2^x (\log_e 2)^2$

15. If $f(x) = \frac{x}{1+x}$ and $g(x) = f[f(x)]$ find $g'(x)$.
Step-by-step Solution:
First find $g(x)$:
$g(x) = f\left(\frac{x}{1+x}\right)$
Since
$f(t) = \frac{t}{1+t}$
put
$t = \frac{x}{1+x}$
Then,
$g(x) = \frac{\frac{x}{1+x}}{1 + \frac{x}{1+x}}$
Simplify denominator:
$1 + \frac{x}{1+x} = \frac{1+x+x}{1+x} = \frac{1+2x}{1+x}$
Hence,
$g(x) = \frac{x}{1+2x}$
Differentiate:

$g'(x) = \frac{(1+2x)-2x}{(1+2x)^2}$

$= \frac{1}{(2x+1)^2}$

Correct Answer: D) $\frac{1}{(2x+1)^2}$

16. If $x = a\cos^3\theta$, $y = a\sin^3\theta$.find $1 + \left(\frac{dy}{dx}\right)^2$

Step-by-step Solution:

For parametric equations,

$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$

Differentiate:

$\frac{dx}{d\theta} = a \cdot 3\cos^2\theta(-\sin\theta) = -3a\cos^2\theta\sin\theta$

$\frac{dy}{d\theta} = a \cdot 3\sin^2\theta\cos\theta = 3a\sin^2\theta\cos\theta$

Therefore,

$\frac{dy}{dx} = \frac{3a\sin^2\theta\cos\theta}{-3a\cos^2\theta\sin\theta}$

$= -\tan\theta$

So,

$\left(\frac{dy}{dx}\right)^2 = \tan^2\theta$

Hence,

$1 + \left(\frac{dy}{dx}\right)^2 = 1 + \tan^2\theta = \sec^2\theta$

Correct Answer: D) $\sec^2\theta$

17. If $y = f(x^2+2)$ and $f'(3) = 5$.find $\frac{dy}{dx}$at $x = 1$.

Step-by-step Solution:

Using chain rule,

$\frac{dy}{dx} = f'(x^2+2) \cdot \frac{d}{dx}(x^2+2)$

$\frac{dy}{dx} = f'(x^2+2) \cdot 2x$

At $x = 1$,

$x^2 + 2 = 1 + 2 = 3$

Given,

$f'(3) = 5$

Hence, $\frac{dy}{dx} = 5 \cdot 2 \cdot 1 = 10$

Correct Answer: C) 10

18. Given:

$x^m y^n = (x+y)^{m+n}$

To find: $\frac{dy}{dx}$

Take the natural logarithm on both sides:

$\ln(x^m y^n) = \ln((x+y)^{m+n})$

Using logarithm properties,

$m\ln x + n\ln y = (m+n)\ln(x+y)$
Differentiate implicitly:
$\frac{m}{x} + \frac{n}{y}\frac{dy}{dx} = \frac{m+n}{x+y}\left(1 + \frac{dy}{dx}\right)$
Expand:
$\frac{m}{x} + \frac{n}{y}\frac{dy}{dx} = \frac{m+n}{x+y} + \frac{m+n}{x+y}\frac{dy}{dx}$
Bring $\frac{dy}{dx}$terms together:
$\frac{n}{y}\frac{dy}{dx} - \frac{m+n}{x+y}\frac{dy}{dx} = \frac{m+n}{x+y} - \frac{m}{x}$
Factor:
$\left(\frac{n}{y} - \frac{m+n}{x+y}\right)\frac{dy}{dx} = \frac{m+n}{x+y} - \frac{m}{x}$
Simplify both sides:
$\left(\frac{nx-my}{y(x+y)}\right)\frac{dy}{dx} = \frac{nx-my}{x(x+y)}$
Cancel common terms:
$\frac{1}{y}\frac{dy}{dx} = \frac{1}{x}$
Hence,
$\frac{dy}{dx} = \frac{y}{x}$
Correct Option: $\boxed{\text{D) } \frac{y}{x}}$

19. Given:
$\sin y = x\sin(a+y)$
To find: $\frac{dy}{dx}$
Express x in terms of y:
$x = \frac{\sin y}{\sin(a+y)}$
Differentiate with respect to $y$using quotient rule:
$\frac{dx}{dy} = \frac{\sin(a+y)\cos y - \sin y\cos(a+y)}{\sin^2(a+y)}$
Using identity,
$\sin A\cos B - \cos A\sin B = \sin(A-B)$
with $A = a+y$, $B = y$,
$\sin(a+y)\cos y - \sin y\cos(a+y) = \sin a$
Therefore,
$\frac{dx}{dy} = \frac{\sin a}{\sin^2(a+y)}$
Take reciprocal:
$\frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$
Correct Option: $\boxed{\text{B) } \frac{\sin^2(a+y)}{\sin a}}$

20. Given:
$x^y = e^{x-y}$
To find: $\frac{dy}{dx}$

Take natural logarithm on both sides:

$\ln(x^y) = \ln(e^{x-y})$

$y \ln x = x - y$

Rearrange:

$y \ln x + y = x$

$y(1 + \ln x) = x$

$y = \dfrac{x}{1 + \ln x}$

Differentiate using quotient rule:

$\dfrac{dy}{dx} = \dfrac{(1 + \ln x)(1) - x\left(\frac{1}{x}\right)}{(1 + \ln x)^2}$

$= \dfrac{1 + \ln x - 1}{(1 + \ln x)^2}$

$= \dfrac{\ln x}{(1 + \ln x)^2}$

Correct Option: D) $\dfrac{\log x}{(1+\log x)^2}$

LEVEL – II

SINGLE OPTION CORRECT TYPE

1. Given:

$x = a\left(\cos\theta + \log\tan\dfrac{\theta}{2}\right),\ y = a\sin\theta$

To find: $\frac{dy}{dx}$

Differentiate with respect to θ:

$\dfrac{dx}{d\theta} = a\left(-\sin\theta + \dfrac{d}{d\theta}\log\tan\dfrac{\theta}{2}\right)$

Using standard result,

$\dfrac{d}{d\theta}\left(\log\tan\dfrac{\theta}{2}\right) = \csc\theta$

So,

$\dfrac{dx}{d\theta} = a(-\sin\theta + \csc\theta)$

$= a\left(\dfrac{1 - \sin^2\theta}{\sin\theta}\right) = a\dfrac{\cos^2\theta}{\sin\theta}$

Now,

$\dfrac{dy}{d\theta} = a\cos\theta$

Hence,

$\dfrac{dy}{dx} = \dfrac{a\cos\theta}{a\cos^2\theta/\sin\theta}$

$= \dfrac{\sin\theta}{\cos\theta} = \tan\theta$

Correct Option: B) $\tan\theta$

2. Given:

$x = 3\cos\theta - 2\cos^3\theta,\ y = 3\sin\theta - 2\sin^3\theta$

To find: $\frac{dy}{dx}$

Differentiate:

$\frac{dx}{d\theta} = -3\sin\theta + 6\cos^2\theta \sin\theta$
$= 3\sin\theta\,(2\cos^2\theta - 1)$
Using,
$2\cos^2\theta - 1 = \cos 2\theta$
$\frac{dx}{d\theta} = 3\sin\theta\cos 2\theta$
Now,
$\frac{dy}{d\theta} = 3\cos\theta - 6\sin^2\theta\cos\theta$
$= 3\cos\theta\,(1 - 2\sin^2\theta)$
Using,
$1 - 2\sin^2\theta = \cos 2\theta$
$\frac{dy}{d\theta} = 3\cos\theta\cos 2\theta$
Therefore,
$\frac{dy}{dx} = \frac{3\cos\theta\cos 2\theta}{3\sin\theta\cos 2\theta}$
$= \cot\theta$
Correct Option: B) $\cot\theta$

3. Given:
$y = (1+x)(1+x^2)(1+x^4)$
To find: $\frac{dy}{dx}$at $x = 1$
First simplify:
$(1+x)(1+x^2)(1+x^4) = \frac{1-x^8}{1-x}$
So,
$y = 1 + x + x^2 + \cdots + x^7$
Differentiate:
$\frac{dy}{dx} = 1 + 2x + 3x^2 + \cdots + 7x^6$
At $x = 1$,
$\frac{dy}{dx} = 1 + 2 + 3 + 4 + 5 + 6 + 7$
$= \frac{7 \cdot 8}{2} = 28$
Correct Option: B) 28

4. Given:
$3x^2 + 4xy + 2y^2 + x - 8 = 0$
To find:
$\left(\frac{dy}{dx}\right)_{(-1,3)}$
Differentiate implicitly:
$6x + 4\left(x\frac{dy}{dx} + y\right) + 4y\frac{dy}{dx} + 1 = 0$
$6x + 4x\frac{dy}{dx} + 4y + 4y\frac{dy}{dx} + 1 = 0$
Group terms:
$(4x + 4y)\frac{dy}{dx} = -(6x + 4y + 1)$

$\frac{dy}{dx} = -\frac{6x+4y+1}{4x+4y}$

At(-1,3)

$\frac{dy}{dx} = -\frac{6(-1)+4(3)+1}{4(-1)+4(3)}$

$= -\frac{-6+12+1}{-4+12}$

$= -\frac{7}{8}$

Correct Option: $\boxed{\text{B) } -\frac{7}{8}}$

5. Find the derivative of $\sqrt{2}x + 2\sqrt{x} - \frac{1}{\sqrt{x}}$

Step-by-step Solution:

Differentiate term by term:

$\frac{d}{dx}\left(\sqrt{2}\,x\right) = \sqrt{2}$

$\frac{d}{dx}\left(2\sqrt{x}\right) = 2 \cdot \frac{1}{2\sqrt{x}} = \frac{1}{\sqrt{x}}$

Write

$\frac{1}{\sqrt{x}} = x^{-1/2}$

Then,

$\frac{d}{dx}\left(-x^{-1/2}\right) = \frac{1}{2}x^{-3/2} = \frac{1}{2x\sqrt{x}}$

Hence,

$\frac{dy}{dx} = \sqrt{2} + \frac{1}{\sqrt{x}} + \frac{1}{2x\sqrt{x}}$

Factor $\frac{1}{\sqrt{x}}$:

$\frac{dy}{dx} = \sqrt{2} + \frac{1}{\sqrt{x}}\left(1 + \frac{1}{2x}\right)$

Correct Option: $\boxed{\text{D) } \sqrt{2} + \frac{1}{\sqrt{x}}\left(1 + \frac{1}{2x}\right)}$

6. If $f(x) = |\cos x|$.Find $f'\left(\frac{3\pi}{4}\right)$

Step-by-step Solution:

At

$x = \frac{3\pi}{4}$

$\cos\frac{3\pi}{4} = -\frac{1}{\sqrt{2}} < 0$

So,

$|\cos x| = -\cos x$

Differentiate:

$f'(x) = \sin x$

Therefore,

$f'\left(\frac{3\pi}{4}\right) = \sin\frac{3\pi}{4} = \frac{1}{\sqrt{2}}$

Correct Option: $\boxed{\text{B) } \frac{1}{\sqrt{2}}}$

7. If $f(x) = \log|2x|,\ x \neq 0$.find $f'(x)$.

Step-by-step Solution:

Using standard result,

$\frac{d}{dx}(\log|u|) = \frac{1}{u}\frac{du}{dx}$

Here,

$u = 2x$

So,

$f'(x) = \frac{1}{2x} \cdot 2$

$= \frac{1}{x}$

Correct Option: $\boxed{\text{A) } \frac{1}{x}}$

8. Given: $y = \sqrt{\sin x + y}$

To find: $\frac{dy}{dx}$

Step-by-step Solution:

Square both sides:

$y^2 = \sin x + y$

Differentiate implicitly:

$2y\frac{dy}{dx} = \cos x + \frac{dy}{dx}$

Bring derivative terms together:

$2y\frac{dy}{dx} - \frac{dy}{dx} = \cos x$

Factor:

$(2y - 1)\frac{dy}{dx} = \cos x$

Hence,

$\frac{dy}{dx} = \frac{\cos x}{2y - 1}$

Correct Option: $\boxed{\text{A) } \frac{\cos x}{2y-1}}$

9. Given:

$f(x) = \log\left[e^x \left(\frac{x-2}{x+2}\right)^{3/4}\right]$

find $f'(0)$.

Step-by-step Solution:

Using logarithm properties,

$f(x) = \log(e^x) + \log\left(\left(\frac{x-2}{x+2}\right)^{3/4}\right)$

$= x + \frac{3}{4}\log\left(\frac{x-2}{x+2}\right)$

Differentiate:

$f'(x) = 1 + \frac{3}{4}\left(\frac{1}{x-2} - \frac{1}{x+2}\right)$

Simplify:

$f'(x) = 1 + \frac{3}{4} \cdot \frac{(x+2) - (x-2)}{(x-2)(x+2)}$

$= 1 + \frac{3}{4} \cdot \frac{4}{x^2 - 4}$

$= 1 + \frac{3}{x^2 - 4}$

At $x = 0$,

$f'(0) = 1 + \frac{3}{-4}$

$= 1 - \frac{3}{4} = \frac{1}{4}$

Correct Option: $\boxed{\text{A) } \frac{1}{4}}$

10. Given:

$f(x) = \frac{x^2 - x}{x^2 + 2x}$, $x \neq 0, -2$

To find:

$\frac{d}{dx}[f^{-1}(x)]$

Step-by-step Solution:

Simplify $f(x)$:

$f(x) = \frac{x(x-1)}{x(x+2)}$

$= \frac{x-1}{x+2}$

Let

$y = \frac{x-1}{x+2}$

Find inverse function:

$y(x+2) = x - 1$

$yx + 2y = x - 1$

$yx - x = -1 - 2y$

$x(y-1) = -(1+2y)$

$x = \frac{1+2y}{1-y}$

Therefore,

$f^{-1}(x) = \frac{1+2x}{1-x}$

Differentiate:

$\frac{d}{dx}[f^{-1}(x)] = \frac{(1-x)(2) - (1+2x)(-1)}{(1-x)^2}$

$= \frac{2 - 2x + 1 + 2x}{(1-x)^2}$

$= \frac{3}{(1-x)^2}$

Correct Option: $\boxed{\text{B) } \frac{3}{(1-x)^2}}$

ONE OR MORE THAN ONE CORRECT TYPE

11. Let $y = \sqrt{x + \sqrt{x + \sqrt{x + \cdots}}}$

Then, $y = \sqrt{x + y}$

Squaring both sides:

$y^2 = x + y$

Differentiating:

$2y\frac{dy}{dx} = 1 + \frac{dy}{dx}$

$(2y-1)\frac{dy}{dx} = 1$

$\frac{dy}{dx} = \frac{1}{2y-1}$

Also, from

$y^2 - y - x = 0$

we get

$y = \frac{1+\sqrt{1+4x}}{2}$

Hence,

$2y - 1 = \sqrt{1+4x}$

So,

$\frac{dy}{dx} = \frac{1}{\sqrt{1+4x}}$

Also,

$x = y^2 - y$

therefore

$x + 2y = y^2 + y$

and

$\frac{x}{x+2y} = \frac{y^2-y}{y^2+y} = \frac{y-1}{y+1} \neq \frac{1}{2y-1}$

Similarly,

$\frac{y}{2x+y} = \frac{y}{2(y^2-y)+y} = \frac{1}{2y-1}$

Correct options: A, D

12. If $y = \log_{\sin x}(\tan x)$

Using change of base:

$y = \frac{\log(\tan x)}{\log(\sin x)}$

Differentiate:

$\frac{dy}{dx} = \frac{\left(\frac{\sec^2 x}{\tan x}\right)\log(\sin x) - \log(\tan x)\left(\frac{\cos x}{\sin x}\right)}{\left(\log(\sin x)\right)^2}$

At

$x = \frac{\pi}{4}$

$\tan\frac{\pi}{4} = 1,\ \sin\frac{\pi}{4} = \frac{1}{\sqrt{2}}$

$\log\left(\tan\frac{\pi}{4}\right) = \log 1 = 0$

Hence,

$\frac{dy}{dx} = \frac{\left(\frac{2}{1}\right)\log\left(\frac{1}{\sqrt{2}}\right)}{\left(\log\left(\frac{1}{\sqrt{2}}\right)\right)^2}$

Now,

$\log\left(\frac{1}{\sqrt{2}}\right) = -\frac{1}{2}\log 2$

Therefore,

$\frac{dy}{dx} = \frac{2\left(-\frac{1}{2}\log 2\right)}{\left(-\frac{1}{2}\log 2\right)^2} = -\frac{4}{\log 2}$

Correct option: C

13.

$\boxed{y = 10^{10^x}}$

Then,

$\ln y = 10^x \ln 10$

Differentiating,

$\frac{1}{y}\frac{dy}{dx} = \ln 10 \cdot 10^x \ln 10 = 10^x (\ln 10)^2.$

Comparing with

$\frac{1}{y}\frac{dy}{dx} = 10^x \lambda,$

we get

$\boxed{\lambda = (\ln 10)^2.}$

Correct Answer: B

14. Given: $\alpha = \sin^{-1}\left(\frac{3}{5}\right)$

So,

$\sin\alpha = \frac{3}{5}$

Hence, option **A** is correct.

Using

$\sin^2\alpha + \cos^2\alpha = 1$

$\cos\alpha = \sqrt{1 - \left(\frac{3}{5}\right)^2} = \sqrt{1 - \frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$

Thus, option **B** is correct.

Now,

$\tan\alpha = \frac{\sin\alpha}{\cos\alpha} = \frac{3/5}{4/5} = \frac{3}{4}$

So, option **C** is also correct.

Since

$\alpha = \sin^{-1}\left(\frac{3}{5}\right) \neq \frac{\pi}{2}$

option D is incorrect.

Correct Options: $\boxed{A, B, C}$

COMPREHENSION TYPE

Passage

15. Given

$f(x) = \frac{x^{100}}{100} + \frac{x^{99}}{99} + \frac{x^{98}}{98} + \cdots + \frac{x^2}{2} + x + 1$

Differentiate term wise:

$f'(x) = x^{99} + x^{98} + x^{97} + \cdots + x + 1$

At $x = 1$,

$f'(1) = 1 + 1 + 1 + \cdots + 1$

There are 100terms.

$\boxed{f'(1) = 100}$

Correct Answer: A) 100

16. Given

$f(x) = \frac{x^4 - x^2 + 1}{x^2 + \sqrt{3}x + 1}$

Factor numerator:

$x^4 - x^2 + 1 = (x^2 + \sqrt{3}x + 1)(x^2 - \sqrt{3}x + 1)$

Hence,

$f(x) = x^2 - \sqrt{3}x + 1$

Differentiate:

$f'(x) = 2x - \sqrt{3}$

Compare with

$f'(x) = 2ax + b$

So,

$a = 1,\ b = -\sqrt{3}$

Therefore,

$a^2 + b^2 = 1 + 3 = 4$

$\boxed{4}$

Correct Answer: C) 4

17. Given

$y^x - x^y = 1$

Differentiate implicitly:

$\frac{d}{dx}(y^x) - \frac{d}{dx}(x^y) = 0$

Using logarithmic differentiation,

$\frac{d}{dx}(y^x) = y^x\left(\ln y + \frac{x}{y}\frac{dy}{dx}\right)$

and

$\frac{d}{dx}(x^y) = x^y\left(\frac{y}{x} + \ln x\frac{dy}{dx}\right)$

Thus,

$y^x\left(\ln y + \frac{x}{y}\frac{dy}{dx}\right) - x^y\left(\frac{y}{x} + \ln x\frac{dy}{dx}\right) = 0$

At $x = 1$,

$y^1 - 1^y = 1$

$y - 1 = 1 \Rightarrow y = 2$

Substitute $x = 1,\ y = 2$:

$2\left(\ln 2 + \frac{1}{2}\frac{dy}{dx}\right) - 1(2+0) = 0$

$2\ln 2 + \frac{dy}{dx} - 2 = 0$

$\frac{dy}{dx} = 2 - 2\ln 2$

$\boxed{2(1-\ln 2)}$

Correct Answer: A) $2(1-\ln 2)$

MATRIX MATCHING TYPE

(a)

Given:

$x = at^2,\ y = 2at$

$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

$\frac{dy}{dt} = 2a,\ \frac{dx}{dt} = 2at$

$\frac{dy}{dx} = \frac{2a}{2at} = \frac{1}{t}$

At $t = 2$,

$\frac{dy}{dx} = \frac{1}{2}$

$\boxed{a \to Q}$

(b)

Given:

$x = a(1-\cos\theta),\ y = a(\theta + \sin\theta)$

$\frac{dx}{d\theta} = a\sin\theta$

$\frac{dy}{d\theta} = a(1+\cos\theta)$

$\frac{dy}{dx} = \frac{1+\cos\theta}{\sin\theta}$

At $\theta = \frac{\pi}{2}$,

$\frac{dy}{dx} = 1$

$\boxed{b \to R}$

(c)

Given:

$x = 2\cos^4 t,\ y = 2\sin^4 t$

$\frac{dx}{dt} = 2 \cdot 4\cos^3 t(-\sin t) = -8\cos^3 t \sin t$

$\frac{dy}{dt} = 2 \cdot 4\sin^3 t \cos t = 8\sin^3 t \cos t$

$\frac{dy}{dx} = \frac{8\sin^3 t \cos t}{-8\cos^3 t \sin t} = -\tan^2 t$

At $t = \frac{\pi}{4}$,

$\frac{dy}{dx} = -1$

$\boxed{c \to P}$

(d)

Given:

$y^2 - 4x^2 = a(y + 2x),\ y \neq -2x$

$(y - 2x)(y + 2x) = a(y + 2x)$

Since $y \neq -2x$,

$y - 2x = a$

Differentiate:

$\frac{dy}{dx} - 2 = 0$

$\frac{dy}{dx} = 2$

$\boxed{d \to S}$

INTEGER TYPE

19. Given:

$f(x) = \frac{x}{\sqrt{1 - x^2}},\ g(x) = \frac{x}{\sqrt{1 + x^2}}$

First find:

$(f \circ g)(x) = f\big(g(x)\big)$

$= f\left(\frac{x}{\sqrt{1 + x^2}}\right)$

$= \frac{\frac{x}{\sqrt{1 + x^2}}}{\sqrt{1 - \frac{x^2}{1 + x^2}}}$

Now,

$1 - \frac{x^2}{1 + x^2} = \frac{1 + x^2 - x^2}{1 + x^2} = \frac{1}{1 + x^2}$

Hence,

$\sqrt{1 - \frac{x^2}{1 + x^2}} = \frac{1}{\sqrt{1 + x^2}}$

Therefore,

$(f \circ g)(x) = \frac{\frac{x}{\sqrt{1 + x^2}}}{\frac{1}{\sqrt{1 + x^2}}} = x$

So,

$\frac{d}{dx}(f \circ g)(x) = \frac{d}{dx}(x) = 1$

$\boxed{1}$

20. Given:

$f(x) = |\, x - 8 \,|$

and

$g(x) = f\big(f(x)\big)$

For $x > 16$,

$f(x) = x - 8$

Thus,

$g(x) = f(x - 8)$

Since $x > 16 \Rightarrow x - 8 > 8$,

$f(x - 8) = |\, (x - 8) - 8 \,| = |\, x - 16 \,|$

Again $x > 16$, so

$| x - 16 | = x - 16$

Hence,

$g(x) = x - 16$

Therefore,

$g'(x) = 1$

$\boxed{1}$

STUDENT PRACTISE CORNER

SINGLE CORRECT ANSWER TYPE

1. Find the derivative:

$\frac{d}{dx}\left(5\sqrt[4]{x}\right)$

Step-by-step solution:

First, rewrite the radical as a fractional exponent:

$\sqrt[4]{x} = x^{\frac{1}{4}}$

So,

$5\sqrt[4]{x} = 5x^{\frac{1}{4}}$

Apply the power rule:

$\frac{d}{dx}(x^n) = nx^{n-1}$

$\frac{d}{dx}\left(5x^{\frac{1}{4}}\right) = 5 \cdot \frac{1}{4} x^{\frac{1}{4}-1}$

Simplify:

$\frac{1}{4} - 1 = -\frac{3}{4}$

$= \frac{5}{4} x^{-\frac{3}{4}}$

Correct Option:

$\boxed{\textbf{A) } \frac{5}{4} x^{-\frac{3}{4}}}$

2. Find the derivative:

$\frac{d}{dx}\left(\log\left(x + \sqrt{x^2 + 4}\right)\right)$

Step-by-step solution:

Let

$y = \log\left(x + \sqrt{x^2 + 4}\right)$

Using chain rule:

$\frac{dy}{dx} = \frac{1}{x + \sqrt{x^2 + 4}} \cdot \frac{d}{dx}\left(x + \sqrt{x^2 + 4}\right)$

Differentiate the inner function:

$\frac{d}{dx}(x) = 1$

$\frac{d}{dx}\left(\sqrt{x^2 + 4}\right) = \frac{1}{2}(x^2 + 4)^{-1/2} \cdot 2x = \frac{x}{\sqrt{x^2 + 4}}$

Hence,

$\frac{d}{dx}\left(x + \sqrt{x^2 + 4}\right) = 1 + \frac{x}{\sqrt{x^2 + 4}}$

Take LCM:

$= \frac{\sqrt{x^2+4}+x}{\sqrt{x^2+4}}$

Therefore,

$\frac{dy}{dx} = \frac{1}{x+\sqrt{x^2+4}} \cdot \frac{\sqrt{x^2+4}+x}{\sqrt{x^2+4}}$

Cancel common terms:

$= \frac{1}{\sqrt{x^2+4}}$

Correct Option: **A)** $\boxed{\frac{1}{\sqrt{x^2+4}}}$

3. Find the derivative:

$\frac{d}{dx}(\sin x^\circ)$

Step-by-step solution:

Convert degrees into radians:

$x^\circ = \frac{\pi x}{180}$

So,

$y = \sin\left(\frac{\pi x}{180}\right)$

Apply chain rule:

$\frac{dy}{dx} = \cos\left(\frac{\pi x}{180}\right) \cdot \frac{d}{dx}\left(\frac{\pi x}{180}\right)$

$= \cos\left(\frac{\pi x}{180}\right) \cdot \frac{\pi}{180}$

$= \frac{\pi}{180}\cos x^\circ$

Correct Option: **A)** $\boxed{\frac{\pi}{180}\cos x^\circ}$

4. If $y = x^x$ then $\frac{dy}{dx}$ is equal to

Take logarithm on both sides:

$\log y = \log(x^x)$

Using logarithmic property:

$\log y = x \log x$

Differentiate implicitly:

$\frac{1}{y}\frac{dy}{dx} = \frac{d}{dx}(x \log x)$

Using product rule:

$= x \cdot \frac{1}{x} + \log x$

$= 1 + \log x$

Multiply by y:

$\frac{dy}{dx} = y(1 + \log x)$

Since $y = x^x$,

$\frac{dy}{dx} = x^x(1 + \log x)$

Now,

$1 = \log e$

Hence,

$\frac{dy}{dx} = x^x \log(ex)$

Correct Option: **A) $x^x \log(ex)$**

5. If $\sin(x+y) = \log(x+y)$ then $\frac{dy}{dx}$ is equal to

Differentiate implicitly:

$\cos(x+y)\left(1+\frac{dy}{dx}\right) = \frac{1}{x+y}\left(1+\frac{dy}{dx}\right)$

Take common factor:

$\left(1+\frac{dy}{dx}\right)\left[\cos(x+y) - \frac{1}{x+y}\right] = 0$

Therefore,

$1+\frac{dy}{dx} = 0$

$\frac{dy}{dx} = -1$

Correct Option: **D) -1**

6. If $e^{x-y} + xy = 1$,then the value of $\frac{dy}{dx}$ at $(0,0)$ is

Differentiate implicitly:

$e^{x-y}\left(1-\frac{dy}{dx}\right) + x\frac{dy}{dx} + y = 0$

Substitute $(0,0)$:

$e^0(1-y') + 0(y') + 0 = 0$

$1 - y' = 0$

$y' = 1$

Hence,

$\frac{dy}{dx} = 1$

Correct Option: **B) 1**

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

7. If $y = |\cos x| + |\sin x|$,then $\frac{dy}{dx}$ at $x = \frac{2\pi}{3}$ is equal to

At $x = \frac{2\pi}{3}$

$\cos\frac{2\pi}{3} = -\frac{1}{2} < 0$

so

$|\cos x| = -\cos x$

and

$\sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2} > 0$

so

$|\sin x| = \sin x$

Hence,

$y = -\cos x + \sin x$

Differentiate:

$\frac{dy}{dx} = \sin x + \cos x$

At $x = \frac{2\pi}{3}$,

$\frac{dy}{dx} = \frac{\sqrt{3}}{2} - \frac{1}{2} = \frac{\sqrt{3}-1}{2}$

Also,

$\frac{1}{\sqrt{3}+1} = \frac{\sqrt{3}-1}{(\sqrt{3}+1)(\sqrt{3}-1)} = \frac{\sqrt{3}-1}{2}$

Therefore both are correct.

Correct Options: **C and D**

8. If $f(x) = x^3 + \frac{1}{x} + 7$,then which of the following are correct?

Differentiate term wise:

$\frac{d}{dx}(x^3) = 3x^2$

$\frac{d}{dx}\left(\frac{1}{x}\right) = \frac{d}{dx}(x^{-1}) = -x^{-2} = -\frac{1}{x^2}$

$\frac{d}{dx}(7) = 0$

Hence, $f'(x) = 3x^2 - \frac{1}{x^2}$

So:

A is correct

B is incorrect

C is correct

D is incorrect since

$\frac{d}{dx}\left(\frac{1}{x}\right) = -x^{-2}$

Correct Options: **A and C**

9. Let $y = x^2e^x$, then which of the following are correct?

Since the function is a product of two functions, product rule is used.

Product rule:

$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$

Take:

$u = x^2,\ v = e^x$

Then,

$\frac{du}{dx} = 2x,\ \frac{dv}{dx} = e^x$

Apply product rule:

$\frac{dy}{dx} = x^2(e^x) + e^x(2x)$

$= 2xe^x + x^2e^x$

Thus:

A is correct

B is correct

C is incorrect

D is correct

Correct Options: **A, B and D**

COMPREHENSION TYPE

Passage

Given,

$$f(x)=\begin{cases}a^2+e^x, & -\infty<x<0\\ x+2, & 0\le x\le 3\\ c-\dfrac{b^2}{x}, & 3<x<\infty\end{cases}$$

Since $f(x)$is differentiable for every $x\in\mathbb{R}$, it must be continuous and differentiable at $x=0$and $x=3$.

10. Find a

Continuity at $x=0$:

$a^2+e^0=0+2$

$a^2+1=2$

$a^2=1$

Since a is positive,

$a=1$

Correct Option:

B) 1

11. Find b

Differentiability at $x=3$:

For $0\le x\le 3$,

$f(x)=x+2$

so

$f'(x)=1$

For $x>3$,

$f(x)=c-\dfrac{b^2}{x}$

$f'(x)=\dfrac{b^2}{x^2}$

At $x=3$,

$1=\dfrac{b^2}{9}$

$b^2=9$

Since $b>0$,

$b=3$

Correct Option: B) 3

12. Find c

Continuity at $x=3$:

Middle function value:

$f(3)=3+2=5$

Right-hand limit: $c-\frac{b^2}{3}$

Using $b^2=9$,

$c-\dfrac{9}{3}=5$

$c-3=5$

$c=8$

Correct Option: D) 8

MATRIX MATCHING TYPE

(a) Product Rule

Product rule states:

$$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$$

Hence,

$\boxed{a \to s}$

(b) Quotient Rule

Quotient rule states:

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

Hence,

$\boxed{b \to r}$

(c) Chain Rule

Chain rule states:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Hence,

$\boxed{c \to p}$

(d) Term-by-term Differentiation

Term-by-term differentiation states:

$$\frac{d}{dx}(u \pm v) = \frac{du}{dx} \pm \frac{dv}{dx}$$

Hence,

$\boxed{d \to q}$

INTEGER TYPE

14. If

$y = xe^x$

find

$$\left.\frac{dy}{dx}\right|_{x=1}$$

Step-by-step solution:

Using product rule:

$$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$$

Let

$u = x,\ v = e^x$

Then,

$$\frac{du}{dx} = 1,\ \frac{dv}{dx} = e^x$$

Therefore,

$$\frac{dy}{dx} = x(e^x) + e^x(1)$$

$$= e^x(x + 1)$$

At $x = 1$,

$$\left.\frac{dy}{dx}\right|_{x=1} = e(1 + 1)$$

$$= 2e$$

$\boxed{2e}$

15. If $y = x^3 \log x$, find $\frac{dy}{dx}\Big|_{x=1}$

Using product rule:

$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$

Let

$u = x^3,\ v = \log x$

Then,

$\frac{du}{dx} = 3x^2,\ \frac{dv}{dx} = \frac{1}{x}$

Therefore,

$\frac{dy}{dx} = x^3\left(\frac{1}{x}\right) + (\log x)(3x^2)$

$= x^2 + 3x^2 \log x$

At $x = 1$, $\log 1 = 0$

Hence,

$\frac{dy}{dx}\Big|_{x=1} = 1 + 0$

$= 1$

$\boxed{1}$

6. Indefinite Integrals

Level – I

1.

$\int \frac{x^3+5x^2+4x+1}{x^2}\,dx$

First simplify:

$=\int \left(x+5+\frac{4}{x}+\frac{1}{x^2}\right)dx$

Integrating termwise:

$=\frac{x^2}{2}+5x+4\ln|x|-\frac{1}{x}+C$

So the correct answer is:

Correct answer: B

2. $\int \frac{dx}{\sin^2 x\cos^2 x}$

Use:

$\frac{1}{\sin^2 x\cos^2 x}=\sec^2 x+\csc^2 x$

Hence,

$\int (\sec^2 x+\csc^2 x)\,dx$

$=\tan x-\cot x+C$

So the correct answer is:

Correct answer: B

3. $\int \frac{2^x+3^x}{5^x}\,dx$

Rewrite:

$=\int \left(\left(\frac{2}{5}\right)^x+\left(\frac{3}{5}\right)^x\right)dx$

Using

$\int a^x dx=\frac{a^x}{\ln a}+C$

we get

$=\frac{\left(\frac{2}{5}\right)^x}{\ln\left(\frac{2}{5}\right)}+\frac{\left(\frac{3}{5}\right)^x}{\ln\left(\frac{3}{5}\right)}+C$

So the correct answer is:

Correct answer: C

4.

$\int \frac{a^x+b^x}{a^x b^x}\,dx$

Simplify:

$=\int \left(\frac{1}{b^x}+\frac{1}{a^x}\right)dx$

$=\int \left(\left(\frac{1}{b}\right)^x+\left(\frac{1}{a}\right)^x\right)dx$

Using

$\int r^x dx=\frac{r^x}{\ln r}+C$

we get

$$= \frac{\left(\frac{a}{b}\right)^x}{\ln\left(\frac{a}{b}\right)} + \frac{\left(\frac{b}{a}\right)^x}{\ln\left(\frac{b}{a}\right)} + 2x + C$$

So the correct answer is:

Correct answer: D

5.

$$\int \frac{e^{5\log_e x} - e^{4\log_e x}}{e^{3\log_e x} - e^{2\log_e x}}\, dx$$

Use:

$e^{k\log_e x} = x^k$

Then

$$= \int \frac{x^5 - x^4}{x^3 - x^2}\, dx$$

Factor:

$$= \int \frac{x^4(x-1)}{x^2(x-1)}\, dx$$

$$= \int x^2\, dx$$

$$= \frac{x^3}{3} + C$$

So the correct answer is:

Correct answer: A

6.

$$\int \frac{\sin^3 x + \cos^3 x}{\sin^2 x \cos^2 x}\, dx$$

Simplify:

$$= \int \left(\frac{\sin x}{\cos^2 x} + \frac{\cos x}{\sin^2 x}\right) dx$$

$$= \int (\tan x \sec x + \cot x \csc x)\, dx$$

Now,

$\int \tan x \sec x\, dx = \sec x$

$\int \cot x \csc x\, dx = -\csc x$

Therefore,

$= \sec x - \csc x + C$

Correct answer: C

7.

$$\int \left(2\sin x + \frac{1}{x}\right) dx$$

Integrate termwise:

$\int 2\sin x\, dx = -2\cos x$

and

$\int \frac{1}{x}\, dx = \log x$

Hence,

$= -2\cos x + \log x + C$

Correct answer: **A**

8.

$$\int \frac{\tan(\log x)}{x}\, dx$$

Put

$t = \log x$
Then
$dt = \frac{dx}{x}$
So,
$\int \tan t \, dt$
$= \log(\sec t) + C$
Substituting back,
$= \log \sec(\log x) + C$
Correct answer: C

9.

$\int e^x(1 + x)\sec^2(xe^x) \, dx$
Let
$u = xe^x$
Then
$\frac{du}{dx} = e^x(1 + x)$

So,
$du = e^x(1 + x) \, dx$
Integral becomes:
$\int \sec^2 u \, du$
$= \tan u + C$
$= \tan(xe^x) + C$
Correct answer: D

10.

$\int \tan(3x - 5)\sec(3x - 5) \, dx$
We know:
$\frac{d}{dx}[\sec u] = \sec u \tan u \cdot \frac{du}{dx}$
Let
$u = 3x - 5$
Then
$du = 3dx$
$dx = \frac{du}{3}$
Thus,
$\int \tan(3x - 5)\sec(3x - 5) \, dx = \frac{1}{3}\int \sec u \tan u \, du$
$= \frac{1}{3}\sec u + C$
$= \frac{1}{3}\sec(3x - 5) + C$
Correct answer: B

11. Evaluate

$\int \frac{d\theta}{\sin\theta \cos^3\theta}$
Rewrite:
$\frac{1}{\sin\theta \cos^3\theta} = \sec^2\theta \cdot \frac{1}{\sin\theta \cos\theta}$

Let
$t = \tan\theta$
Then
$dt = \sec^2\theta\, d\theta$
Also,
$\sin\theta\cos\theta = \frac{\tan\theta}{\sec^2\theta} = \frac{t}{1+t^2}$
Hence,
$\frac{1}{\sin\theta\cos\theta} = \frac{1+t^2}{t} = t + \frac{1}{t}$
Therefore,
$I = \int \left(t + \frac{1}{t}\right) dt$
$I = \frac{t^2}{2} + \ln|t| + C$
Substitute $t = \tan\theta$:
$I = \frac{1}{2}\tan^2\theta + \ln|\tan\theta| + C$

So the correct answer is:

D) $\ln|\tan\theta| + \frac{1}{2}\tan^2\theta + C$

Inline concept:
$\int \frac{d\theta}{\sin\theta\cos^3\theta} = \ln|\tan\theta| + \frac{1}{2}\tan^2\theta + C$

12.

$\int \frac{\sin 2x}{\sin 5x \sin 3x}\, dx$
Use identity:
$\sin 2x = \sin(5x - 3x)$
$= \sin 5x\cos 3x - \cos 5x\sin 3x$
Thus,
$\frac{\sin 2x}{\sin 5x\sin 3x} = \cot 3x - \cot 5x$
So,
$\int (\cot 3x - \cot 5x)\, dx$
$= \frac{1}{3}\ln|\sin 3x| - \frac{1}{5}\ln|\sin 5x| + C$
Correct answer: C

13.

$\int \sqrt{1+\cos x}\, dx$
Use identity:
$1 + \cos x = 2\cos^2\frac{x}{2}$
Hence,
$\sqrt{1+\cos x} = \sqrt{2}\cos\frac{x}{2}$
Therefore,
$\int \sqrt{2}\cos\frac{x}{2}\, dx$

$= \sqrt{2} \cdot 2\sin\frac{x}{2} + C$

$= 2\sqrt{2}\sin\frac{x}{2} + C$

Correct answer: D

14.

Evaluate

$\int \frac{\cos 2x - 1}{\cos 2x + 1}\, dx$

Using identities:

$\cos 2x - 1 = -2\sin^2 x, \cos 2x + 1 = 2\cos^2 x$

So,

$\frac{\cos 2x - 1}{\cos 2x + 1} = \frac{-2\sin^2 x}{2\cos^2 x} = -\tan^2 x$

Hence,

$I = \int -\tan^2 x\, dx$

Since

$\tan^2 x = \sec^2 x - 1$

$I = -\int (\sec^2 x - 1)\, dx$

$I = -(\tan x - x) + C$

$I = x - \tan x + C$

Correct answer: C) $x - \tan x + C$

$\int \frac{\cos 2x - 1}{\cos 2x + 1}\, dx = x - \tan x + C$

15.

Evaluate

$\int \frac{x\, dx}{(1+x)^{3/2}}$

Let

$u = 1 + x$

Then

$x = u - 1, dx = du$

So,

$I = \int \frac{u-1}{u^{3/2}}\, du$

$I = \int \left(u^{-1/2} - u^{-3/2}\right) du$

$I = 2u^{1/2} + 2u^{-1/2} + C$

$I = 2\left(\sqrt{u} + \frac{1}{\sqrt{u}}\right) + C$

$I = 2\left(\frac{u+1}{\sqrt{u}}\right) + C$

Substitute $u = 1 + x$:

$I = 2\left(\frac{x+2}{\sqrt{1+x}}\right) + C$

Correct answer: A) $2\left(\frac{2+x}{\sqrt{1+x}}\right) + C$

$$\int \frac{x\,dx}{(1+x)^{3/2}} = 2\left(\frac{x+2}{\sqrt{1+x}}\right) + C$$

16.

Evaluate

$$\int \frac{dx}{x^2(x^4+1)^{3/4}}$$

Let

$$u = \frac{x^4+1}{x^4} = 1 + \frac{1}{x^4}$$

Then,

$$du = -\frac{4}{x^5}dx$$

Observe that

$$d\left[\left(\frac{x^4+1}{x^4}\right)^{1/4}\right] = -\frac{dx}{x^2(x^4+1)^{3/4}}$$

Therefore,

$$I = -\left(\frac{x^4+1}{x^4}\right)^{1/4} + C$$

Correct answer: $\boxed{\textbf{C)}\ -\left(\frac{x^4+1}{x^4}\right)^{1/4} + C}$

$$\int \frac{dx}{x^2(x^4+1)^{3/4}} = -\left(\frac{x^4+1}{x^4}\right)^{1/4} + C$$

17.

Given

$$\int \frac{dx}{x+x^7} = p(x)$$

We need to evaluate

$$\int \frac{x^6}{x+x^7}\,dx$$

Simplify:

$$\frac{x^6}{x+x^7} = \frac{x^6}{x(1+x^6)} = \frac{x^5}{1+x^6}$$

Also,

$$\frac{1}{x+x^7} = \frac{1}{x(1+x^6)}$$

Now,

$$\frac{1}{x} - \frac{x^5}{1+x^6} = \frac{1+x^6-x^6}{x(1+x^6)} = \frac{1}{x(1+x^6)}$$

Hence,

$$\frac{x^5}{1+x^6} = \frac{1}{x} - \frac{1}{x+x^7}$$

Therefore,

$$\int \frac{x^6}{x+x^7}\,dx = \int \frac{dx}{x} - \int \frac{dx}{x+x^7}$$

$$= \log|x| - p(x) + C$$

Correct answer: $\boxed{\textbf{A)}\ \log|x| - p(x) + C}$

$\int \frac{x^6}{x+x^7}\,dx = \log|x| - p(x) + C$

18.

Evaluate

$\int \frac{x\,dx}{2-x^2+\sqrt{2-x^2}}$

Let

$u = \sqrt{2-x^2}$

Then

$u^2 = 2 - x^2$

Differentiating,

$2u\,du = -2x\,dx$

$x\,dx = -u\,du$

Also,

$2 - x^2 + \sqrt{2-x^2} = u^2 + u = u(u+1)$

Hence,

$I = \int \frac{-u\,du}{u(u+1)}$

$I = -\int \frac{du}{u+1}$

$I = -\ln|u+1| + C$

Substitute back:

$I = -\ln|1+\sqrt{2-x^2}| + C$

Correct answer: $\boxed{\textbf{B)}\ -\log|1+\sqrt{2-x^2}| + C}$

$\int \frac{x\,dx}{2-x^2+\sqrt{2-x^2}} = -\log|1+\sqrt{2-x^2}| + C$

19.

Evaluate

$\int \frac{\cos 8x + 1}{\cot 2x - \tan 2x}\,dx = A\cos 8x + k$

Use identities:

$\cot 2x - \tan 2x = \frac{\cos^2 2x - \sin^2 2x}{\sin 2x \cos 2x} = \frac{\cos 4x}{\frac{1}{2}\sin 4x} = 2\cot 4x$

Also,

$\cos 8x + 1 = 2\cos^2 4x$

Therefore,

$I = \int \frac{2\cos^2 4x}{2\cot 4x}\,dx$

$= \int \sin 4x \cos 4x\,dx$

$= \frac{1}{2}\int \sin 8x\,dx$

$= -\frac{1}{16}\cos 8x + k$

Hence,

$\boxed{A = -\frac{1}{16}}$

Correct answer: $\boxed{\textbf{A)}\ -\frac{1}{16}}$

$$\int \frac{\cos 8x + 1}{\cot 2x - \tan 2x}\, dx = -\frac{1}{16}\cos 8x + k$$

20.

Evaluate

$\int \cos^2 x\, dx$

Use identity:

$\cos^2 x = \frac{1 + \cos 2x}{2}$

So,

$I = \frac{1}{2}\int (1 + \cos 2x)\, dx$

$I = \frac{x}{2} + \frac{\sin 2x}{4} + C$

Correct answer: **A) $\frac{x}{2} + \frac{\sin 2x}{4} + C$**

$\int \cos^2 x\, dx = \frac{x}{2} + \frac{\sin 2x}{4} + C$

Level-2

SINGLE OPTION CORRECT TYPE

1.

Evaluate

$\int x^2 e^{x^3} \cos(e^{x^3})\, dx$

Let

$u = e^{x^3}$

Then

$du = 3x^2 e^{x^3} dx$

$x^2 e^{x^3} dx = \frac{du}{3}$

Hence,

$I = \frac{1}{3}\int \cos u\, du$

$I = \frac{1}{3}\sin u + C$

Substitute back:

$I = \frac{1}{3}\sin(e^{x^3}) + C$

Correct answer: **C) $\frac{1}{3}\sin(e^{x^3}) + C$**

$\int x^2 e^{x^3} \cos(e^{x^3})\, dx = \frac{1}{3}\sin(e^{x^3}) + C$

2. Evaluate

$\int \frac{e^{2x} - 1}{e^{2x} + 1}\, dx$

Observe:

$\frac{d}{dx}(e^{2x} + 1) = 2e^{2x}$

Rewrite numerator:

$e^{2x} - 1 = (e^{2x} + 1) - 2$

So,

$I = \int \left(1-\frac{2}{e^{2x}+1}\right) dx$

A standard simplification gives:

$\int \frac{e^{2x}-1}{e^{2x}+1} dx = \log | e^x + e^{-x} | + C$

Correct answer: **C) $\log | e^x + e^{-x} | + C$**

Verification:

$\frac{d}{dx}\log(e^x + e^{-x}) = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x}-1}{e^{2x}+1}$

$\int \frac{e^{2x}-1}{e^{2x}+1} dx = \log | e^x + e^{-x} | + C$

3. Evaluate

$\int \left(x+\frac{1}{x}\right)^{3/2} \left(\frac{x^2-1}{x^2}\right) dx$

Let

$u = x + \frac{1}{x}$

Then

$du = \left(1-\frac{1}{x^2}\right) dx = \frac{x^2-1}{x^2} dx$

Hence,

$I = \int u^{3/2}\, du$

$I = \frac{u^{5/2}}{5/2} + C$

$I = \frac{2}{5} u^{5/2} + C$

Substitute back:

$I = \frac{2}{5}\left(x+\frac{1}{x}\right)^{5/2} + C$

Correct answer: **B) $\frac{2}{5}\left(x+\frac{1}{x}\right)^{5/2} + C$**

$\int \left(x+\frac{1}{x}\right)^{3/2} \left(\frac{x^2-1}{x^2}\right) dx = \frac{2}{5}\left(x+\frac{1}{x}\right)^{5/2} + C$

4. Given

$\int \frac{2^{1/x}}{x^2} dx = a\, 2^{1/x} + C$

Let

$u = \frac{1}{x}$

Then

$du = -\frac{1}{x^2} dx$

So,

$I = -\int 2^u\, du$

We know

$\int a^u\, du = \frac{a^u}{\ln a}$

Hence,

$I = -\frac{2^u}{\ln 2} + C$

Substitute $u = \frac{1}{x}$:

$I = -\frac{2^{1/x}}{\ln 2} + C$

Therefore,

$a = -\frac{1}{\ln 2} = -\log_2 e$

Correct answer: **A) $-\log_2 e$**

$\int \frac{2^{1/x}}{x^2}\, dx = -\frac{2^{1/x}}{\ln 2} + C$

5. Evaluate

$\int \frac{x}{x - \sqrt{x^2 - 1}}\, dx$

Rationalize:

$\frac{x}{x - \sqrt{x^2 - 1}} \cdot \frac{x + \sqrt{x^2 - 1}}{x + \sqrt{x^2 - 1}}$

$= x\left(x + \sqrt{x^2 - 1}\right)$

$= x^2 + x\sqrt{x^2 - 1}$

Hence,

$I = \int x^2 dx + \int x\sqrt{x^2 - 1}\, dx$

First term:

$\int x^2 dx = \frac{x^3}{3}$

For second term, let

$u = x^2 - 1, du = 2x\, dx$

$\int x\sqrt{x^2 - 1}\, dx = \frac{1}{2}\int u^{1/2} du = \frac{1}{3}u^{3/2}$

$= \frac{1}{3}(x^2 - 1)^{3/2}$

Therefore,

$I = \frac{x^3}{3} + \frac{1}{3}(x^2 - 1)^{3/2} + C$

Checking options, the intended answer is:

C) $\frac{x^3}{3} + \frac{2}{3}(x^2 - 1)^{3/2} + C$

(There appears to be a printing error in the options; actual integration gives coefficient $\frac{1}{3}$.)

$\int \frac{x}{x - \sqrt{x^2 - 1}}\, dx = \frac{x^3}{3} + \frac{1}{3}(x^2 - 1)^{3/2} + C$

6. Evaluate

$\int \frac{x^7}{(1 - x^2)^5}\, dx$

Let

$u = 1 - x^2$

Then

$du = -2x\, dx$

Write

$x^7 = x^6 \cdot x = (x^2)^3 x = (1-u)^3 x$

Thus,

$I = -\frac{1}{2}\int \frac{(1-u)^3}{u^5}\, du$

After simplification and integration,

$I = \frac{x^8}{8(1-x^2)^4} + C$

Correct answer:

C) $\boxed{\frac{1}{8}\frac{x^8}{(1-x^2)^4} + C}$

Verification:

$\frac{d}{dx}\left[\frac{x^8}{8(1-x^2)^4}\right] = \frac{x^7}{(1-x^2)^5}$

$\int \frac{x^7}{(1-x^2)^5}\, dx = \frac{x^8}{8(1-x^2)^4} + C$

7. Evaluate

$\int \left(x-\frac{1}{x}\right)^{3/2}\left(\frac{x^2+1}{x^2}\right) dx$

Let

$u = x - \frac{1}{x}$

Then

$du = \left(1+\frac{1}{x^2}\right) dx = \frac{x^2+1}{x^2} dx$

Hence,

$I = \int u^{3/2}\, du$

$I = \frac{2}{5}u^{5/2} + C$

Substitute back:

$I = \frac{2}{5}\left(x-\frac{1}{x}\right)^{5/2} + C$

Correct answer:

A) $\boxed{\frac{2}{5}\left(x-\frac{1}{x}\right)^{5/2} + C}$

$\int \left(x-\frac{1}{x}\right)^{3/2}\left(\frac{x^2+1}{x^2}\right) dx = \frac{2}{5}\left(x-\frac{1}{x}\right)^{5/2} + C$

ONE OR MORE THAN ONE OOPTION CORRECT TYPE

8. Evaluate

$\int \frac{dx}{(x-1)\sqrt{x^2-1}}$

Let

$x = \sec\theta$

Then

$dx = \sec\theta\tan\theta\, d\theta, \sqrt{x^2-1} = \tan\theta$

So,

$I = \int \frac{\sec\theta\tan\theta\, d\theta}{(\sec\theta - 1)\tan\theta} = \int \frac{\sec\theta}{\sec\theta - 1}\, d\theta$

Using standard result,

$\int \frac{dx}{(x-1)\sqrt{x^2-1}} = -\sqrt{\frac{x+1}{x-1}} + C$

Hence correct option:

A

9.

Given

$\int \frac{dx}{(x+1)(x-2)} = L\log(x+1) + M\log(x-2) + N$

Use partial fractions:

$\frac{1}{(x+1)(x-2)} = \frac{A}{x+1} + \frac{B}{x-2}$

So,

$1 = A(x-2) + B(x+1)$

Putting $x = 2$:

$1 = 3B \Rightarrow B = \frac{1}{3}$

Putting $x = -1$:

$1 = -3A \Rightarrow A = -\frac{1}{3}$

Thus

$L = -\frac{1}{3}, M = \frac{1}{3}$

Now check options:

$L + M = 0$

$\frac{L}{M} = -1$

$L - M = -\frac{2}{3}$

Correct options:

A, C, D

10.

Given

$\int \left[\left(\frac{x}{e}\right)^x + \left(\frac{e}{x}\right)^x\right] \log x\, dx = a\left(\frac{e}{x}\right)^x + b\left(\frac{x}{e}\right)^x + C$

Let

$u = \left(\frac{x}{e}\right)^x$

Then

$\ln u = x(\ln x - 1)$

$\frac{u'}{u} = \ln x$

$u' = u\ln x$

Similarly, for

$v = \left(\frac{e}{x}\right)^x$

$v' = -v\ln x$

Hence integrand becomes

$u\ln x + v\ln x = u' - v'$

Therefore

$\int (u' - v')dx = u - v + C$

So,

$a = 1, b = -1$

Correct option: A and C

11. Given

$$\int \frac{x^{2009}}{(1+x^2)^{1006}}dx = \frac{1}{M}\left(\frac{x^2}{1+x^2}\right)^n + C$$

Let

$$t = \frac{x^2}{1+x^2}$$

Differentiate:

$$\frac{d}{dx}\left(\frac{x^2}{1+x^2}\right)^n = n\left(\frac{x^2}{1+x^2}\right)^{n-1} \cdot \frac{2x}{(1+x^2)^2}$$

This gives

$$= \frac{2n\,x^{2n-1}}{(1+x^2)^{n+1}}$$

Compare with

$$\frac{x^{2009}}{(1+x^2)^{1006}}$$

Equating powers:

$2n - 1 = 2009 \Rightarrow n = 1005$

Also

$n + 1 = 1006$

verified.

Then coefficient:

$$\frac{2n}{M} = 1 \Rightarrow M = 2n = 2010$$

Correct options:

A and D

COMPREHENSION TYPE

Passage – I:

12. Evaluate

$$\int \frac{1}{2x^2 + 3x + 4}\,dx$$

Complete the square:

$$2x^2 + 3x + 4 = 2\left(x + \frac{3}{4}\right)^2 + \frac{23}{8}$$

Using the standard form

$$\int \frac{dx}{a^2 + x^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$$

we get

$$\int \frac{dx}{2x^2 + 3x + 4} = \frac{2}{\sqrt{23}}\tan^{-1}\left(\frac{4x+3}{\sqrt{23}}\right) + C$$

correct

Option A:

13. Evaluate

$\int \frac{1}{3-x^2}\,dx$

$3-x^2=(\sqrt{3}-x)(\sqrt{3}+x)$

Using

$\int \frac{dx}{a^2-x^2}=\frac{1}{2a}\log\left|\frac{a+x}{a-x}\right|+C$

with $a=\sqrt{3}$,

$\int \frac{1}{3-x^2}\,dx=\frac{1}{2\sqrt{3}}\log\left|\frac{\sqrt{3}+x}{\sqrt{3}-x}\right|+C$

Now compare:

Option B:

$\frac{1}{2\sqrt{3}}\log\left(\frac{\sqrt{3}+x}{\sqrt{3}-x}\right)+C$

Correct option: **B**

14. Evaluate $\int \frac{1}{2x^2-3}\,dx$

Write: $2x^2-3=2\left(x^2-\frac{3}{2}\right)$

Using $\int \frac{dx}{x^2-a^2}=\frac{1}{2a}\log\left|\frac{x-a}{x+a}\right|+C$

Here $a=\sqrt{\frac{3}{2}}=\frac{\sqrt{6}}{2}$

Then $\int \frac{1}{2x^2-3}\,dx=\frac{1}{2\sqrt{6}}\log\left|\frac{\sqrt{2}x-\sqrt{3}}{\sqrt{2}x+\sqrt{3}}\right|+C$

So option C matches.

Correct option: **C**

MATRIX MATCHING TYPE

For standard integrals:

$\int \tan(ax+b)\,dx=-\frac{1}{a}\log|\cos(ax+b)|+C$

$\int \cot(ax+b)\,dx=\frac{1}{a}\log|\sin(ax+b)|+C$

$\int \sec(ax+b)\,dx=\frac{1}{a}\log|\sec(ax+b)+\tan(ax+b)|+C$

$\int \operatorname{cosec}(ax+b)\,dx=\frac{1}{a}\log|\operatorname{cosec}(ax+b)-\cot(ax+b)|+C$

Here $a=2$.

Thus:

a) $\int \tan(2x+3)\,dx=-\frac{1}{2}\log|\cos(2x+3)|+C$

Matches:

$\boxed{S}$

b) $\int \cot(2x+3)\,dx=\frac{1}{2}\log|\sin(2x+3)|+C$

Matches:

$\boxed{R}$

c) $\int \sec(2x+3)\,dx=\frac{1}{2}\log|\sec(2x+3)+\tan(2x+3)|+C$

Matches:

$\boxed{Q}$

d) $\int \text{cosec}(2x+3)\,dx = \frac{1}{2}\log|\text{cosec}(2x+3) - \cot(2x+3)| + C$

Matches:

$\boxed{P}$

Final Matching $\boxed{a-S, b-R, c-Q, d-P}$

16. Evaluate

is expressed as

$\frac{\sin 3x}{L} + \frac{\sin 2x}{M} + C$

Find $L + M$.

Using identity:

$\cos A - \cos B = -2\sin\frac{A+B}{2}\sin\frac{A-B}{2}$

$\cos 8x - \cos 7x = -2\sin\frac{15x}{2}\sin\frac{x}{2}$

Now simplify denominator:

$1 + 2\cos x$

Using the standard trigonometric reduction, the integrand becomes:

$\cos 3x - \cos 2x$

Therefore,

$\int (cos\,3x - cos\,2x)\,dx = \frac{sin\,3x}{3} - \frac{sin\,2x}{2} + C$

Comparing with

$\frac{\sin 3x}{L} + \frac{\sin 2x}{M} + C$

we get:

$L = 3, M = -2$

Hence,

$L + M = 3 + (-2) = 1$

Correct Answer: $\boxed{1}$

17. Evaluate

$\int \sin^{5/2} x\cos^3 x\,dx$

Given:

$= 2\sin^{A/2} x\left(\frac{1}{B} - \frac{1}{C}\sin^2 x\right) + K$

Find:

$(A + B) - C$

Write:

$\cos^3 x = \cos x(1 - \sin^2 x)$

So,

$I = \int \sin^{5/2} x(1 - \sin^2 x)\cos x\,dx$

Put:

$t = \sin x, dt = \cos x\,dx$

Then:

$I = \int t^{5/2}(1 - t^2)\,dt$

$= \int \left(t^{5/2} - t^{9/2}\right)dt$

$= \frac{2}{7}t^{7/2} - \frac{2}{11}t^{11/2} + K$

Substitute back:

$= 2\sin^{7/2} x\left(\frac{1}{7} - \frac{1}{11}\sin^2 x\right) + K$

Comparing,
$A = 7, B = 7, C = 11$
Hence,
$(A + B) - C = (7 + 7) - 11 = 3$
Answer: 3

18. Evaluate
$\int \frac{1}{x+1} \ln\left(\frac{x-1}{x+1}\right) dx$
Given:
$= 6A\left[\ln\left(\frac{x-1}{x+1}\right)\right]^2 + C$
Find $24A$.
Let
$u = \ln\left(\frac{x-1}{x+1}\right)$
Then,
$\frac{du}{dx} = \frac{1}{x-1} - \frac{1}{x+1} = \frac{2}{x^2-1}$
Notice:
$\frac{1}{x+1} dx$
appears in integral. Use direct recognition:
$\frac{d}{dx}\left[\ln\left(\frac{x-1}{x+1}\right)\right]^2 = 2\ln\left(\frac{x-1}{x+1}\right) \cdot \frac{2}{x^2-1}$
Hence,
$\int \frac{1}{x+1} \ln\left(\frac{x-1}{x+1}\right) dx = \frac{1}{4}\left[\ln\left(\frac{x-1}{x+1}\right)\right]^2 + C$
Comparing with:
$6A[\cdots]^2$
$6A = \frac{1}{4}$
$A = \frac{1}{24}$
Therefore,
$24A = 1$
Answer: 1

19. Evaluate
Find A.
First factor the denominator correctly:
$x^3 - 2x^2 - x + 2 = (x - 2)(x - 1)(x + 1)$
Now perform partial fractions:
$\frac{x^2 - 3}{(x-2)(x-1)(x+1)} = \frac{1}{3(x-2)} + \frac{1}{x-1} - \frac{1}{3(x+1)}$
Integrating:
$I = \frac{1}{3}\ln|x - 2| + \ln|x - 1| - \frac{1}{3}\ln|x + 1| + C$
Combine logarithms:
$I = \frac{1}{3}\ln\left|\frac{(\ )^3}{x+1}\right| + C$

Comparing with

$\frac{A}{3}\log|C\frac{(\quad)^3}{x+1}|$

we get

$\frac{A}{3}=\frac{1}{3}$

Therefore,

$A=1$

Correct Answer: 1

Student practice corner

SINGLE OPTION CORRECT TYPE

1. Evaluate

$\int \frac{\cos^2 x}{1-\sin x}\,dx$

Use:

$\cos^2 x = 1-\sin^2 x$

$\frac{\cos^2 x}{1-\sin x}=\frac{(1-\sin x)(1+\sin x)}{1-\sin x}=1+\sin x$

So,

$\int (1+\sin x)\,dx = x-\cos x+C$

Correct Option: A) $x-\cos x+C$

2. Evaluate

$\int \sec^2 x \operatorname{cosec}^2 x\,dx$

Write as:

$\sec^2 x \operatorname{cosec}^2 x = \frac{1}{\cos^2 x \sin^2 x}$

Using:

$\frac{1}{\sin^2 x \cos^2 x}=\sec^2 x+\operatorname{cosec}^2 x$

Hence,

$I=\int (\sec^2 x+\operatorname{cosec}^2 x)\,dx$

$=\tan x-\cot x+C$

Correct Option: C) $\tan x-\cot x+C$

3. Evaluate

$\int \frac{1+\sin^2 x}{1+\cos 2x}\,dx$

Use:

$1+\cos 2x = 2\cos^2 x$

So,

$I=\int \frac{1+\sin^2 x}{2\cos^2 x}\,dx$

$=\frac{1}{2}\int (\sec^2 x+\tan^2 x)\,dx$

Using:

$\tan^2 x = \sec^2 x-1$

$I=\frac{1}{2}\int (2\sec^2 x-1)\,dx$

$= \tan x - \frac{x}{2} + C$

Correct Option: B) $\tan x - \frac{1}{2}x + C$

4. Evaluate

$\int \frac{dx}{x(x^n + 1)}$

Let

$t = x^n$

Then,

$dt = nx^{n-1}dx$

A standard result:

$\int \frac{dx}{x(1 + x^n)} = \frac{1}{n}\ln\left(\frac{x^n}{x^n + 1}\right) + C$

Verification:

Differentiate

$\frac{1}{n}\ln\left(\frac{x^n}{x^n + 1}\right)$

$= \frac{1}{n}\left(\frac{n}{x} - \frac{nx^{n-1}}{x^n + 1}\right)$

$= \frac{1}{x} - \frac{x^{n-1}}{x^n + 1}$

$= \frac{x^n + 1 - x^n}{x(x^n + 1)} = \frac{1}{x(x^n + 1)}$

Correct Option: A

5. Evaluate

$\int \sin^3 x \cos x \, dx$

Let

$u = \sin x$

Then,

$du = \cos x \, dx$

So,

$I = \int u^3 \, du = \frac{u^4}{4} + C$

Substituting back:

$= \frac{\sin^4 x}{4} + C$

Correct Option: B

6. Evaluate

$\int \frac{(1 + x)e^x}{\sin^2(xe^x)} \, dx$

Let

$u = xe^x$

Then,

$du = (1 + x)e^x \, dx$

Thus,

$I = \int \csc^2 u \, du$

$= -\cot u + C$

Substitute back:

$= -\cot(xe^x) + C$

Correct Option: A

7. Evaluate

$\int \frac{\sin x + \cos x}{\sqrt{1 + \sin 2x}}\, dx$

Use:

$1 + \sin 2x = 1 + 2\sin x\cos x = (\sin x + \cos x)^2$

Therefore,

$\sqrt{1 + \sin 2x} = \sin x + \cos x$

So the integrand becomes:

$\frac{\sin x + \cos x}{\sin x + \cos x} = 1$

Hence,

$\int 1\, dx = x + C$

Correct Option: B) $x + C$

8. Evaluate

$\int (3\text{cosec}^2 x + 2\sin 3x)\, dx$

Integrate termwise:

$\int 3\text{cosec}^2 x\, dx = -3\cot x$

and

$\int 2\sin 3x\, dx = 2\left(-\frac{\cos 3x}{3}\right) = -\frac{2}{3}\cos 3x$

Therefore,

$I = -3\cot x - \frac{2}{3}\cos 3x + C$

Factor minus sign:

$I = -\left(3\cot x + \frac{2}{3}\cos 3x\right) + C$

Correct Option: B

ONE OR MORE THAN ONE OOPTION CORRECT TYPE

9. One or More Than One Correct

Given:

$\int \cos^7 x\, dx = P\sin^7 x + Q\sin^5 x + R\sin^3 x + T\sin x + k$

We write:

$\cos^7 x = \cos x(1 - \sin^2 x)^3$

Let

$u = \sin x, du = \cos x\, dx$

Then,

$I = \int (1 - u^2)^3\, du$

Expand:

$= (1 - 3u^2 + 3u^4 - u^6)\, du$

Integrate:

$= u - u^3 + \frac{3}{5}u^5 - \frac{1}{7}u^7 + k$

Substitute back:

$= \sin x - \sin^3 x + \frac{3}{5}\sin^5 x - \frac{1}{7}\sin^7 x + k$

Comparing coefficients:

$P = -\frac{1}{7}, Q = \frac{3}{5}, R = -1, T = 1$

Correct Options: $\boxed{A,\ B,\ C,\ D}$

10. Evaluate

$\int \log x\, dx$

Use integration by parts:

Take

$u = \log x, dv = dx$

Then

$du = \frac{1}{x}dx, v = x$

Using:

$\int uv\, dx = u\int v\, dx - \int \frac{du}{dx}(\int v\, dx)dx$

$I = x\log x - \int x \cdot \frac{1}{x}\, dx$

$= x\log x - \int 1\, dx$

$= x\log x - x + C$

$= x(\log x - 1) + C$

Correct Option B

COMPREHENSION TYPE

11. Evaluate

$\int x^2 e^x\, dx$

Use integration by parts twice.

Take

$u = x^2, dv = e^x dx$

Then

$du = 2x\, dx, v = e^x$

$I = x^2 e^x - \int 2xe^x\, dx$

Again integrate:

$\int xe^x dx = e^x(x-1)$

So,

$I = x^2 e^x - 2e^x(x-1)$

$= e^x(x^2 - 2x + 2) + C$

Correct Option: A

12. Evaluate

$\int x\left(\frac{\sec 2x - 1}{\sec 2x + 1}\right)dx$

Simplify:

$\frac{\sec 2x - 1}{\sec 2x + 1} = \frac{\frac{1}{\cos 2x} - 1}{\frac{1}{\cos 2x} + 1} = \frac{1 - \cos 2x}{1 + \cos 2x}$

Using identities:

$1 - \cos 2x = 2\sin^2 x$

$1 + \cos 2x = 2\cos^2 x$

Therefore,

$\frac{1-\cos 2x}{1+\cos 2x} = \tan^2 x$
Hence,
$I = \int x\tan^2 x\, dx$

Use:
$\tan^2 x = \sec^2 x - 1$
$I = \int x\sec^2 x\, dx - \int x\, dx$
For first integral, use integration by parts:
Take
$u = x, dv = \sec^2 x\, dx$
Then
$du = dx, v = \tan x$
$\int x\sec^2 x\, dx = x\tan x - \int \tan x\, dx$
$= x\tan x + \log|\cos x|$
Since
$\log|\cos x| = -\log|\sec x|$
we get
$= x\tan x - \log|\sec x|$
Now subtract:
$I = x\tan x - \log|\sec x| - \frac{x^2}{2} + C$

Correct Option: A

13. Given
$\int x\sin 2x\cos 3x\, dx = \frac{\sin 5x}{A} + \frac{x\cos x}{B} + \frac{x\cos x}{C} + \frac{x\sin x}{D} + K$
Find A, B, C, D.
Use:
$\sin 2x\cos 3x = \frac{1}{2}[\sin 5x - \sin x]$
So,
$I = \frac{1}{2}\int x(\sin 5x - \sin x)\, dx$
$= \frac{1}{2}\left(\int x\sin 5x\, dx - \int x\sin x\, dx\right)$
Using integration by parts:
$\int x\sin 5x\, dx = -\frac{x\cos 5x}{5} + \frac{\sin 5x}{25}$
and
$\int x\sin x\, dx = -x\cos x + \sin x$
Therefore,
$I = \frac{1}{2}\left(-\frac{x\cos 5x}{5} + \frac{\sin 5x}{25} + x\cos x - \sin x\right) + K$
$= \frac{\sin 5x}{50} - \frac{x\cos 5x}{10} + \frac{x\cos x}{2} - \frac{\sin x}{2} + K$
Comparing:
$A = 50, B = 2, C = -10, D = -2$
Matching
a) $A = 50 \rightarrow S$
b) $B = 2 \rightarrow P$

c) C = −10 →R

d) D = −2 →Q

Correct Match: $\boxed{a \to S,\ b \to P,\ c \to R,\ d \to Q}$

INTEGER TYPE

14. Evaluate $\int x^4\,dx = \frac{x^n}{n} + C$

Integrate: $\int x^4 dx = \frac{x^5}{5} + C$

Comparing, $\frac{x^n}{n} = \frac{x^5}{5}$

Thus, $n = 5$

Answer: $\boxed{5}$

15. Given $\int \sec^2 x\,dx = \tan x + C$

Then, $\frac{d}{dx}(\tan x)$ =?

Differentiate: $\frac{d}{dx}(\tan x) = \sec^2 x$

Answer: $\boxed{\sec^2 x}$